OPTIMAL FEEDBACK LAW RECOVERY BY GRADIENT-AUGMENTED NEURAL NETWORK WITH NONCONVEX REGULARIZATION ON THE UNBOUNDED PARAMETER SPACE

Abstract. We recover optimal feedback laws for nonlinear control problems by approximating Hamilton–Jacobi–Bellman value functions with sparse shallow neural networks. Value and gradient data are generated from the open-loop problem by the Pontryagin maximum principle, and the network is trained with a gradient-augmented H^1 fidelity. For nonhomogeneous activation functions on the unbounded parameter space $\Omega = \mathbb{R}^{d+1}$, fix $p > 0$, set $w_p(\omega) = 1 + |\omega|^p$ and $\mu_p = w_p\mu$, and normalize the ridge atom K as $K_p = K/w_p$. We study the normalized objective

$$J(\mu) = L(\mu) + \alpha\Phi_\phi(\mu_p),$$

where the concave scalar penalty has finite positive right slope at zero. Normalization makes the dictionary vanish at infinity under a polynomial growth gap. This yields existence by normalized weak-* compactness. Under strict bounded-interval curvature, every finite-objective local minimizer has finite support and satisfies a covering-number bound, while the global minimum is attained by a finite network. Exact maximization of the weighted Gâteaux derivative at each insertion, followed by a nonincreasing correction, drives its excess above the insertion threshold to zero, with a best-iterate rate of order $n^{-1/2}$. For positively k-homogeneous activations, we retain the equivalent formulation on the unit sphere with its induced fractional-power penalty on the outer coefficients. The resulting algorithms are studied on the Van der Pol oscillator and the pendulum swing-up problem, whose value function has a gradient discontinuity across a switching set.

Contents

1. Introduction

The problem under consideration is the dynamic optimal control problem in the space-time domain

$$(P)\qquad \begin{aligned} &\inf_{u\in L^2(t_0,T;\mathbb{R}^{m_u})} \mathcal{C}(u;t_0,x) := \int_{t_0}^{T} \left(h(y(t)) + \eta|u(t)|^2\right) dt, \quad \eta>0,\\ &\text{s.t.}\\ &\dot{y} = f(y) + g(y)u,\\ &y(t_0) = x, \end{aligned}$$

We assume $y(t)\in\mathbb{R}^d$, $u(t)\in\mathbb{R}^{m_u}$, $h\in C^1(\mathbb{R}^d)$, $f\in C^1(\mathbb{R}^d;\mathbb{R}^d)$, and $g\in C^1(\mathbb{R}^d;\mathbb{R}^{d\times m_u})$. We assume throughout that a minimizer of (P) exists.

One approach computes an open-loop state-costate-control path for a fixed initial condition x. The Pontryagin maximum principle (PMP) yields a coupled forward–backward two-point boundary value problem:

$$(\text{TPBVP})\qquad \begin{cases} \dfrac{d}{dt}y^* = f(y^*) + g(y^*)u^*,\\ y^*(t_0) = x,\\ -\dfrac{d}{dt}p^* = \partial_y\Big(f(y^*) + g(y^*)u^*\Big)^\top p^* + \partial_y h(y^*),\\ p^*(T) = 0. \end{cases}$$

The stationarity condition with respect to the control gives

$$u^* = -\frac{1}{2\eta} g(y^*)^\top p^* \quad \forall t\in(t_0,T).$$

The open-loop control is not robust with respect to disturbances of the state [1]: it is computed for a single initial condition, and deviations from the optimal path are not corrected. In addition, numerical methods for the two-point boundary value problem are sensitive to the initial guess and may fail to converge. The dynamic programming principle instead introduces the value function

$$V(t,x) := \inf_u \mathcal{C}(u;t,x)$$

which satisfies the Hamilton–Jacobi–Bellman (HJB) equation:

$$(\text{HJB})\qquad \begin{cases} \partial_t V - \dfrac{1}{4\eta}\nabla V(t,x)^\top g(x)g(x)^\top \nabla V(t,x) + \nabla V(t,x)^\top f(x) + h(x) = 0,\\ V(T,x) = 0. \end{cases}$$

The value function encodes the optimal cost from every state. The feedback law synthesized from it corrects deviations from a nominal path.

For nonlinear control problems one usually seeks a feedback law, which requires solving the HJB equation. Computing this function is notoriously difficult because of the curse of dimensionality. Alongside classical methods, data-driven methods have therefore become increasingly common. One line of work, introduced in [20], generates the training data with (TPBVP) and approximates the optimal value function from it. The optimal feedback law is then synthesized from the fitted value function after validation. Its samples are causality-free: generating one does not require approximate values at neighbouring grid points. Moreover, Azmi, Kalise and Kunisch [1] enrich the dataset with the gradient of the value function: a state-adjoint-control triple is generated by solving the open-loop problem, and the adjoint supplies the gradient. The value function is then approximated by hyperbolic-cross basis functions with an ℓ^1 penalty, and including the adjoint data targets an H^1 approximation. Neural-network approximations form another growing line of work [18, 28].

For $\boldsymbol{\omega} = (\omega_n)_{n=1}^N$, $\omega_n = (a_n, b_n)$, and $c = (c_n)_{n=1}^N$, we use the two-layer network

$$\mathcal{N}_{\boldsymbol{\omega},c}(x) := \sum_{n=1}^{N} c_n \rho(a_n \cdot x + b_n) \in \mathbb{R}, \tag{1}$$

where ρ is a sufficiently smooth activation function and N the width of the network. We associate it with the finite signed measure $\mu_{\boldsymbol{\omega},c} := \sum_{n=1}^N c_n \delta_{\omega_n}$. Its gradient is

$$\nabla_x \mathcal{N}_{\boldsymbol{\omega},c}(x) = \sum_{n=1}^{N} c_n \nabla_x \rho(a_n \cdot x + b_n) \in \mathbb{R}^d.$$

The training data are generated by solving (TPBVP). The adjoint state p^* is obtained as a byproduct and provides the gradient information of the optimal value function: $p^*(t) = \nabla V(t, y^*(t))$ wherever $V(t,\cdot)$ is differentiable at $y^*(t)$ [9, 7, 8]. Where V is not differentiable — on the switching sets studied in Section 6 — competing optimal paths may have distinct costates, and each sampled p^* gives the corresponding one-sided gradient of V. Thus the data consist of triples $\{x^i, V(x^i), p^{*,i}\}_{i=1}^M$; wherever V is differentiable, $p^{*,i} = \nabla V(x^i)$. For the finite-horizon problem, we write $V(x) := V(0,x)$. The infinite-horizon pendulum value considered below is stationary, so it has no suppressed time argument. We have the minimization problem

$$\min_{N,\boldsymbol{\omega},c} l^M(\mu_{\boldsymbol{\omega},c}), \qquad l^M(\mu_{\boldsymbol{\omega},c}) := \frac{1}{2M}\sum_{m=1}^{M} \left(|V(x^m) - \mathcal{N}_{\boldsymbol{\omega},c}(x^m)|^2 + |\nabla V(x^m) - \nabla \mathcal{N}_{\boldsymbol{\omega},c}(x^m)|^2 \right).$$

Section 3 introduces the corresponding population fidelity L with respect to Lebesgue measure on D.

A traditional sparse shallow network uses the ReLU activation and an ℓ^1 penalty on the outer coefficients [5, 2]. Under gradient-augmented fitting, this formulation has two limitations. 1. The convex penalty may retain redundant neurons whose inner parameters cluster. 2. The second-order weak derivative of ReLU does not exist as a function while it is a distribution, and we cannot use it to approximate high-order derivatives.

Sparsity beyond the convex penalty. The number of neurons N is itself an optimization variable. Although the number of neurons does not act as the inductive bias of the model in general [21], sparsity is crucial for computational efficiency in many problems. Furthermore, sparsity is necessary for certain applications, such as knowledge distillation and deconvolution of point-like objects. A standard sparse-learning procedure uses a convex reformulation with a sparsity-promoting penalty on the outer weights, most commonly the ℓ^1 penalty (LASSO). Pieper and Petrosyan [24] showed that convex penalties cannot effectively eliminate this redundancy when neurons cluster in a small neighborhood and introduced a concave nonconvex functional for this purpose. We extend the relaxed concave measure formulation to gradient-augmented fitting with an unbounded, nonhomogeneous dictionary. Placing the parameter weight inside the scalar penalty preserves its strict curvature after normalization and yields both finite support and a quantitative neuron bound. For positively k-homogeneous activations, radial rescaling instead induces a fractional exponent and its corresponding width bound.

Activation functions beyond ReLU. We therefore consider smoother activation functions. This extension requires a treatment of the unbounded inner-parameter space for gradient training. For positively k-homogeneous activations, radial scaling can be absorbed into the outer coefficient, and the network has an equivalent representation with inner parameters on the unit sphere. For nonhomogeneous activations such as tanh, softplus, and the Gaussian, radial scaling changes the atom and cannot be removed by normalization. Without normalization, sequences with $|\omega| \to \infty$ may retain a nonzero contribution to the represented function, as shown in Section 3.3. We therefore introduce

$$w_p(\omega) = 1 + |\omega|^p, \qquad \mu_p = w_p\mu, \qquad K_p = \frac{K}{w_p}.$$

Then $\mathcal{N}\mu = \int_\Omega K_p \, d\mu_p$, and K_p vanishes at infinity when p exceeds the polynomial growth of K. We analyze

$$J(\mu) = L(\mu) + \alpha\Phi_\phi(\mu_p),$$

whose finite-network penalty is $\alpha \sum_n \phi(w_p(\omega_n)|c_n|)$ after repeated locations are merged.

Contributions. The contributions are the following.

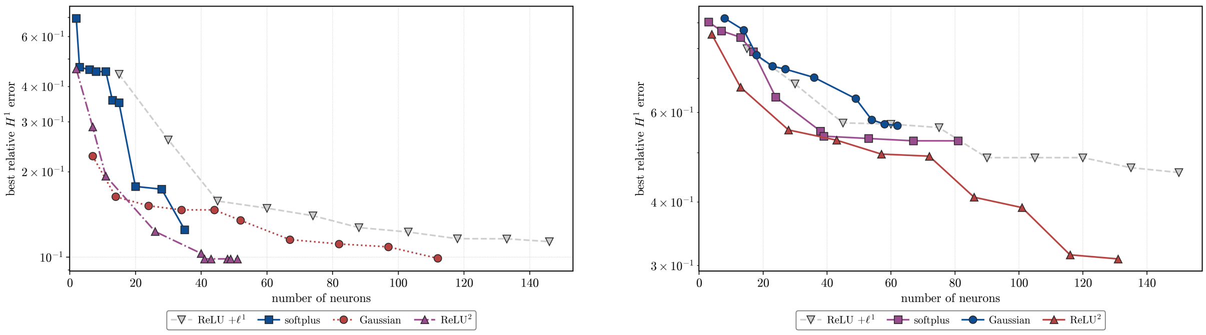
(1) *Nonhomogeneous activations on the unbounded parameter domain.* For concave scalar penalties with finite positive right slope, we prove a disjoint-test representation and derive weak-* lower semicontinuity and, under coercivity, bounded total-variation sublevels (Theorem 3.9 and Corollary 3.10). The weighted-measure isometry in Theorem 3.13 moves the parameter growth into the normalized atom K_p. For continuous activations of polynomial value growth $s_0 < p$, this gives a global minimizer on $\Omega = \mathbb{R}^{d+1}$ (Theorem 3.16). The unnormalized comparison problem can still lose lower semicontinuity or fail to attain its infimum (Examples 3.11 and 3.12). Under full H^1 growth $s_1 < p$ and strict bounded-interval curvature, every finite-objective local minimizer is finite atomic, its normalized atoms are quantitatively separated, and its width is bounded by a covering number (Theorem 3.21). Corollary 3.22 gives an attained finite-network problem and an a priori neuron bound. The normalized optimality system is Theorem 3.18, while Theorem 5.1 gives squared descent and an $n^{-1/2}$ best-iterate rate for the excess above the insertion threshold.

(2) *The k-homogeneous formulation.* For positively k-homogeneous activations, minimizing a separable penalty on the inner and outer parameters over radial representations induces the fractional-power penalty $|c|^{2/(k+1)}$ on the unit sphere (Lemma 4.3), and the resulting problem is posed over all of $\mathcal{M}(\mathbb{S}^d)$. Finite atomic truncations show that this problem and the finite-network problem have the same infimum (Proposition 4.10). We derive the atomwise optimality conditions and a positive lower bound on the weights of every total-variation local minimizer (Theorem 4.11), and prove that a global minimizer exists, that every global minimizer is a finite network obeying an explicit a priori width bound (Theorem 4.13), and that every total-variation local minimizer with finite objective has finite support (Corollary 4.14).

(3) *Gradient-augmented algorithms and feedback recovery.* The first insertion algorithm places the polynomial parameter weight inside the log penalty and searches $|P^M_{p,\mu}|$; the second uses the fractional-exponent penalty on the unit sphere. The numerical examples compare activation regularity, support size, and feedback recovery for a smooth value function and for a value function whose gradient is discontinuous across a switching set.

Figure 1 compares the traditional ReLU network with an ℓ^1 penalty against representative models from the two formulations. The Gaussian and ReLU^2 reduce the relative H^1 error on the Van der Pol problem, and ReLU^2 gives the lowest error on the pendulum problem.

The thesis is organized as follows. Section 2 summarizes the general measure-theoretic and approximation results needed for the proofs. Section 3 introduces the learning problem over measures, exhibits escape in the unnormalized problem, and develops the normalization, existence, optimality, finite-support, neuron-bound, and insertion-rate theory for nonhomogeneous activations. Section 4 treats the k-homogeneous reformulation and the fractional-exponent penalty. Section 5 describes the data generation and the two insertion algorithms, and Section 6 reports the numerical studies.



(A) Van der Pol oscillator.

(B) Pendulum swing-up.

FIGURE 1. Running best relative H^1 validation error as neurons are inserted. Van der Pol parameters: ReLU with ℓ^1, $\alpha = 10^{-5}$; softplus, $(\alpha, \beta, p, \gamma) = (10^{-5}, 10^{-10}, 2.01, 1)$; Gaussian, $(10^{-5}, 10^{-10}, 3, 1)$; and ReLU^2, $\alpha = 10^{-5}$. Pendulum parameters: ReLU with ℓ^1, $\alpha = 10^{-5}$; softplus, $(10^{-4}, 10^{-10}, 2.01, 1)$; Gaussian, $(10^{-4}, 10^{-4}, 2.01, 0)$; and ReLU^2, $\alpha = 10^{-6}$. Here β is the coefficient of the additional moment penalty used in these frozen numerical runs; it is not part of the revised objective analyzed in Sections 3–5. All curves use seed 42.

The overall pipeline has four steps:

(1) **data generation**: initially, a dataset of causality-free state-value-gradient triples must be generated. Many methods are discussed in the literature [14, 11]. The process is highly parallelizable and adaptive: dense samples can be placed in regions of interest.
(2) **model selection:** depending on the underlying system, different activation functions and penalties can be chosen. The activation function and the regularization affect the approximation error and support size. We can also incorporate other prior information such as semiconcavity of the value function or the physics-informed loss [10, 19, 26, 29, 17].
(3) **training and validation**: this step follows the standard machine-learning pipeline. A neural network is trained to approximate the optimal value function, and the performance of the model is subsequently evaluated on a separate validation set. The validation set can be chosen to probe specific regions, such as the switching set.
(4) **feedback control**: once the value function is trained, the feedback law follows from the Hamiltonian minimization. Sparsity keeps the evaluation of the law cheap.

2. PRELIMINARIES

This section reviews general theories used in the thesis. The learning objectives and problem-specific assumptions are introduced in Section 3.

2.1. **Finite signed Radon measures.** Let Ω be a locally compact Polish space. We write $\mathcal{M}(\Omega)$ for the Banach space of finite signed Radon measures on Ω, equipped with the total-variation norm

$$\|\mu\|_{\mathcal{M}(\Omega)} := |\mu|(\Omega).$$

By the Riesz representation theorem, $\mathcal{M}(\Omega) = C_0(\Omega)^*$ under the pairing

$$\langle f, \mu \rangle := \int_\Omega f \, d\mu.$$

Here $C_0(\Omega)$ denotes the continuous functions vanishing at infinity.

Let $\mathcal{M}^0(\Omega)$ and $\mathcal{M}^\sharp(\Omega)$ denote, respectively, the nonatomic and purely atomic measures.

For $\mu \in \mathcal{M}(\Omega)$ we define the *atom set* $\operatorname{atom}\mu := \{\omega \in \Omega : |\mu|(\{\omega\}) > 0\}$, which is at most countable, the purely atomic part $\mu_{\text{atom}} := \sum_{\omega \in \operatorname{atom}\mu} \mu(\{\omega\})\, \delta_\omega \in \mathcal{M}^\sharp(\Omega)$, and the nonatomic part $\mu_{\mathrm{cont}} := \mu - \mu_{\mathrm{atom}} \in \mathcal{M}^0(\Omega)$ [6]. This gives the decomposition

$$\mathcal{M}(\Omega) = \mathcal{M}^0(\Omega) \oplus \mathcal{M}^\sharp(\Omega).$$

The atom set need not be closed because atoms may accumulate at a point that carries no mass. The polar decomposition of a signed measure is written as

$$\mu = \mathrm{sign}(\mu)\,|\mu|, \qquad \mathrm{sign}(\mu) := \frac{d\mu}{d|\mu|} \in \{-1, 1\} \quad |\mu|\text{-almost everywhere.}$$

2.2. **Measures with a finite moment.** In this subsection let $\Omega = \mathbb{R}^{d_\Omega}$, where $d_\Omega \geq 1$. For $p > 0$, set

$$w_p(\omega) := 1 + |\omega|^p$$

and define

$$\mathcal{M}_p(\Omega) := \left\{ \mu \in \mathcal{M}(\Omega) : \int_\Omega w_p\, d|\mu| < \infty \right\}, \qquad \|\mu\|_{\mathcal{M}_p(\Omega)} := \int_\Omega w_p\, d|\mu|. \tag{2}$$

The associated space of continuous functions is

$$C_{w_p^{-1}}(\Omega) := \left\{ f \in C(\Omega) : \frac{f}{w_p} \in C_0(\Omega) \right\}, \qquad \|f\|_{C_{w_p^{-1}}} := \sup_{\omega \in \Omega} \frac{|f(\omega)|}{w_p(\omega)}.$$

Under the dual pairing, $\mathcal{M}_p(\Omega) = C_{w_p^{-1}}(\Omega)^*$ isometrically [4, Section 5]. For an atomic measure,

$$\left\| \sum_{j=1}^N c_j \delta_{\omega_j} \right\|_{\mathcal{M}_p(\Omega)} = \sum_{j=1}^N |c_j| \big(1 + |\omega_j|^p\big).$$

2.3. **Barron spaces.** In view of the approximation by neural networks, the seminal paper [3] develops a universal approximation theorem on function approximation by feedforward neural networks with sigmoidal activation functions. Later, due to the popularity of ReLU, the approximation properties of networks with ReLU activation are investigated in the subsequent studies [22, 27, 5, 23]. In particular, it is argued that generalization is achieved not by limiting the size of the network but by controlling the magnitude of the weights. Let $\mathcal{W}(D)$ be the space of functions f on D that can be represented by $f(x) = \int_{\mathbb{S}^d} (a \cdot x + b)_+ \, d\mu(a, b)$ for every $x \in D$ and some $\mu \in \mathcal{M}(\mathbb{S}^d)$. For $f \in \mathcal{W}(D)$, we define the norm

$$\|f\|_{\mathcal{W}(D)} = \inf \left\{ \|\mu\|_{\mathcal{M}(\mathbb{S}^d)} : \ f(x) = \int_{\mathbb{S}^d} (a \cdot x + b)_+ \, d\mu(a, b) \text{ for every } x \in D \right\}.$$

It is shown that this norm is closely related to a fractional Laplacian of f [22, 27].

For ReLU^k with $k \geq 2$, we follow the extended Barron spaces of [16]. There, a function of the form

$$f(x) = \int c\,(a \cdot x + b)_+^k \, d\varrho(a, b, c),$$

with ϱ a *probability* measure over the parameters $(a, b) \in \mathbb{R}^{d+1}$ and the outer weight $c \in \mathbb{R}$, is assigned the representation cost

$$\|f\|_{B^k_{1,\varrho}} := \int |c| \left(\|a\|_1 + |b| \right)^k d\varrho(a, b, c),$$

and the Barron norm is the infimum over all representing probabilities,

$$\|f\|_{B^k_1} := \inf_\varrho \|f\|_{B^k_{1,\varrho}}.$$

The space $B^k_1(D)$ of functions on D with finite Barron norm is a normed vector space [16, Theorem 6].

Proposition 2.1 (Barron–Sobolev embedding, [16, Theorem 7])**.** *Given any fixed integer k, there holds $B_1^k(D) \subset H^k(D)$; moreover there is a constant $C(D,d,k)$ such that*

$$\|f\|_{H^m(D)} \le C(D,d,k)\, \|f\|_{B_1^k}, \qquad 0 \le m \le k,$$

for all $f \in B_1^k(D)$.

For a Lipschitz activation ρ, let

$$\mathcal{G}_{\rho,f} := \left\{ \mu \in \mathcal{M}(\mathbb{R}^{d+1}) : f(x) = \int_{\mathbb{R}^{d+1}} \rho(a\cdot x + b)\, d\mu(a,b) \text{ for every } x \in D \right\}.$$

Following [12], define

$$\|f\|_{B_\rho} := \inf_{\mu \in \mathcal{G}_{\rho,f}} \int_{\mathbb{R}^{d+1}} (1 + \|a\|_1 + |b|)\, d|\mu|(a,b), \qquad B_\rho := \{f : \|f\|_{B_\rho} < \infty\}.$$

Proposition 2.2 (Embeddings into ReLU^k Barron spaces, [12, Theorem 1])**.** *Let ρ be a Lipschitz activation function and let D be bounded.*

1. *If $\rho \in C^1(\mathbb{R})$ and $\partial^2\rho \in L^1(\mathbb{R})$, then $B_\rho \hookrightarrow B_{\mathrm{ReLU}}$.*
2. *If $k \ge 1$, $\rho \in C^k(\mathbb{R})$, and the one-sided (Caputo) derivatives of order $k+1$ satisfy $\partial_+^{k+1}\rho \in L^1((0,\infty))$ and $\partial_-^{k+1}\rho \in L^1((-\infty,0))$ — for which it suffices that the $(k+1)$-st distributional derivative of ρ exists as a finite measure — then*

$$B_\rho \hookrightarrow B_{\mathrm{ReLU}^k}.$$

Notation. Throughout the remainder of the thesis, $D \subset \mathbb{R}^d$ is a bounded Lipschitz domain, $\nu := \mathcal{L}^d|_D$ is the population measure, and $L^2(D)$ means $L^2(D,\nu)$. The symbol M denotes the number of samples in an empirical loss. The inner-parameter domain is $\Omega = \mathbb{R}^{d+1}$, and $\rho \in C(\mathbb{R})$ denotes the fixed activation function of (1), with

$$\omega = (a,b), \qquad [K(\omega)](x) := \rho(a\cdot x + b).$$

For a finite network, we write

$$\boldsymbol{\omega} = (\omega_n)_{n=1}^N, \qquad c = (c_n)_{n=1}^N, \qquad \mu_{\boldsymbol{\omega},c} := \sum_{n=1}^N c_n \delta_{\omega_n}, \qquad \mathcal{N}_{\boldsymbol{\omega},c} := \sum_{n=1}^N c_n K(\omega_n).$$

The symbol δ_ω denotes the Dirac measure at ω, while μ' denotes a general signed-measure direction. We reserve p for the moment order.

3. The learning problem

3.1. The value operator and the gradient-augmented loss.

Whenever the integral exists, we define the *value operator* of a measure $\mu \in \mathcal{M}(\Omega)$ by

$$[\mathcal{N}\mu](x) := \int_\Omega [K(\omega)](x)\, d\mu(\omega).$$

In particular, $\mathcal{N}\mu_{\boldsymbol{\omega},c} = \mathcal{N}_{\boldsymbol{\omega},c}$.

Assumption 3.1 (Regularity and polynomial growth of ρ)**.** The activation satisfies $\rho \in C^1(\mathbb{R})$. There are $C_\rho > 0$, $s_0 \ge 0$, and $s_1 \ge \max\{s_0, 1\}$ such that

1. $|\rho(z)| \le C_\rho(1+|z|)^{s_0}$ for every $z \in \mathbb{R}$;
2. $|\rho'(z)| \le C_\rho(1+|z|)^{s_1-1}$ for every $z \in \mathbb{R}$.

Since D is bounded, Assumption 3.1(1) gives a constant C_D with

$$\|K(\omega)\|_{L^2(D)} \le C_D(1+|\omega|)^{s_0}. \tag{3}$$

Thus $\mathcal{N}\mu$ is a well-defined $L^2(D)$-valued Bochner integral if $\int_\Omega (1+|\omega|)^{s_0}\, d|\mu|(\omega) < \infty$; in particular this holds on $\mathcal{M}_p(\Omega)$ when $p > s_0$. The map $K : \Omega \to L^2(D)$ is continuous: if $\omega_n \to \omega$, all pre-activations $a_n \cdot x + b_n$ remain in one compact interval for $x \in D$, and uniform continuity of ρ on that interval gives $K(\omega_n) \to K(\omega)$ uniformly on D.

The gradient of the operator is understood distributionally; whenever

$$\int_\Omega \|\nabla K(\omega)\|_{L^2(D)}\, d|\mu|(\omega) < \infty,$$

the map $\omega \mapsto \partial_{x_i} K(\omega)$ is Bochner integrable into $L^2(D)$. Fubini's theorem against test functions gives

$$\nabla(\mathcal{N}\mu)(x) = \int_\Omega \nabla K(\omega)(x)\, d\mu(\omega).$$

We write $\nabla\mathcal{N}\mu := \nabla(\mathcal{N}\mu)$ in this case and set $r_\mu := \mathcal{N}\mu - V$.

The fidelity term is the gradient-augmented loss with the distributional gradient convention,

$$L(\mu) := \begin{cases} \frac{1}{2}\|\mathcal{N}\mu - V\|^2_{L^2(D)} + \frac{1}{2}\|\nabla(\mathcal{N}\mu) - \nabla V\|^2_{L^2(D)}, & \mathcal{N}\mu \in H^1(D), \\ +\infty, & \text{otherwise.} \end{cases}$$

For sample points $x^1, \dots, x^M \in D$ and a finite network for which the displayed point values exist, the corresponding empirical fidelity is

$$l^M(\mu) := \frac{1}{2M}\sum_{m=1}^{M} \left(|r_\mu(x^m)|^2 + |\nabla r_\mu(x^m)|^2 \right). \tag{4}$$

3.2. **The nonconvex functional.** Let $\phi : \mathbb{R}_+ \to \mathbb{R}_+$ satisfy the following assumptions.

Assumption 3.2. The finite right derivative

$$L_\phi := \phi'(0+)$$

exists and satisfies $0 < L_\phi < \infty$. We use the convention $\phi'(0) := L_\phi$.

Assumption 3.3. The function ϕ is concave, nondecreasing, and coercive, with $\phi(0) = 0$, and belongs to $C^1((0,\infty))$.

Assumptions 3.2 and 3.3 imply that $\phi'(t) > 0$ for every $t > 0$. Indeed, if $\phi'(t_0) = 0$ at some $t_0 > 0$, concavity and monotonicity would give $\phi'(t) = 0$ for every $t \geq t_0$, making ϕ constant there and contradicting coercivity. Hence ϕ is continuous and strictly increasing from $[0,\infty)$ onto $[0,\infty)$. We denote its inverse by ϕ^{-1}.

Assumption 3.4 (Local curvature)**.** The function ϕ belongs to $C^2((0,\infty))$ and satisfies both of the following conditions:

(1) there are $\gamma_1 \geq 0$ and $\hat{z}_1 > 0$ such that $-\gamma_1(z_2 - z_1) \leq \phi'(z_2) - \phi'(z_1) \leq 0$ for $0 \leq z_1 \leq z_2 \leq \hat{z}_1$;

(2) for every $A > 0$,

$$\gamma_A := -\sup_{0<t\leq A} \phi''(t) > 0.$$

Remark 3.5. The lower curvature control in Assumption 3.4(1) is local and does not follow from global concavity. For example, $\phi(z) = z - \frac{2}{3}z^{3/2}$ near the origin has $\phi''(z) = -\frac{1}{2}z^{-1/2}$; it can be extended linearly after a small positive point so as to remain nondecreasing, concave, and coercive.

Lemma 3.6. *Let $\phi : [0,\infty) \to [0,\infty)$ be nondecreasing and concave with $\phi(0) = 0$. Then*

(1) $\phi(z_1 + z_2) \leq \phi(z_1) + \phi(z_2)$ *for all* $z_1, z_2 \geq 0$.

If in addition $L_\phi := \phi'(0+)$ is finite, then

(2) $\phi(z) \leq L_\phi z$ *for all* $z \geq 0$*;*

(3) $\phi(z)/z \longrightarrow L_\phi$ *as* $z \downarrow 0$*;*

(4) *$\phi\big(\sum_{j\geq 1} t_j\big) \leq \sum_{j\geq 1}\phi(t_j)$ for every sequence $(t_j)_{j\geq 1}$ of nonnegative numbers with $\sum_{j\geq 1} t_j < \infty$.*

Proof. Concavity and $\phi(0) = 0$ make $z \mapsto \phi(z)/z$ nonincreasing on $(0,\infty)$: for $0 < z_1 \leq z_2$, writing $z_1 = \frac{z_1}{z_2} z_2 + \big(1 - \frac{z_1}{z_2}\big)\cdot 0$ and using concavity together with $\phi(0) = 0$ gives $\phi(z_1) \geq \frac{z_1}{z_2}\phi(z_2)$.

(1) Suppose without loss of generality that $0 < z_1 \leq z_2$. Then

$$\begin{aligned} \phi(z_1+z_2) &\leq \frac{z_1+z_2}{z_2}\phi(z_2) && \text{because } \phi(z)/z \text{ is nonincreasing} \\ &= \phi(z_2) + z_1\frac{\phi(z_2)}{z_2} \\ &\leq \phi(z_2) + \phi(z_1) && \text{by the same monotonicity.} \end{aligned}$$

The cases with $z_1 = 0$ or $z_2 = 0$ follow from $\phi(0) = 0$.

(2) and (3) By the same monotonicity and $\phi(0) = 0$,

$$\frac{\phi(z)}{z} \leq \lim_{s\downarrow 0}\frac{\phi(s)}{s} = \lim_{s\downarrow 0}\frac{\phi(s)-\phi(0)}{s} = L_\phi \qquad (z>0),$$

and the displayed limit is precisely (3).

(4) By (2) the function ϕ is continuous at 0, and a finite concave function is continuous on $(0,\infty)$. Iterating (1) over partial sums gives

$$\phi\Big(\sum_{j\leq N} t_j\Big) \leq \sum_{j\leq N}\phi(t_j) \leq \sum_{j\geq 1}\phi(t_j),$$

the last step because the terms are nonnegative. Letting $N\to\infty$ and using continuity of ϕ at $\sum_{j\geq 1} t_j$ proves the claim. □

Assumption 3.3 implies the standing hypotheses of Lemma 3.6, and Assumption 3.2 supplies the finite right slope needed for parts (2)–(4). Neither coercivity nor differentiability on $(0,\infty)$ is used, so the lemma also applies under the weaker hypotheses of Theorem 3.9.

Proposition 3.7 (Second-order bounds)**.** *Under Assumptions 3.2–3.4,*

(1) *$\phi(z) \geq L_\phi z - \frac{\gamma_1}{2}z^2$ for $0 \leq z \leq \hat z_1$;*
(2) *for every $A > 0$, $\phi(z) \leq L_\phi z - \frac{\gamma_A}{2}z^2$ for $0 \leq z \leq A$.*

Proof. For $\varepsilon > 0$, apply the fundamental theorem of calculus on $[\varepsilon, z]$ and then let $\varepsilon \downarrow 0$. The definition of the finite right derivative and $\phi(0) = 0$ give

$$\phi(z) - L_\phi z = \int_0^z [\phi'(s) - L_\phi]\,ds.$$

The first conclusion follows from Assumption 3.4(1). For the second, Assumption 3.4(2) gives $\phi''(t) \leq -\gamma_A$ on $(0, A]$. Integrating from ε to s and then letting $\varepsilon\downarrow 0$ gives $\phi'(s) - L_\phi \leq -\gamma_A s$ for $0 < s \leq A$. Integration proves the claim. □

For an open set $U \subset \Omega$, let $C_c(U)$ denote the continuous functions with compact support in U. Define

$$\Phi_\phi(\mu) := L_\phi\|\mu_{\mathrm{cont}}\|_{\mathcal{M}(\Omega)} + \sum_{\omega\in\mathrm{atom}\,\mu}\phi\big(|\mu(\{\omega\})|\big), \qquad \mu\in\mathcal{M}(\Omega). \tag{5}$$

Lemma 3.8. *Let $U \subset \Omega = \mathbb{R}^{d+1}$ be open, let $\mu \in \mathcal{M}(U)$, and let $\delta > 0$. Suppose that*

$$|\mu(\{\omega\})| \leq \frac{\delta}{4} \qquad \text{for every } \omega\in U.$$

Then, for every $\varepsilon > 0$, there are $m \geq 1$ and functions $h_1, \dots, h_m \in C_c(U)$ with pairwise disjoint supports and $|h_i| \leq 1$ such that

$$\left| \int_U h_i \, d\mu \right| \leq \delta \quad (i = 1, \dots, m), \qquad \sum_{i=1}^{m} \left| \int_U h_i \, d\mu \right| \geq \|\mu\|_{\mathcal{M}(U)} - \varepsilon.$$

Proof. Let $U = P \cup N$ be a Hahn decomposition of μ. By inner regularity, there are disjoint compact sets $K_+ \subset P$ and $K_- \subset N$ such that

$$|\mu|\big(U \setminus (K_+ \cup K_-)\big) < \frac{\varepsilon}{4}. \tag{6}$$

We now construct the required finite small-mass partition, including both the atomic and nonatomic parts. For every $\omega \in K_+ \cup K_-$, continuity from above gives

$$|\mu|\big(B_r(\omega)\big) \longrightarrow |\mu|(\{\omega\}) \leq \frac{\delta}{4} \qquad (r \downarrow 0).$$

Choose $r_\omega > 0$ such that $|\mu|(B_{r_\omega}(\omega)) < \delta/2$. Compactness gives a finite subcover $B_1, \dots, B_\ell$ of $K_+ \cup K_-$. Intersect these balls separately with K_+ and K_- and disjointize the resulting finite covers by replacing the i-th set with the part not covered by the preceding sets. This gives, for some $m_0 \geq 1$, a finite Borel partition $E_1, \dots, E_{m_0}$ of $K_+ \cup K_-$ such that each E_i lies in either P or N and $|\mu|(E_i) < \delta/2$.

By inner regularity, choose compact sets $C_i \subset E_i$ whose total omitted mass is less than $\varepsilon/4$, and discard the empty sets. At least one set remains because $\varepsilon < M_\mu$ and (6). Relabel the remaining sets $C_1, \dots, C_m$. They are pairwise disjoint and satisfy

$$\sum_{i=1}^{m} |\mu|(C_i) \geq |\mu|(K_+ \cup K_-) - \frac{\varepsilon}{4}, \qquad |\mu|(C_i) \leq \frac{\delta}{2}. \tag{7}$$

Each C_i lies entirely in either P or N.

Because the compact sets C_i are finitely many, pairwise disjoint, and contained in the open set U, they have pairwise disjoint relatively compact open neighborhoods. Outer regularity lets us shrink these neighborhoods, while keeping each C_i inside, to obtain open sets U_i satisfying

$$\sum_{i=1}^{m} |\mu|(U_i \setminus C_i) < \min\left\{\frac{\varepsilon}{4}, \frac{\delta}{2}\right\}. \tag{8}$$

For each i, the continuous function

$$\chi_i(\omega) := \frac{\operatorname{dist}(\omega, U_i^c)}{\operatorname{dist}(\omega, C_i) + \operatorname{dist}(\omega, U_i^c)}$$

equals one on C_i, vanishes outside U_i, and has compact support in U. Set $h_i = \chi_i$ when $C_i \subset P$ and $h_i = -\chi_i$ when $C_i \subset N$. Their supports are pairwise disjoint and $|h_i| \leq 1$. Moreover,

$$\left| \int_U h_i \, d\mu \right| \leq |\mu|(U_i) \leq |\mu|(C_i) + |\mu|(U_i \setminus C_i) \leq \delta, \qquad \text{by (7)–(8)},$$

and

$$\begin{aligned} \sum_{i=1}^{m} \left| \int_U h_i \, d\mu \right| &\geq \sum_{i=1}^{m} \big(|\mu|(C_i) - |\mu|(U_i \setminus C_i)\big) \\ &\geq \|\mu\|_{\mathcal{M}(U)} - \frac{3\varepsilon}{4} \qquad \text{by (6)–(8).} \end{aligned}$$

This is stronger than the required lower bound. $\square$

Theorem 3.9 (representation formula of the nonconvex functional)**.** *Suppose that $\phi : [0, \infty) \to [0, \infty)$ is nondecreasing and concave, $\phi(0) = 0$, and*

$$L_\phi := \phi'(0+) \in (0, \infty).$$

Then, for every $\mu \in \mathcal{M}(\Omega)$,

$$\Phi_\phi(\mu) = \sup\left\{ \sum_{i=1}^m \phi\left(\left| \int_\Omega f_i \, d\mu \right| \right) : \begin{array}{l} m \geq 1, \quad f_i \in C_c(\Omega), \quad |f_i| \leq 1, \\ \operatorname{supp}(f_i) \textit{ are pairwise disjoint} \end{array} \right\}. \tag{9}$$

Proof. We prove the representation by first bounding every admissible finite sum from above and then constructing sums that approach $\Phi_\phi(\mu)$ from below. The hypotheses are exactly the standing hypotheses of Lemma 3.6 together with $L_\phi \in (0, \infty)$, so parts (1)–(3) of that lemma give

$$\phi(r+s) \leq \phi(r) + \phi(s), \qquad \phi(r) \leq L_\phi r, \qquad \frac{\phi(r)}{r} \longrightarrow L_\phi \quad (r \downarrow 0), \tag{10}$$

and part (4) gives the countable form $\phi\big(\sum_{j\geq 1} t_j\big) \leq \sum_{j \geq 1} \phi(t_j)$.

Write the atomic part of μ as

$$\mu_{\text{atom}} = \sum_{j \geq 1} c_j \delta_{\omega_j},$$

where the ω_j are distinct and $c_j \neq 0$. For an admissible family $(f_i)_{i=1}^m$, set

$$A_i := \sum_{j\geq 1} |c_j| \, |f_i(\omega_j)|, \qquad B_i := \int_\Omega |f_i| \, d|\mu_{\text{cont}}|.$$

$$\begin{aligned} \phi\left(\left| \int_\Omega f_i \, d\mu \right| \right) &\leq \phi(A_i + B_i) && \text{by the triangle inequality} \\ &\leq \sum_{\omega_j \in \operatorname{supp}(f_i)} \phi(|c_j|) + L_\phi B_i && \text{by Lemma 3.6(2), (4) and } |f_i| \leq 1. \end{aligned}$$

The disjoint supports count each atom at most once and satisfy $\sum_i |f_i| \leq 1$. Therefore

$$\begin{aligned} \sum_{i=1}^m \phi\left(\left| \int_\Omega f_i \, d\mu \right| \right) &\leq \sum_{j \geq 1} \phi(|c_j|) + L_\phi \int_\Omega \sum_{i=1}^m |f_i| \, d|\mu_{\text{cont}}| \\ &\leq \sum_{j\geq 1} \phi(|c_j|) + L_\phi \|\mu_{\text{cont}}\|_{\mathcal{M}(\Omega)} = \Phi_\phi(\mu). \end{aligned}$$

This proves the upper bound in (9).

For the reverse inequality, the claim is immediate when $\mu = 0$. Otherwise, fix $\varepsilon > 0$. Choose $\theta \in (0, L_\phi)$ such that $\theta \|\mu\|_{\mathcal{M}(\Omega)} < \varepsilon/4$. By the last limit in (10), choose $\delta > 0$ such that

$$\phi(r) \geq (L_\phi - \theta) r \qquad (0 \leq r \leq \delta). \tag{11}$$

Let

$$F := \{ j \geq 1 : |c_j| > \delta/8 \}, \qquad \mu_{\text{rem}} := \mu - \sum_{j \in F} c_j \delta_{\omega_j}.$$

The set F is finite and every point mass of μ_{rem} has magnitude at most $\delta/8$.

Because F is finite, its points have pairwise disjoint relatively compact neighborhoods. Moreover, $\mu_{\text{rem}}(\{\omega_j\}) = 0$ for $j \in F$, so continuity of the measure from above allows these neighborhoods to be chosen with arbitrarily small total $|\mu_{\text{rem}}|$-mass. Using the distance-function construction from Lemma 3.8, choose $g_j \in C_c(\Omega)$, $j \in F$, with pairwise disjoint supports, $|g_j| \leq 1$, and

$$g_j(\omega_j) = \operatorname{sign}(c_j), \qquad \sum_{j \in F} |\mu_{\text{rem}}|(\operatorname{supp}(g_j)) < \tau,$$

where $\tau > 0$ can be chosen arbitrarily small. For every $j \in F$,

$$\left| \left| \int_\Omega g_j \, d\mu \right| - |c_j| \right| \leq |\mu_{\text{rem}}|(\operatorname{supp}(g_j)).$$

By continuity of ϕ and finiteness of F, first choose τ sufficiently small and then choose the supports so that

$$\sum_{j\in F}\phi\left(\left|\int_\Omega g_j\,d\mu\right|\right) \geq \sum_{j\in F}\phi(|c_j|)-\frac{\varepsilon}{4}, \qquad 2L_\phi\tau<\frac{\varepsilon}{4}. \tag{12}$$

Set

$$U := \Omega\setminus\bigcup_{j\in F}\operatorname{supp}(g_j) \qquad \text{and} \qquad \lambda := \mu_{\mathrm{rem}}|_U.$$

Lemma 3.8 applied on the open set U gives $h_1,\ldots,h_m\in C_c(U)$ whose zero extensions belong to $C_c(\Omega)$, have pairwise disjoint supports, and satisfy

$$\left|\int_U h_i\,d\lambda\right|\leq\delta, \qquad \sum_{i=1}^m\left|\int_U h_i\,d\lambda\right|\geq\|\lambda\|_{\mathcal{M}(U)}-\tau\geq\|\mu_{\mathrm{rem}}\|_{\mathcal{M}(\Omega)}-2\tau.$$

Together with the g_j, the zero extensions form an admissible family in (9). Since every h_i is supported in U and $\mu|_U=\lambda$, its integrals against μ and λ agree. Therefore (11) and (12) yield

$$\begin{aligned}
&\sum_{j\in F}\phi\left(\left|\int_\Omega g_j\,d\mu\right|\right)+\sum_{i=1}^m\phi\left(\left|\int_\Omega h_i\,d\mu\right|\right)\\
&\quad\geq\sum_{j\in F}\phi(|c_j|)-\frac{\varepsilon}{4}+(L_\phi-\theta)\big(\|\mu_{\mathrm{rem}}\|_{\mathcal{M}(\Omega)}-2\tau\big)\\
&\quad\geq\Phi_\phi(\mu)-\frac{\varepsilon}{4}-\theta\|\mu_{\mathrm{rem}}\|_{\mathcal{M}(\Omega)}-2L_\phi\tau && \text{by } \Phi_\phi(\mu_{\mathrm{rem}})\leq L_\phi\|\mu_{\mathrm{rem}}\|_{\mathcal{M}(\Omega)}\\
&\quad>\Phi_\phi(\mu)-\varepsilon.
\end{aligned}$$

Taking the supremum and then letting $\varepsilon\downarrow 0$ proves the reverse inequality. □

Corollary 3.10. *Under the assumptions of Theorem 3.9, the functional Φ_ϕ is weak-* lower semicontinuous on $\mathcal{M}(\Omega)=C_0(\Omega)^*$. If ϕ is also coercive, then every sublevel of Φ_ϕ is bounded in total variation.*

Proof. We use the representation in Theorem 3.9 for lower semicontinuity and the atomic–continuous decomposition for the sublevel bound. For each fixed admissible family in (9), weak-* convergence gives convergence of all its integrals, and continuity of ϕ gives convergence of the corresponding finite sum. The representation is a supremum of weak-* continuous functions and is therefore weak-* lower semicontinuous.

For the sublevel bound, apply repeated subadditivity to the first N atoms and let $N\to\infty$. The atom magnitudes are summable, so continuity of ϕ gives

$$\phi\big(\|\mu_{\mathrm{atom}}\|_{\mathcal{M}(\Omega)}\big)\leq\sum_{\omega\in\operatorname{atom}\mu}\phi\big(|\mu(\{\omega\})|\big)\leq\Phi_\phi(\mu), \qquad L_\phi\|\mu_{\mathrm{cont}}\|_{\mathcal{M}(\Omega)}\leq\Phi_\phi(\mu).$$

If $\Phi_\phi(\mu)\leq C$, coercivity bounds $\|\mu_{\mathrm{atom}}\|_{\mathcal{M}(\Omega)}$ in terms of C, while $\|\mu_{\mathrm{cont}}\|_{\mathcal{M}(\Omega)}\leq C/L_\phi$. The atomic part is supported on the countable atom set, while the continuous part assigns that set zero variation. The two parts are therefore mutually singular, so their total-variation norms add to $\|\mu\|_{\mathcal{M}(\Omega)}$. Hence the sublevel is bounded in total variation. □

For $\alpha>0$, define

$$J_0(\mu) := L(\mu)+\alpha\Phi_\phi(\mu)$$

for the problem without the moment term.

3.3. **The normalized problem.**

Escape on the unbounded parameter domain. For positively homogeneous profiles the radial scale of ω is absorbed by the outer coefficient, and the parameter can be restricted to the unit sphere $\mathbb{S}^d$ (Section 4). But in general the domain is $\Omega = \mathbb{R}^{d+1}$, which is not compact. The functional J_0 need not be well posed. The regularizer Φ_ϕ depends on the sizes of the coefficients but not on the locations of the atoms. Along a sequence with $|\omega| \to \infty$, the fidelity need not be lower semicontinuous.

Example 3.11 (Failure of weak-* lower semicontinuity for tanh). Let $\rho = \tanh$, $V \equiv 1$, fix $a \in \mathbb{R}^d$, and set $\mu_n = \delta_{(a,b_n)}$ with $b_n \to \infty$. Then $\mu_n \xrightarrow{*} 0$, yet

$$\mathcal{N}\mu_n = \tanh(a \cdot x + b_n) \to 1$$

strongly in $H^1(D)$. Hence $\liminf_{n\to\infty} L(\mu_n) = 0 < L(0)$, so the fidelity term is not weak-* lower semicontinuous. Mass escaping to infinity carries a non-vanishing value contribution unless $K(\omega)$ tends to zero as $|\omega| \to \infty$. Note that $\Phi_\phi(\mu_n) = \phi(1)$ is constant along this sequence, so the unnormalized coefficient penalty does not prevent this escape. The defect is that the unnormalized kernel does not vanish at parameter infinity. The normalized atom introduced in Theorem 3.13 does vanish under the corresponding polynomial growth gap.

Example 3.12 (Nonattainment for softplus). For the profile $\rho(z) = \log(1 + e^z)$ (softplus), let $b_n \to \infty$. The measures $\mu_n = \rho(b_n)^{-1}\delta_{(0,b_n)}$ fit $V \equiv 1$ exactly for every n at penalty $\alpha\,\phi(1/\rho(b_n)) \to 0$, so $\inf J_0 = 0$ is not attained. Indeed, if $J_0(\mu) = 0$, then $\Phi_\phi(\mu) = 0$. Strict positivity of ϕ on $(0,\infty)$ and $L_\phi > 0$ force both the atomic and continuous parts of μ to vanish. Thus $\mu = 0$, but $L(0) = \frac{1}{2}\|1\|^2_{H^1(D)} > 0$, a contradiction.

The boundary behavior can also differ from the interior behavior. Let $F = \partial D \cap \{x : n_F \cdot x = c_F\}$ be a flat face of positive surface measure, assume that D lies on one side of its supporting hyperplane, and let $\rho \in C_0(\mathbb{R})$. Choose z_* with $\rho(z_*) \neq 0$, let $t_j \to \infty$, and set

$$a_j = t_j n_F, \qquad b_j = -t_j c_F + z_*.$$

Then

$$[K(a_j, b_j)](x) = \rho\big(t_j(n_F \cdot x - c_F) + z_*\big).$$

Since D lies on one side of the supporting hyperplane, $n_F \cdot x - c_F$ has a constant sign on D and vanishes only on $D \cap \{x : n_F \cdot x = c_F\} \subset \partial D$, a Lebesgue-null set. Hence for almost every $x \in D$ the argument $t_j(n_F \cdot x - c_F) + z_*$ tends to a single infinity, with the sign fixed by the orientation of n_F. Because ρ is bounded and D is bounded, the constant $\|\rho\|_\infty$ is an integrable dominating function, and since ρ vanishes at infinity, dominated convergence gives $K(a_j, b_j) \to 0$ in $L^2(D)$. The trace on the face F, however, satisfies $[K(a_j, b_j)]|_F = \rho(z_*)$ for every j.

Therefore, for $p > 0$, let w_p and $\mathcal{M}_p(\Omega)$ be as in (2). For $\mu \in \mathcal{M}_p(\Omega)$, we define the normalized measure and normalized atom by

$$d\mu_p := w_p\, d\mu, \qquad K_p(\omega) := \frac{K(\omega)}{w_p(\omega)}.$$

Theorem 3.13. *Multiplication by w_p is a linear isometric bijection*

$$\mathcal{M}_p(\Omega) \longrightarrow \mathcal{M}(\Omega), \qquad \mu \longmapsto \mu_p,$$

with inverse $d\mu = w_p^{-1} d\mu_p$, and

$$\|\mu\|_{\mathcal{M}_p(\Omega)} = \|\mu_p\|_{\mathcal{M}(\Omega)}. \tag{13}$$

Moreover, μ_n converges weak- to μ in $C_{w_p^{-1}}(\Omega)^*$ if and only if $(\mu_n)_p \xrightarrow{*} \mu_p$ in $C_0(\Omega)^*$. The multiplication preserves the atom set and the polar sign, and*

$$(\mu_p)_{\text{atom}} = w_p \mu_{\text{atom}}, \qquad (\mu_p)_{\text{cont}} = w_p \mu_{\text{cont}}. \tag{14}$$

If part (1) of Assumption 3.1 holds and $p > s_0$, then $K_p \in C_0(\Omega; L^2(D))$ and

$$\mathcal{N}\mu = \int_\Omega K \, d\mu = \int_\Omega K_p \, d\mu_p. \tag{15}$$

Finally,

$$\Phi_\phi(\mu_p) = L_\phi \int_\Omega w_p \, d|\mu_{\mathrm{cont}}| + \sum_{\omega \in \mathrm{atom}\, \mu} \phi\big(w_p(\omega)|\mu(\{\omega\})|\big). \tag{16}$$

Proof. We use multiplication by a positive continuous weight to prove every claim. If $\mu = \mathrm{sign}(\mu)|\mu|$, then

$$\mu_p = \mathrm{sign}(\mu) w_p |\mu|, \qquad |\mu_p| = w_p|\mu|.$$

This proves (13), preserves the polar sign, and shows that the inverse is $w_p^{-1}\mu_p$. Conversely, if $\xi \in \mathcal{M}(\Omega)$ and $d\mu = w_p^{-1} d\xi$, then

$$\int_\Omega w_p \, d|\mu| = |\xi|(\Omega),$$

so $\mu \in \mathcal{M}_p(\Omega)$ and the two multiplication maps are inverse.

For $f \in C_{w_p^{-1}}(\Omega)$, set $h = f/w_p \in C_0(\Omega)$. Then

$$\int_\Omega f \, d\mu = \int_\Omega h \, d\mu_p.$$

The bijection $f = w_p h$ between the two test-function spaces proves the weak-* equivalence.

At each $\omega \in \Omega$,

$$\mu_p(\{\omega\}) = w_p(\omega)\mu(\{\omega\}).$$

Since w_p is strictly positive, the atom sets agree. Multiplication by w_p also preserves nonatomicity, which proves (14).

It remains to treat the network map. The continuity of K_p follows from that of K and w_p. For $r = |\omega| \geq 1$, the growth estimate (3) gives

$$\begin{aligned} \|K_p(\omega)\|_{L^2(D)} &\leq C_D \frac{(1+r)^{s_0}}{1+r^p} \\ &\leq 2^{s_0} C_D r^{s_0 - p} \longrightarrow 0, \qquad r \to \infty, \end{aligned}$$

because $p > s_0$. Thus $K_p \in C_0(\Omega; L^2(D))$. The change-of-measure identity for the Bochner integral now gives (15). Finally, (14) and $|w_p \mu_{\mathrm{cont}}| = w_p|\mu_{\mathrm{cont}}|$ give (16) directly from (5). □

For $\alpha > 0$, define the adopted objective

$$J(\mu) := L(\mu) + \alpha \Phi_\phi(\mu_p), \qquad \mu \in \mathcal{M}_p(\Omega).$$

For a finite atomic measure $\mu_{\boldsymbol{\omega},c} = \sum_{n=1}^N c_n \delta_{\omega_n}$ with distinct support points, this becomes

$$J(\mu_{\boldsymbol{\omega},c}) = L(\mu_{\boldsymbol{\omega},c}) + \alpha \sum_{n=1}^N \phi\big(w_p(\omega_n)|c_n|\big).$$

Definition 3.14 (Local minimizer)**.** A measure $\bar{\mu} \in \mathcal{M}_p(\Omega)$ is a local minimizer of J if there is $\varepsilon > 0$ such that

$$J(\bar{\mu}) \leq J(\mu) \qquad \text{for all} \qquad \mu \in \mathcal{M}_p(\Omega), \quad \|\mu - \bar{\mu}\|_{\mathcal{M}_p(\Omega)} \leq \varepsilon.$$

By Theorem 3.13, this is exactly total-variation local minimality in the normalized coordinate μ_p.

3.4. **Existence of a global minimizer.** We apply the direct method in the normalized coordinate. Weak-* compactness controls μ_p, Corollary 3.10 supplies the penalty lower semicontinuity and sublevel bound, and Lemma 3.15 gives the strong convergence needed for the value part of the fidelity.

Lemma 3.15. *Assume* (3)*, let* $p > s_0$*, and suppose that*

$$(\mu_n)_p \overset{*}{\rightharpoonup} \mu_p \quad \text{in } \mathcal{M}(\Omega), \qquad \sup_n \|(\mu_n)_p\|_{\mathcal{M}(\Omega)} < \infty.$$

Then $\mathcal{N}\mu_n \to \mathcal{N}\mu$ *strongly in* $L^2(D)$*.*

Proof. We use a compactly supported cutoff to pass to the weak-* limit locally and then control the tails uniformly. By Theorem 3.13,

$$\mathcal{N}\mu_n = \int_\Omega K_p\, d(\mu_n)_p, \qquad \mathcal{N}\mu = \int_\Omega K_p\, d\mu_p.$$

Set

$$C := \sup_n \|(\mu_n)_p\|_{\mathcal{M}(\Omega)}.$$

Weak-* lower semicontinuity of total variation gives $\|\mu_p\|_{\mathcal{M}(\Omega)} \le C$.

For $R \ge 1$, choose $\chi_R \in C_c(\Omega)$ with $0 \le \chi_R \le 1$, $\chi_R = 1$ on B_R, and $\operatorname{supp}\chi_R \subset B_{R+1}$. For fixed R and $x \in D$, the function

$$\omega \longmapsto \chi_R(\omega)[K_p(\omega)](x)$$

belongs to $C_c(\Omega)$. Hence weak-* convergence gives

$$\int_\Omega \chi_R(\omega)[K_p(\omega)](x)\, d(\mu_n)_p(\omega) \longrightarrow \int_\Omega \chi_R(\omega)[K_p(\omega)](x)\, d\mu_p(\omega)$$

for every $x \in D$.

Since D is bounded and $\operatorname{supp}\chi_R$ is compact,

$$M_R := \sup_{\omega\in\operatorname{supp}\chi_R,\ x\in D} |[K_p(\omega)](x)| < \infty.$$

The absolute values of both truncated integrals are bounded by CM_R. The constant function $2CM_R$ belongs to $L^2(D)$ because D is bounded. Dominated convergence therefore yields

$$\int_\Omega \chi_R K_p\, d(\mu_n)_p \longrightarrow \int_\Omega \chi_R K_p\, d\mu_p \qquad \text{strongly in } L^2(D).$$

It remains to remove the cutoff. If $r := |\omega| > R \ge 1$, then

$$\begin{aligned} \|K_p(\omega)\|_{L^2(D)} &\le C_D \frac{(1+r)^{s_0}}{1+r^p} && \text{by (3)} \\ &\le 2^{s_0} C_D r^{s_0-p} \le 2^{s_0} C_D R^{s_0-p}, && \text{since } p > s_0 \text{ and } r > R. \end{aligned}$$

Consequently,

$$\left\| \int_\Omega (1-\chi_R) K_p\, d(\mu_n)_p \right\|_{L^2(D)} \le 2^{s_0} C_D C R^{s_0-p},$$
$$\left\| \int_\Omega (1-\chi_R) K_p\, d\mu_p \right\|_{L^2(D)} \le 2^{s_0} C_D C R^{s_0-p}.$$

For fixed R, let $n \to \infty$ in the truncated integrals. Then let $R \to \infty$; the two tail bounds vanish because $s_0 - p < 0$. This proves the claim. □

Theorem 3.16 (Existence on the unbounded parameter domain)**.** *Let $D \subset \mathbb{R}^d$ be bounded and Lipschitz, let $V \in H^1(D)$, let ϕ satisfy Assumptions 3.2 and 3.3, and let the activation satisfy Assumption 3.1. If $p > s_0$ and $\alpha > 0$, then*

$$\inf_{\mu \in \mathcal{M}_p(\Omega)} J(\mu)$$

admits at least one global minimizer.

Proof. We use the direct method in the normalized measure coordinate. Only the value-growth bound in Assumption 3.1(1) is needed in this proof. Since $V \in H^1(D)$,

$$J(0) = \frac{1}{2}\|V\|^2_{H^1(D)} =: C < \infty.$$

Choose a minimizing sequence $(\mu_n) \subset \mathcal{M}_p(\Omega)$ with $J(\mu_n) \leq C + 1$. The fidelity and penalty are nonnegative, so

$$\Phi_\phi((\mu_n)_p) \leq \frac{C+1}{\alpha}.$$

The total-variation sublevel bound in Corollary 3.10 shows that $((\mu_n)_p)$ is bounded in $\mathcal{M}(\Omega)$. Weak-* compactness and separability of $C_0(\Omega)$ give, after passing to a subsequence,

$$(\mu_n)_p \overset{*}{\rightharpoonup} \bar{\mu}_p.$$

Theorem 3.13 defines a unique $\bar{\mu} \in \mathcal{M}_p(\Omega)$ with normalized coordinate $\bar{\mu}_p$. Lemma 3.15 gives

$$\mathcal{N}\mu_n \longrightarrow \mathcal{N}\bar{\mu} \qquad \text{strongly in } L^2(D). \tag{17}$$

The fidelity bound gives

$$\sup_n \|\nabla(\mathcal{N}\mu_n)\|_{L^2(D)} \leq \|\nabla V\|_{L^2(D)} + \sqrt{2(C+1)}.$$

After another subsequence, $\nabla(\mathcal{N}\mu_n) \rightharpoonup g$ weakly in $L^2(D)^d$. For every $\psi \in C_c^\infty(D)^d$,

$$\int_D g \cdot \psi \, dx = \lim_{n\to\infty} \int_D \nabla(\mathcal{N}\mu_n) \cdot \psi \, dx = -\lim_{n\to\infty} \int_D \mathcal{N}\mu_n \operatorname{div} \psi \, dx = -\int_D \mathcal{N}\bar{\mu} \operatorname{div} \psi \, dx,$$

the last step by (17). Hence $g = \nabla(\mathcal{N}\bar{\mu})$ distributionally and $\mathcal{N}\bar{\mu} \in H^1(D)$.

The value part of the fidelity is continuous by (17), and the gradient part is weakly lower semicontinuous. Hence

$$L(\bar{\mu}) \leq \liminf_{n\to\infty} L(\mu_n).$$

Corollary 3.10 gives

$$\Phi_\phi(\bar{\mu}_p) \leq \liminf_{n\to\infty} \Phi_\phi((\mu_n)_p).$$

Combining the two inequalities proves $J(\bar{\mu}) \leq \liminf_n J(\mu_n) = \inf J$. □

3.5. **Optimality conditions and finite support.** Let $\bar{\mu}$ be a finite-objective local minimizer. For a measure μ and a direction μ', the Gâteaux derivative is

$$DL(\mu)[\mu'] := \lim_{t\to 0} \frac{L(\mu + t\mu') - L(\mu)}{t} = \langle r_\mu, \mathcal{N}\mu' \rangle_{H^1(D)}.$$

At such a local minimizer, define

$$\bar{P}(\omega) := DL(\bar{\mu})[\delta_\omega] = \langle r_{\bar{\mu}}, K(\omega) \rangle_{H^1(D)}.$$

Its weighted form is

$$\bar{P}_p(\omega) := \frac{\bar{P}(\omega)}{w_p(\omega)} = \langle r_{\bar{\mu}}, K_p(\omega) \rangle_{H^1(D)}.$$

Thus $\bar{P}$ is the function representing the Gâteaux derivative. For differentiable ρ,

$$\bar{P}(a, b) = \int_D r_{\bar{\mu}}(x) \rho(a \cdot x + b) \, dx + \int_D \nabla r_{\bar{\mu}}(x) \cdot a \rho'(a \cdot x + b) \, dx.$$

The second term carries the factor a, so on the unbounded domain $\bar{P}$ is defined pointwise but need not be bounded.

The activation bounds in Assumption 3.1 imply the full $H^1(D)$ properties needed below.

Lemma 3.17. *Under Assumption 3.1, the map $K : \Omega \to H^1(D)$ is continuous. Moreover, with*

$$A_D := \max\left\{1, \sup_{x\in D} \sqrt{1+|x|^2}\right\}$$

and

$$C_K := C_D + C_\rho |D|^{1/2} A_D^{s_1-1},$$

$$\|K(\omega)\|_{H^1(D)} \le C_K(1+|\omega|)^{s_1} \qquad (\omega \in \Omega). \tag{18}$$

Proof. We estimate the value and gradient parts separately and then verify continuity. The constant A_D is finite because D is bounded. For $\omega = (a, b) \in \Omega$ and $x \in D$, Cauchy–Schwarz in $\mathbb{R}^{d+1}$ gives

$$1 + |a \cdot x + b| \le 1 + |\omega|\sqrt{1+|x|^2} \le A_D(1+|\omega|).$$

The value estimate (3), the identity $\nabla K(a, b)(x) = a\rho'(a \cdot x + b)$, and Assumption 3.1(2) therefore give

$$\begin{aligned}
\|K(\omega)\|_{L^2(D)} &\le C_D(1+|\omega|)^{s_0} \le C_D(1+|\omega|)^{s_1},\\
\|\nabla K(\omega)\|_{L^2(D)} &= |a| \left(\int_D |\rho'(a\cdot x+b)|^2\, dx\right)^{1/2}\\
&\le C_\rho |D|^{1/2} A_D^{s_1-1}(1+|\omega|)^{s_1},
\end{aligned}$$

where the last step uses $|a| \le |\omega| \le 1 + |\omega|$. Since the $H^1(D)$ norm is at most the sum of its value and gradient components, (18) follows from the definition of C_K.

It remains to prove continuity. Let $\omega_n = (a_n, b_n) \to \omega = (a, b)$. The pre-activations $a_n \cdot x + b_n$ converge uniformly on D and remain in one compact interval. Since $\rho, \rho' \in C(\mathbb{R})$, both are uniformly continuous on that interval. Consequently,

$$\rho(a_n \cdot x + b_n) \longrightarrow \rho(a \cdot x + b), \qquad a_n\rho'(a_n \cdot x + b_n) \longrightarrow a\rho'(a \cdot x + b)$$

uniformly for $x \in D$. Hence $K(\omega_n) \to K(\omega)$ in $H^1(D)$. □

By Cauchy–Schwarz, (18) gives

$$|\bar{P}(\omega)| \le \|r_{\bar\mu}\|_{H^1(D)} \|K(\omega)\|_{H^1(D)} \le C_P(1+|\omega|)^{s_1}, \qquad C_P := C_K\|r_{\bar\mu}\|_{H^1(D)}.$$

The constant C_P depends on the fixed minimizer through its residual, but not on ω. If $p > s_1$, then

$$K_p \in C_0(\Omega; H^1(D)), \qquad \bar{P}_p \in C_0(\Omega), \qquad \bar{P} \in C_{w_p^{-1}}(\Omega).$$

Consequently, for every $\mu' \in \mathcal{M}_p(\Omega)$,

$$DL(\bar\mu)[\mu'] = \int_\Omega \bar{P}\, d\mu' = \int_\Omega \bar{P}_p\, d\mu'_p.$$

The pointwise definition of $\bar{P}$ as a Gâteaux derivative does not require $\bar{P} \in C_0(\Omega)$.

Theorem 3.18 (The optimality condition)**.** *Let ϕ satisfy Assumptions 3.2 and 3.3, let $\bar\mu \in \mathcal{M}_p(\Omega)$ be a local minimizer of J with $J(\bar\mu) < \infty$, and suppose that Assumption 3.1 holds and $p > s_1$. Then*

$$|\bar{P}_p(\omega)| \le \alpha L_\phi \qquad \text{for every } \omega \in \Omega, \tag{19}$$

$$\bar{P}_p(\omega) = -\alpha\phi'\big(w_p(\omega)|\bar\mu(\{\omega\})|\big)\operatorname{sign}\big(\bar\mu(\{\omega\})\big) \qquad \text{for } \omega \in \operatorname{atom}\bar\mu. \tag{20}$$

Moreover,

$$\bar{P}_p = -\alpha L_\phi \operatorname{sign}(\bar\mu_{\mathrm{cont}}) \qquad |(\bar\mu_p)_{\mathrm{cont}}|\text{-almost everywhere.} \tag{21}$$

Proof. We use Theorem 3.13 to work with $\bar{\mu}_p$ in the total-variation norm. The normalized fidelity derivative in the direction $\xi \in \mathcal{M}(\Omega)$ is

$$\left\langle r_{\bar{\mu}}, \int_\Omega K_p \, d\xi \right\rangle_{H^1(D)} = \int_\Omega \bar{P}_p \, d\xi.$$

Fix $\omega \in \operatorname{atom} \bar{\mu}$ and put $q := \bar{\mu}_p(\{\omega\}) = w_p(\omega)\bar{\mu}(\{\omega\}) \neq 0$. For $s \in \{-1, 1\}$ and sufficiently small $t > 0$, the sign of $q + ts$ equals the sign of q. Dividing the local-minimality inequality by t and letting $t \downarrow 0$ gives

$$0 \leq s \left[\bar{P}_p(\omega) + \alpha\phi'(|q|) \operatorname{sign}(q)\right].$$

Using both choices of s makes the bracket vanish, and hence

$$\bar{P}_p(\omega) = -\alpha\phi'(|q|) \operatorname{sign}(q).$$

Because $w_p > 0$, $\operatorname{sign}(q) = \operatorname{sign}(\bar{\mu}(\{\omega\}))$, so this is (20). Concavity and monotonicity give $0 \leq \phi'(|q|) \leq L_\phi$, and hence $|\bar{P}_p(\omega)| \leq \alpha L_\phi$ at every atom.

If $\omega \notin \operatorname{atom} \bar{\mu}$, insertion of $ts\delta_\omega$ into the normalized measure changes the penalty by $\phi(t)$. Dividing the local minimality inequality by $t > 0$ and letting $t \downarrow 0$ yields

$$0 \leq s\bar{P}_p(\omega) + \alpha L_\phi.$$

The two choices of s prove (19) away from the atoms, and the preceding atom estimate completes its proof.

It remains to identify the continuous part. Set $\xi := (\bar{\mu}_p)_{\text{cont}}$. Since $K_p \in C_0(\Omega; H^1(D))$, the directions $\pm\xi$ are admissible. For $s \in \{-1, 1\}$ and $0 < t < 1$, the perturbation $\bar{\mu}_p + ts\xi$ multiplies the continuous part by $1 + ts > 0$. Its one-sided directional derivative therefore gives

$$0 \leq s \left[\int_\Omega \bar{P}_p \, d\xi + \alpha L_\phi \|\xi\|_{\mathcal{M}(\Omega)}\right].$$

Using both signs shows that the bracket vanishes. On the other hand, (19) makes

$$\bar{P}_p \operatorname{sign}(\xi) + \alpha L_\phi \geq 0$$

$|\xi|$-almost everywhere. Its integral is zero, so the integrand vanishes $|\xi|$-almost everywhere. The sign preservation in Theorem 3.13 gives (21). □

Remark 3.19. Because $\bar{P}_p \in C_0(\Omega)$ and $\alpha L_\phi > 0$, every closed positive superlevel set of $|\bar{P}_p|$ is compact. Thus any violation of (19) is confined to a compact set, even when the unnormalized function $\bar{P}$ does not decay at parameter infinity. Lemma 3.20 uses a lower superlevel to localize the entire support.

Lemma 3.20. *Let ϕ satisfy Assumptions 3.2 and 3.3, let $p > s_1$, and suppose that Assumption 3.1 holds. Suppose that $\bar{\mu} \in \mathcal{M}_p(\Omega)$ is an $\mathcal{M}_p(\Omega)$-local minimizer of J such that $0 < J(\bar{\mu}) < \infty$. Define*

$$T(\bar{\mu}) := \phi^{-1}\left(\frac{J(\bar{\mu})}{\alpha}\right)$$

and

$$\mathcal{K}(\bar{\mu}) := \left\{\omega \in \Omega : |\bar{P}_p(\omega)| \geq \alpha\phi'\big(T(\bar{\mu})\big)\right\}.$$

Then $\mathcal{K}(\bar{\mu})$ is compact and

$$\operatorname{supp} \bar{\mu}_{\text{atom}} \cup \operatorname{supp} \bar{\mu}_{\text{cont}} \subset \mathcal{K}(\bar{\mu}). \tag{22}$$

Moreover, every atom $\omega \in \operatorname{atom} \bar{\mu}$ satisfies

$$w_p(\omega)|\bar{\mu}(\{\omega\})| \leq T(\bar{\mu}).$$

Proof. We use Theorem 3.18 and the fact that $\bar{P}_p$ vanishes at infinity to prove the localization. Concavity and monotonicity make ϕ' positive and nonincreasing, as observed after Assumption 3.3. Since $\bar{P}_p \in C_0(\Omega)$, its closed superlevel set at the strictly positive level $\alpha\phi'(T(\bar{\mu}))$ is compact. This proves the first assertion.

Let $\omega \in \operatorname{atom}\bar{\mu}$ and put

$$t_\omega := |\bar{\mu}_p(\{\omega\})| = w_p(\omega)|\bar{\mu}(\{\omega\})|.$$

The contribution of this atom to the objective gives

$$\begin{aligned}\alpha\phi(t_\omega) &\le \alpha\Phi_\phi(\bar{\mu}_p) && \text{by nonnegativity of the other terms}\\ &\le J(\bar{\mu}) = \alpha\phi\big(T(\bar{\mu})\big).\end{aligned}$$

Strict monotonicity of ϕ gives $t_\omega \le T(\bar{\mu})$. The atom condition in Theorem 3.18 and monotonicity of ϕ' now give

$$|\bar{P}_p(\omega)| = \alpha\phi'(t_\omega) \ge \alpha\phi'\big(T(\bar{\mu})\big).$$

Thus every atom lies in $\mathcal{K}(\bar{\mu})$.

On the continuous part, Theorem 3.18 gives

$$|\bar{P}_p(\omega)| = \alpha L_\phi \ge \alpha\phi'\big(T(\bar{\mu})\big) \qquad |(\bar{\mu}_p)_{\mathrm{cont}}|\text{-almost everywhere.}$$

Hence $(\bar{\mu}_p)_{\mathrm{cont}}$ is concentrated on the closed set $\mathcal{K}(\bar{\mu})$. Theorem 3.13 preserves the atomic and continuous decompositions. Since w_p is strictly positive, each component of $\bar{\mu}_p = w_p\bar{\mu}$ has the same null sets, and hence the same support, as the corresponding component of $\bar{\mu}$. Taking closures for the atomic part and supports for the continuous part proves (22). $\square$

Theorem 3.21 (finite support of the optimal solution)**.** *Let ϕ satisfy Assumptions 3.2–3.4, let $p > s_1$, and suppose that Assumption 3.1 holds. Then every $\mathcal{M}_p(\Omega)$-local minimizer $\bar{\mu} \in \mathcal{M}_p(\Omega)$ of J with $J(\bar{\mu}) < \infty$ is finite atomic. If $J(\bar{\mu}) > 0$, set*

$$T := T(\bar{\mu}), \qquad \mathcal{K} := \mathcal{K}(\bar{\mu}).$$

Then $\bar{\mu}_{\mathrm{cont}} = 0$, every atom belongs to the compact set $\mathcal{K}$, and any two distinct atoms ω_i and ω_j satisfy

$$\|K_p(\omega_i) - K_p(\omega_j)\|^2_{H^1(D)} \ge 2\alpha\gamma_T. \tag{23}$$

Consequently,

$$\#\operatorname{atom}(\bar{\mu}) \le N_{\mathrm{cov}}\left(K_p(\mathcal{K}), \frac{1}{2}\sqrt{\alpha\gamma_T}\right) < \infty, \tag{24}$$

where the covering number $N_{\mathrm{cov}}(A, r)$ is the least number of balls of radius r needed to cover A.

Proof. We use the isometry of Theorem 3.13, the first-order identities of Theorem 3.18, and local coefficient variations to prove the assertions. Concavity, coercivity, and $L_\phi > 0$ imply that $\phi(t) > 0$ for $t > 0$.

Suppose $J(\bar{\mu}) > 0$. Lemma 3.20 gives the compact set $\mathcal{K}$ and already localizes the atomic and continuous supports. Every normalized atom magnitude

$$t_i := |\bar{\mu}_p(\{\omega_i\})|$$

satisfies $t_i \le T$ by Lemma 3.20. The curvature assumption gives $\phi''(t) \le -\gamma_T$ for $0 < t \le T$, while Proposition 3.7 gives

$$L_\phi t - \phi(t) \ge \frac{\gamma_T}{2}t^2 \qquad (0 \le t \le T). \tag{25}$$

We first separate distinct atoms. For two such atoms, Theorem 3.13 defines a unique measure $\mu_s \in \mathcal{M}_p(\Omega)$ by

$$(\mu_s)_p := \bar{\mu}_p + s\delta_{\omega_i} - s\delta_{\omega_j}.$$

For sufficiently small $|s|$, the signs of both atom coefficients remain fixed, and

$$\begin{aligned} \|\mu_s - \bar\mu\|_{\mathcal{M}_p(\Omega)} &= 2|s|, && \text{by Theorem 3.13,} \\ \mathcal{N}\mu_s &= \mathcal{N}\bar\mu + s\big[K_p(\omega_i) - K_p(\omega_j)\big], && \text{by the normalized representation of } \mathcal{N}. \end{aligned}$$

Thus $s \mapsto J(\mu_s)$ is twice differentiable near zero and has a local minimum there. Its second derivative satisfies

$$\begin{aligned} 0 &\le \frac{d^2}{ds^2} J(\mu_s)\Big|_{s=0} \\ &= \|K_p(\omega_i) - K_p(\omega_j)\|^2_{H^1(D)} + \alpha\phi''(t_i) + \alpha\phi''(t_j) && \text{by the quadratic fidelity} \\ &\le \|K_p(\omega_i) - K_p(\omega_j)\|^2_{H^1(D)} - 2\alpha\gamma_T, && \text{since } t_i, t_j \le T. \end{aligned}$$

This proves (23).

We next rule out the continuous part by concentrating a local piece of one sign. Suppose $(\bar\mu_p)_{\mathrm{cont}} \neq 0$ and write

$$d(\bar\mu_p)_{\mathrm{cont}} = \sigma\, d\lambda, \qquad \lambda := |(\bar\mu_p)_{\mathrm{cont}}|,$$

where σ takes the values -1 and 1 λ-almost everywhere. Choose $\sigma_0 \in \{-1, 1\}$ such that the restriction of λ to $\{\sigma = \sigma_0\}$ is nonzero, and denote that restriction again by λ. Theorem 3.18 gives $\bar P_p = -\alpha L_\phi \sigma_0$ on a set of full λ-measure. The support of a Radon measure has full measure, while the countable set $\operatorname{atom}\bar\mu$ has zero λ-measure because λ is nonatomic. The intersection of these three full-measure sets is nonempty, so choose

$$\bar\omega \in \operatorname{supp}\lambda \setminus \operatorname{atom}\bar\mu \qquad \text{such that} \qquad \bar P_p(\bar\omega) = -\alpha L_\phi \sigma_0.$$

For $\delta > 0$, put

$$r_\delta := \lambda(B_\delta(\bar\omega)), \qquad \varepsilon_\delta := \sup_{\omega \in B_\delta(\bar\omega)} \|K_p(\omega) - K_p(\bar\omega)\|_{H^1(D)}.$$

Because $\bar\omega \in \operatorname{supp}\lambda$, one has $r_\delta > 0$. Moreover, $r_\delta \to \lambda(\{\bar\omega\}) = 0$ by continuity from above and the nonatomicity of λ, while $\varepsilon_\delta \to 0$ by continuity of K_p at $\bar\omega$. Define μ_δ through its normalized coordinate by

$$(\mu_\delta)_p := \bar\mu_p - \sigma_0 \lambda|_{B_\delta(\bar\omega)} + \sigma_0 r_\delta \delta_{\bar\omega}.$$

The target point is not an atom, and Theorem 3.13 gives

$$\|\mu_\delta - \bar\mu\|_{\mathcal{M}_p(\Omega)} = 2r_\delta \longrightarrow 0.$$

If $e_\delta := \mathcal{N}(\mu_\delta - \bar\mu)$, then

$$\begin{aligned} \|e_\delta\|_{H^1(D)} &\le \varepsilon_\delta r_\delta, && \text{by the Bochner-integral triangle inequality,} \\ \langle r_{\bar\mu}, e_\delta\rangle_{H^1(D)} &= \sigma_0 r_\delta \bar P_p(\bar\omega) - \sigma_0 \int_{B_\delta(\bar\omega)} \bar P_p\, d\lambda = 0, && \text{by Theorem 3.18.} \end{aligned}$$

The exact quadratic expansion of the fidelity and the definition of Φ_ϕ therefore give, for all sufficiently small δ such that $r_\delta \le T$,

$$\begin{aligned} J(\mu_\delta) - J(\bar\mu) &= \frac{1}{2}\|e_\delta\|^2_{H^1(D)} + \alpha\big[\phi(r_\delta) - L_\phi r_\delta\big] \\ &\le \frac{1}{2}\big(\varepsilon_\delta^2 - \alpha\gamma_T\big) r_\delta^2 && \text{by (25)} \\ &< 0, && \text{for sufficiently small } \delta. \end{aligned}$$

This contradicts local minimality. Hence $\bar\mu_{\mathrm{cont}} = 0$.

It remains to count the atoms. The set $K_p(\mathcal{K})$ is compact because $\mathcal{K}$ is compact and K_p is continuous. Cover it by balls of radius $\frac{1}{2}\sqrt{\alpha\gamma_T}$. Each such ball has diameter $\sqrt{\alpha\gamma_T}$, which is strictly smaller than the separation $\sqrt{2\alpha\gamma_T}$ in (23). Hence each ball contains the image of at most one atom. A compact set has a finite cover at every positive radius, and (24) follows. $\square$

Corollary 3.22. *Assume the hypotheses of Theorems 3.16 and 3.21. Then J admits a global minimizer, every global minimizer is finite atomic, and*

$$\min_{\mu\in\mathcal{M}_p(\Omega)} J(\mu) = \min_{\substack{N\geq 1,\ \omega_1,\dots,\omega_N\in\Omega \\ c_1,\dots,c_N\in\mathbb{R}}} \left\{ \frac{1}{2}\left\| \sum_{n=1}^{N} c_n K(\omega_n) - V \right\|_{H^1(D)}^2 + \alpha \sum_{n=1}^{N} \phi\big(w_p(\omega_n)|c_n|\big) \right\}. \tag{26}$$

The minimum on the right is attained. If μ_* is any global minimizer with $J(\mu_*) > 0$, then it satisfies the data-dependent bound

$$\#\,\mathrm{atom}(\mu_*) \leq N_{\mathrm{cov}}\left(K_p\big(\mathcal{K}(\mu_*)\big), \frac{1}{2}\sqrt{\alpha\gamma_{T(\mu_*)}} \right). \tag{27}$$

If $V \neq 0$, define

$$T_0 := \phi^{-1}\left(\frac{\|V\|_{H^1(D)}^2}{2\alpha} \right)$$

and

$$\mathcal{K}_0 := \left\{ \omega \in \Omega : \|V\|_{H^1(D)} \|K_p(\omega)\|_{H^1(D)} \geq \alpha\phi'(T_0) \right\}. \tag{28}$$

Then $\mathcal{K}_0$ is compact, every atom of every global minimizer lies in $\mathcal{K}_0$, and

$$\#\,\mathrm{atom}(\mu_*) \leq N_{\mathrm{cov}}\left(K_p(\mathcal{K}_0), \frac{1}{2}\sqrt{\alpha\gamma_{T_0}} \right) < \infty. \tag{29}$$

If $V = 0$, then the zero measure is the unique global minimizer.

Proof. We combine the direct method in Theorem 3.16 with the local result in Theorem 3.21. Theorem 3.16 gives a global minimizer. Every global minimizer is $\mathcal{M}_p(\Omega)$-local, so Theorem 3.21 makes it finite atomic; when its objective is positive, Theorem 3.21 also gives (27). Its distinct atom representation has objective equal to the expression on the right of (26), so the right-hand infimum is no larger than the measure minimum.

Conversely, consider an arbitrary finite network, possibly with repeated locations. For each distinct location η_j, merge its coefficients into $d_j := \sum_{n:\,\omega_n=\eta_j} c_n$. The represented function, and hence the fidelity, is unchanged. Monotonicity and repeated subadditivity of ϕ give

$$\phi\big(w_p(\eta_j)|d_j|\big) \leq \phi\left(w_p(\eta_j) \sum_{n:\,\omega_n=\eta_j} |c_n| \right) \leq \sum_{n:\,\omega_n=\eta_j} \phi\big(w_p(\eta_j)|c_n|\big).$$

After summing over the distinct locations, the measure objective is no larger than the displayed finite-network objective. Thus the measure minimum is no larger than the right-hand infimum in (26). The two infima are equal, and the finite atom representation of a global minimizer attains the right-hand side, so both infima are minima.

Suppose $V \neq 0$. Then a global minimizer μ_* has positive objective, because Theorem 3.21 would otherwise give $\mu_* = 0$, whose fidelity is positive. By comparison with the zero measure,

$$\begin{aligned} J(\mu_*) &\leq J(0) = \frac{1}{2}\|V\|_{H^1(D)}^2 = \alpha\phi(T_0), \\ \|r_{\mu_*}\|_{H^1(D)} &\leq \|V\|_{H^1(D)}, && \text{since } L(\mu_*) \leq J(\mu_*). \end{aligned}$$

Hence

$$T(\mu_*) \leq T_0, \qquad \gamma_{T(\mu_*)} \geq \gamma_{T_0}. \tag{30}$$

For an atom ω_i of normalized magnitude $t_i = |\mu_{*,p}(\{\omega_i\})|$, Lemma 3.20 gives $t_i \leq T(\mu_*)$. Theorem 3.18 then gives

$$\begin{aligned} \alpha\phi'(T_0) \leq \alpha\phi'(t_i) &= |\langle r_{\mu_*}, K_p(\omega_i)\rangle_{H^1(D)}| && \text{since } t_i \leq T(\mu_*) \leq T_0 \\ &\leq \|r_{\mu_*}\|_{H^1(D)} \|K_p(\omega_i)\|_{H^1(D)} && \text{by Cauchy–Schwarz} \\ &\leq \|V\|_{H^1(D)} \|K_p(\omega_i)\|_{H^1(D)}. \end{aligned}$$

Thus every atom belongs to $\mathcal{K}_0$. The level in (28) is strictly positive, while $K_p \in C_0(\Omega; H^1(D))$, so $\mathcal{K}_0$ is compact. Combining (30) with (23) gives pairwise separation at least $\sqrt{2\alpha\gamma_{T_0}}$. The covering argument from Theorem 3.21 on $K_p(\mathcal{K}_0)$ proves (29).

If $V = 0$, then $J(0) = 0$. Nonnegativity makes zero the minimum, and the zero-objective conclusion of Theorem 3.21 shows that no other global minimizer exists. □

The hypotheses used so far are jointly satisfiable, and the following family makes the constants explicit.

Example 3.23 (Admissible penalty family). For $\gamma \geq 0$, define

$$\phi_\gamma(z) := \begin{cases} \dfrac{z}{2} + \dfrac{1}{4\gamma}\log(1+2\gamma z), & \gamma > 0, \\ z, & \gamma = 0. \end{cases} \tag{31}$$

Then $\phi_\gamma(0) = 0$ and, for $\gamma > 0$,

$$\phi_\gamma'(z) = \frac{1}{2} + \frac{1}{2(1+2\gamma z)} \in \left(\tfrac{1}{2}, 1\right], \qquad \phi_\gamma''(t) = -\frac{\gamma}{(1+2\gamma t)^2}.$$

Hence $L_{\phi_\gamma} = \phi_\gamma'(0) = 1 \in (0,\infty)$, which is Assumption 3.2; the derivative is positive and nonincreasing and $\phi_\gamma(z) \geq z/2 \to \infty$, which is Assumption 3.3. For Assumption 3.4, the second derivative is continuous and negative on $(0,\infty)$, part (1) holds with $\gamma_1 = \gamma$ and any $\hat{z}_1 > 0$, and part (2) holds with

$$\gamma_A = \frac{\gamma}{(1+2\gamma A)^2} > 0. \tag{32}$$

Thus ϕ_γ with $\gamma > 0$ satisfies Assumptions 3.2–3.4, and (32) turns the atom separation $\sqrt{2\alpha\gamma_{T_0}}$ and the neuron bound (29) into computable quantities. This is the penalty used by Algorithm 1.

The limit $\gamma \to 0$ is the convex endpoint. Assumption 3.4 fails at $\gamma = 0$, so Theorem 3.21 and Corollary 3.22 do not apply; the weaker conclusions of Theorems 3.16, 3.18, and 5.1 remain available. The degeneration is quantitative rather than abrupt: by (32), $\gamma_{T_0} \to 0$ as $\gamma \to 0$, so the guaranteed separation between atoms tends to zero and the bound (29) tends to $+\infty$.

For fixed, distinct locations $\boldsymbol{\omega} = (\omega_n)_{n=1}^N \in \Omega^N$, define the reduced coefficient objective

$$J_{\boldsymbol{\omega}}(c) := L\left(\sum_{n=1}^N c_n \delta_{\omega_n}\right) + \alpha \sum_{n=1}^N \phi\big(w_p(\omega_n)|c_n|\big), \qquad c \in \mathbb{R}^N. \tag{33}$$

Theorem 3.24 (Sufficient condition for local optimality)**.** *Let ϕ satisfy Assumptions 3.2–3.4, let $p > s_1$, let $V \in H^1(D)$, and suppose that Assumption 3.1 holds. Let*

$$\bar{\boldsymbol{\omega}} := (\bar{\omega}_n)_{n=1}^N, \qquad \bar{c} := (\bar{c}_n)_{n=1}^N \in (\mathbb{R}\setminus\{0\})^N, \qquad \bar{\mu} := \sum_{n=1}^N \bar{c}_n \delta_{\bar{\omega}_n},$$

where the points $\bar{\omega}_n$ are distinct and every $\bar{c}_n$ is nonzero. Define

$$\bar{P}_p(\omega) := \langle \mathcal{N}\bar{\mu} - V, K_p(\omega)\rangle_{H^1(D)}.$$

Assume that

1. *$\bar{c} \in (\mathbb{R}\setminus\{0\})^N$ is a local minimizer of $J_{\bar{\boldsymbol{\omega}}}$;*
2. *the strict inequality*
$$|\bar{P}_p(\omega)| < \alpha L_\phi$$
holds for every $\omega \in \Omega \setminus \{\bar{\omega}_1, \ldots, \bar{\omega}_N\}$.

Then $\bar{\mu}$ is an $\mathcal{M}_p(\Omega)$-local minimizer of J.

Proof. We use the normalized coordinate to reduce the claim to total-variation estimates. Set

$$\bar q := (\bar q_n)_{n=1}^N, \qquad \bar q_n := w_p(\bar\omega_n)\bar c_n.$$

For $q = (q_n)_{n=1}^N \in \mathbb{R}^N$, define

$$\widehat J(q) := \frac{1}{2}\left\|\sum_{n=1}^N q_n K_p(\bar\omega_n) - V\right\|_{H^1(D)}^2 + \alpha\sum_{n=1}^N \phi(|q_n|).$$

The invertible diagonal change $q_n = w_p(\bar\omega_n)c_n$ shows that $\bar q$ is a local minimizer of $\widehat J$.

We first obtain a uniform strict gap. Stationarity of $J_{\bar{\boldsymbol\omega}}$ at $\bar c$ gives

$$\bar P_p(\bar\omega_n) = -\alpha\phi'(|\bar q_n|)\,\mathrm{sign}(\bar q_n).$$

Assumption 3.4(2) implies $\phi'(t) < L_\phi$ for every $t > 0$. Together with condition (2) of Theorem 3.24, this gives

$$|\bar P_p(\omega)| < \alpha L_\phi \qquad (\omega\in\Omega).$$

Since $\bar P_p \in C_0(\Omega)$, the positive continuous function $\alpha L_\phi - |\bar P_p|$ tends to αL_ϕ at infinity. Hence there is $m > 0$ such that

$$|\bar P_p(\omega)| \le \alpha L_\phi - m \qquad (\omega\in\Omega). \tag{34}$$

Choose $\varepsilon_1 > 0$ such that $\widehat J(q) \ge \widehat J(\bar q)$ whenever $\|q-\bar q\|_{\ell^1} \le \varepsilon_1$. Put

$$B := \sup_{\omega\in\Omega}\|K_p(\omega)\|_{H^1(D)}, \qquad B_{\bar\omega} := \max_{1\le n\le N}\|K_p(\bar\omega_n)\|_{H^1(D)}.$$

Choose $\varepsilon > 0$ so that

$$\varepsilon \le \varepsilon_1, \qquad \varepsilon \le \hat z_1, \qquad B\,B_{\bar\omega}\varepsilon \le \frac{m}{2}, \qquad \frac{\alpha\gamma_1\varepsilon}{2} \le \frac{m}{4}.$$

A condition with zero left-hand side is void.

Let $\mu \in \mathcal{M}_p(\Omega)$ satisfy $\|\mu - \bar\mu\|_{\mathcal{M}_p(\Omega)} \le \varepsilon$. There is nothing to prove if $J(\mu) = +\infty$. Otherwise, write $\xi := \mu_p$ and, for $1 \le n \le N$, set

$$q_n := \xi(\{\bar\omega_n\}), \qquad q := (q_n)_{n=1}^N, \qquad \xi_0 := \sum_{n=1}^N q_n\delta_{\bar\omega_n}, \qquad \xi = \xi_0 + \widetilde\xi.$$

Then $\widetilde\xi$ has no mass at the support points, and the normalization isometry gives

$$\|q-\bar q\|_{\ell^1} + \|\widetilde\xi\|_{\mathcal{M}(\Omega)} = \|\mu-\bar\mu\|_{\mathcal{M}_p(\Omega)} \le \varepsilon.$$

Define

$$r' := \sum_{n=1}^N (q_n - \bar q_n)K_p(\bar\omega_n), \qquad Q_p(\omega) := \langle r', K_p(\omega)\rangle_{H^1(D)}.$$

Then

$$\|r'\|_{H^1(D)} \le B_{\bar\omega}\|q-\bar q\|_{\ell^1} \le B_{\bar\omega}\varepsilon,$$
$$|Q_p(\omega)| \le B\,B_{\bar\omega}\varepsilon \le \frac{m}{2}.$$

Combining this with (34) gives

$$|\bar P_p(\omega) + Q_p(\omega)| \le \alpha L_\phi - \frac{m}{2} \qquad (\omega\in\Omega). \tag{35}$$

The exact quadratic expansion of the fidelity in the direction $\widetilde\xi$ and the nonnegativity of its quadratic remainder give

$$L(\mu) \ge \frac{1}{2}\left\|\sum_{n=1}^N q_n K_p(\bar\omega_n) - V\right\|_{H^1(D)}^2 + \int_\Omega (\bar P_p + Q_p)\,d\widetilde\xi. \tag{36}$$

The measures ξ_0 and $\widetilde{\xi}$ are mutually singular. Every atom of $\widetilde{\xi}$ has magnitude at most $\|\widetilde{\xi}\|_{\mathcal{M}(\Omega)} \le \varepsilon \le \hat{z}_1$. Moreover,

$$\Phi_\phi(\xi) = \sum_{n=1}^{N} \phi(|q_n|) + \Phi_\phi(\widetilde{\xi}).$$

Proposition 3.7(1), applied to the atoms and combined with the definition on the continuous part, gives the linear term $L_\phi\|\widetilde{\xi}\|_{\mathcal{M}(\Omega)}$ and an atomic remainder bounded below by

$$-\frac{\gamma_1}{2} \sum_{\omega \in \operatorname{atom} \widetilde{\xi}} |\widetilde{\xi}(\{\omega\})|^2 \ge -\frac{\gamma_1 \varepsilon}{2} \sum_{\omega \in \operatorname{atom} \widetilde{\xi}} |\widetilde{\xi}(\{\omega\})|,$$

because every atom magnitude is at most ε. Therefore

$$\Phi_\phi(\widetilde{\xi}) \ge \left(L_\phi - \frac{\gamma_1 \varepsilon}{2}\right) \|\widetilde{\xi}\|_{\mathcal{M}(\Omega)}. \tag{37}$$

Using (36), (35), and (37), we obtain

$$\begin{aligned} J(\mu) &\ge \widehat{J}(q) + \int_\Omega (\bar{P}_p + Q_p)\, d\widetilde{\xi} + \alpha \Phi_\phi(\widetilde{\xi}) \\ &\ge \widehat{J}(q) + \left(\frac{m}{2} - \frac{\alpha\gamma_1\varepsilon}{2}\right) \|\widetilde{\xi}\|_{\mathcal{M}(\Omega)} \\ &\ge \widehat{J}(\bar{q}) + \frac{m}{4}\|\widetilde{\xi}\|_{\mathcal{M}(\Omega)} \ge J(\bar{\mu}). \end{aligned}$$

Thus $\bar{\mu}$ is an $\mathcal{M}_p(\Omega)$-local minimizer. □

3.6. **Admissible activations.** The two exponents in Assumption 3.1 play different roles: s_0 controls the value atom in $L^2(D)$ and is enough for Theorem 3.16; s_1 controls the full atom in $H^1(D)$ and is used by Theorems 3.18, 3.21, and 5.1. Since $\|K(\omega)\|_{L^2(D)} \le \|K(\omega)\|_{H^1(D)}$, it is consistent to impose $s_0 \le s_1$, as Assumption 3.1 does. The exponent s_1 is obtained by estimating $a\rho'(a \cdot x + b)$ separately. The condition $p > s_1$ normalizes the full H^1 atom; together with Assumption 3.4, it is sufficient for the finite-support conclusion in Theorem 3.21. The following exponents satisfy the pointwise bounds on a bounded domain.

activation	s_0	s_1	sufficient condition
$\tanh z$	0	1	$p > 1$
e^{-z^2}	0	1	$p > 1$
$\log(1 + e^z)$	1	1	$p > 1$
$(z_+)^k$	k	k	$p > k$

For bounded activations such as tanh and the Gaussian, existence alone requires only $p > 0$; the condition $p > 1$ in the table is the one that also dominates the elementary global H^1 growth estimate. The last row is listed for comparison with the positively homogeneous formulation of Section 4.

4. The k-homogeneous formulation

Section 3 treats concave penalties whose right slope $L_\phi = \phi'(0+)$ is finite and positive. In this section we observe the case where this assumption does not hold.

This section treats activations for which radial scaling can be absorbed into the outer coefficient. Positive homogeneity makes the radial part of an inner parameter redundant: every atom can be moved onto the unit sphere $\mathbb{S}^d \subset \mathbb{R}^{d+1}$ at the cost of rescaling its outer coefficient. The parameter domain of the resulting problem is compact, so the dictionary is bounded, and neither the polynomial weight w_p nor the weighted space $\mathcal{M}_p(\Omega)$ of Section 3 is required. Throughout this section we work on $\mathcal{M}(\mathbb{S}^d)$ with the total-variation norm $\|\cdot\|_{\mathcal{M}(\mathbb{S}^d)}$, which we identify with the subspace of $\mathcal{M}(\Omega)$ of measures vanishing outside $\mathbb{S}^d$; the value operator $\mathcal{N}$, the residual

$r_\mu = \mathcal{N}\mu - V$, and the fidelities L and l^M are those of Section 3. Whenever $K : \mathbb{S}^d \to H^1(D)$ is continuous, $\mathcal{N}\mu$ is read as the $H^1(D)$-valued Bochner integral $\int_{\mathbb{S}^d} K\, d\mu$, which exists because a continuous map on a compact metric space is bounded and has compact, hence separable, range; by the Pettis measurability theorem it is strongly measurable, and $\int_{\mathbb{S}^d} \|K\|_{H^1(D)}\, d|\mu| \leq \big(\sup_{\mathbb{S}^d} \|K\|_{H^1(D)}\big)\|\mu\|_{\mathcal{M}(\mathbb{S}^d)} < \infty$; for a purely atomic $\mu = \sum_j c_j \delta_{\omega_j}$ with $\sum_j |c_j| < \infty$ it is the absolutely convergent series $\sum_j c_j K(\omega_j)$, and it agrees with the pointwise definition of Section 3 whenever both are defined. We use the inner product $\langle u, v\rangle_{H^1(D)} = \langle u, v\rangle_{L^2(D)} + \langle \nabla u, \nabla v\rangle_{L^2(D)}$, so that $L(\mu) = \frac{1}{2}\|r_\mu\|^2_{H^1(D)}$ whenever $\mathcal{N}\mu \in H^1(D)$.

For the $k \geq 2$ gradient-fitting case below, the induced exponent is $q = 2/(k+1) < 1$, so $\psi_k(t) = t^q$ has infinite right slope at zero. The homogeneous formulation is therefore the infinite-slope endpoint of the same structural penalty picture. It does not extend the finite-slope results: Theorem 3.9, Theorem 3.16, Theorem 3.18, Theorem 3.21, Theorem 5.1, and Theorem 3.24 cannot be invoked by substituting $L_\phi = \infty$ in their hypotheses. The measure penalty of this section is instead constructed directly in Definition 4.6, and every result below is proved from that definition.

Assumption 4.1 (k-homogeneity)**.** For some $k \geq 1$,

$$K(\lambda\omega) = \lambda^k K(\omega) \qquad (\lambda > 0,\ \omega \in \mathbb{R}^{d+1}).$$

Remark 4.2. The algebraic rescaling result below holds for every $k \geq 1$. For gradient fitting with ReLU^k we use $k \geq 2$, so that $K(\omega) \in C^1(\overline{D})$ and the induced exponent $q = 2/(k+1)$ lies strictly between 0 and 1. The derivative of $K(\omega)$ is, up to the factor k, a ReLU^{k-1} ridge function: the value term contains powers of order k, whereas the gradient term contains powers of order $k-1$. When $k = 2$, the gradient is continuous but is not differentiable on the hyperplane $a \cdot x + b = 0$.

4.1. **Radial rescaling and the induced exponent.**

Lemma 4.3 (Rescaling of a k-homogeneous network)**.** *Let Assumption 4.1 hold, and let $\xi, \zeta \geq 1$. For a fixed represented coefficient $\widetilde{c}$ and a unit direction $\widehat{\omega} \in \mathbb{S}^d$ with $K(\widehat{\omega}) \neq 0$, minimize*

$$\frac{|c|^\xi}{\xi} + \frac{|\omega|^\zeta}{\zeta}$$

over radial representations

$$\omega = \lambda\widehat{\omega}, \qquad \lambda \geq 0, \qquad c\,K(\omega) = \widetilde{c}\,K(\widehat{\omega}).$$

The minimum equals

$$C_{k,\xi,\zeta}|\widetilde{c}|^q, \qquad q := \frac{\xi\zeta}{k\xi + \zeta}, \qquad C_{k,\xi,\zeta} := \frac{1}{q}\,k^{-k\xi/(k\xi+\zeta)}. \tag{38}$$

Consequently, an objective with the separable penalty above is equivalent to an objective on $\mathbb{S}^d$ with the induced penalty $C_{k,\xi,\zeta}\sum_n |c_n|^q$.

Proof. If $\widetilde{c} = 0$, the zero representation $c = 0$, $\omega = 0$ has penalty zero, in agreement with (38). We therefore assume $\widetilde{c} \neq 0$.

Write $\omega = \lambda\widehat{\omega}$, where $\lambda = |\omega| > 0$ and $\widehat{\omega} \in \mathbb{S}^d$. By k-homogeneity,

$$c\,K(\omega) = c\lambda^k K(\widehat{\omega}) = \widetilde{c}\,K(\widehat{\omega}).$$

Since $K(\widehat{\omega}) \neq 0$, the representation constraint is equivalent to $c\lambda^k = \widetilde{c}$. For fixed $\widetilde{c}$, the penalty becomes

$$g(\lambda) = \frac{|\widetilde{c}|^\xi}{\xi\lambda^{k\xi}} + \frac{\lambda^\zeta}{\zeta}.$$

The function g tends to $+\infty$ as $\lambda \downarrow 0$ and as $\lambda \to \infty$. Moreover,

$$g'(\lambda) = -k|\widetilde{c}|^\xi\lambda^{-k\xi-1} + \lambda^{\zeta-1},$$

so

$$g'(\lambda) = 0 \quad \iff \quad \lambda^{k\xi+\zeta} = k|\widetilde{c}|^{\xi}.$$

This equation has the unique solution

$$\lambda_* = \left(k|\widetilde{c}|^{\xi}\right)^{1/(k\xi+\zeta)}.$$

The derivative is negative before λ_* and positive after λ_*, so this point is the global minimizer. The first term at λ_* is

$$\begin{aligned}\frac{|\widetilde{c}|^{\xi}}{\xi\lambda_*^{k\xi}} &= \frac{1}{\xi}|\widetilde{c}|^{\xi}\left(k|\widetilde{c}|^{\xi}\right)^{-k\xi/(k\xi+\zeta)} \\ &= \frac{1}{\xi}k^{-k\xi/(k\xi+\zeta)}|\widetilde{c}|^{\xi-k\xi^2/(k\xi+\zeta)} \\ &= \frac{1}{\xi}k^{-k\xi/(k\xi+\zeta)}|\widetilde{c}|^{\xi\zeta/(k\xi+\zeta)}.\end{aligned}$$

The second term is

$$\begin{aligned}\frac{\lambda_*^{\zeta}}{\zeta} &= \frac{1}{\zeta}\left(k|\widetilde{c}|^{\xi}\right)^{\zeta/(k\xi+\zeta)} \\ &= \frac{k}{\zeta}k^{-k\xi/(k\xi+\zeta)}|\widetilde{c}|^{\xi\zeta/(k\xi+\zeta)}.\end{aligned}$$

Since

$$q = \frac{\xi\zeta}{k\xi+\zeta}, \qquad \frac{1}{\xi}+\frac{k}{\zeta} = \frac{k\xi+\zeta}{\xi\zeta} = \frac{1}{q},$$

adding the two terms gives

$$\begin{aligned}g(\lambda_*) &= k^{-k\xi/(k\xi+\zeta)}\left(\frac{1}{\xi}+\frac{k}{\zeta}\right)|\widetilde{c}|^{q} \\ &= \frac{1}{q}\,k^{-k\xi/(k\xi+\zeta)}|\widetilde{c}|^{q} = C_{k,\xi,\zeta}|\widetilde{c}|^{q}.\end{aligned}$$

This calculation is performed independently for every neuron. A neuron with $\omega_n = 0$, or with $K(\omega_n/|\omega_n|) = 0$, represents the zero function, and both the separable penalty and the induced penalty $C_{k,\xi,\zeta}|\widetilde{c}|^q$ are then minimized by the zero representation with common value zero, so such neurons do not affect the equivalence. The rescaling preserves each represented function $c_nK(\omega_n)$, and hence preserves the network and either the population fidelity L or the empirical fidelity l^M. Summing the minimum costs proves the stated equivalence. $\square$

Remark 4.4 (The quadratic case and the meaning of α). For $\xi = \zeta = 2$,

$$q = \frac{2}{k+1}, \qquad C_k := C_{k,2,2} = \frac{k+1}{2}\,k^{-k/(k+1)}.$$

In particular $C_1 = 1$ and $C_2 = \frac{3}{2}\,2^{-2/3}$. If the separable penalty of Lemma 4.3 carries a coefficient $\alpha_{\text{orig}} > 0$, the induced sphere-normalized penalty is $\alpha\sum_n |c_n|^q$ with the effective coefficient

$$\alpha := C_k\,\alpha_{\text{orig}}.$$

From here on α always denotes this effective coefficient, which is also the coefficient used by the numerical implementation. Since C_k depends on k, equal nominal values of α for different powers k do *not* correspond to equal coefficients of the original separable penalty.

Lemma 4.5 (The ReLU^k dictionary). *Let $k \geq 2$, let $D \subset \mathbb{R}^d$ be bounded with positive Lebesgue measure, let $\rho(z) = (z_+)^k$, and let*

$$[K(\omega)](x) = (a\cdot x+b)_+^k, \qquad \omega = (a,b) \in \mathbb{R}^{d+1}.$$

Then

(1) *K satisfies Assumption 4.1 with degree k;*

(2) $K(\omega) \in C^1(\mathbb{R}^d)$ *with* $\nabla K(\omega)(x) = k(a \cdot x + b)_+^{k-1} a$*, and this classical gradient is also the distributional gradient of* $K(\omega)$ *on* D*;*
(3) $K : \mathbb{S}^d \to H^1(D)$ *is continuous; and*
(4) $C_S := \sup_{\omega \in \mathbb{S}^d} \|K(\omega)\|_{H^1(D)}$ *satisfies* $0 < C_S < \infty$*.*

Proof. (1) For $\lambda > 0$ one has $(\lambda z)_+ = \lambda z_+$, hence $(\lambda a \cdot x + \lambda b)_+^k = \lambda^k (a \cdot x + b)_+^k$ for every x.

(2) Set $h_k(z) := (z_+)^k$. On $(0, \infty)$ and on $(-\infty, 0)$ the function h_k is differentiable with $h_k'(z) = k(z_+)^{k-1}$. At $z = 0$ both one-sided difference quotients are $O(|z|^{k-1})$ and vanish in the limit because $k \geq 2$, so $h_k'(0) = 0 = k(0_+)^{k-1}$. The formula $h_k'(z) = k(z_+)^{k-1}$ therefore holds on all of $\mathbb{R}$, and h_k' is continuous because $k - 1 \geq 1$. The chain rule applied to $x \mapsto a \cdot x + b$ gives $\nabla K(\omega)(x) = k(a \cdot x + b)_+^{k-1} a$. Since D is bounded, both $K(\omega)$ and $\nabla K(\omega)$ are bounded and continuous on D; a continuously differentiable function on a neighbourhood of D lies in $H^1(D)$ and its distributional gradient is its classical gradient.

(3) Let $\omega_j = (a_j, b_j) \to \omega = (a, b)$ in $\mathbb{S}^d$ and put $R_D := \sup_{x \in D} |x| < \infty$. Then

$$\sup_{x \in D} \big| (a_j \cdot x + b_j) - (a \cdot x + b) \big| \leq |a_j - a| \, R_D + |b_j - b| \longrightarrow 0,$$

so the pre-activations converge uniformly on D. All of them lie in the compact interval $I := [-(R_D + 1), R_D + 1]$, because $|a_j \cdot x + b_j| \leq R_D + 1$ for every j and every $x \in D$ by $|a_j| \leq 1$, $|b_j| \leq 1$. The functions h_k and h_k' are uniformly continuous on I, whence

$$K(\omega_j) \to K(\omega) \quad \text{and} \quad h_k'(a_j \cdot x + b_j) \to h_k'(a \cdot x + b) \qquad \text{uniformly on } D.$$

Since $\nabla K(\omega_j)(x) = h_k'(a_j \cdot x + b_j)\, a_j$, $a_j \to a$, and the scalar factors are uniformly bounded on D, the gradients converge uniformly on D as well. As D has finite Lebesgue measure, uniform convergence implies convergence in $L^2(D)$, so $K(\omega_j) \to K(\omega)$ in $H^1(D)$.

(4) Finiteness follows from (3) and compactness of $\mathbb{S}^d$, since a continuous real function attains a finite maximum on a compact set. For positivity take $\widehat{\omega} = (0, 1) \in \mathbb{S}^d$, for which $[K(\widehat{\omega})](x) = 1$ for every x; hence $C_S \geq \|K(\widehat{\omega})\|_{H^1(D)} = |D|^{1/2} > 0$. ☐

The power k in Assumption 4.1 may be any real number $k \geq 1$; the Barron–Sobolev embedding of Proposition 2.1 applies to integer k. Since these activations are not polynomials, the corresponding networks are dense in $C(Q)$ for every compact set $Q \subset \mathbb{R}^d$ [15, Theorem 1]. The restriction $k \geq 2$ is what makes gradient-augmented fitting available: the second-order weak derivative of ReLU is a distribution and not a function, whereas for $k \geq 2$ the first derivative of $(z_+)^k$ is continuous, so the pointwise gradient values entering l^M are defined.

In the remainder of the section $K : \mathbb{S}^d \to H^1(D)$ is continuous and we write

$$C_S := \sup_{\omega \in \mathbb{S}^d} \|K(\omega)\|_{H^1(D)},$$

which is finite because a continuous real function attains a finite maximum on the compact set $\mathbb{S}^d$. Lemma 4.5 verifies both properties for ReLU^k with $k \geq 2$.

4.2. **The measure problem on the sphere.** For $k \geq 2$ define the scalar penalty

$$\psi_k(t) := t^q, \qquad q := \frac{2}{k+1} \in (0, 1), \qquad t \geq 0,$$

which is the exponent of Lemma 4.3 for $\xi = \zeta = 2$. It is continuous and strictly increasing with $\psi_k(0) = 0$ and $\psi_k'(0^+) = +\infty$, and it is subadditive:

$$\psi_k(s + t) \leq \psi_k(s) + \psi_k(t) \qquad (s, t \geq 0). \tag{39}$$

Indeed, for $s + t > 0$ the numbers $u := s/(s + t)$ and $1 - u = t/(s + t)$ lie in $[0, 1]$, where $v^q \geq v$; hence $\psi_k(s) + \psi_k(t) = (s + t)^q \big(u^q + (1 - u)^q\big) \geq (s + t)^q$. Iterating (39) and passing to the limit in the finite partial sums gives, for every sequence $(t_j)_{j\geq 1}$ of nonnegative numbers,

$$\psi_k\Big(\sum_{j\geq 1} t_j\Big) \leq \sum_{j\geq 1} \psi_k(t_j). \tag{40}$$

Definition 4.6 (Homogeneous measure penalty)**.** For $\mu \in \mathcal{M}(\mathbb{S}^d)$ with atomic and continuous parts μ_{atom} and μ_{cont} as in Section 2.1, set

$$\Phi_{\psi_k}(\mu) := \begin{cases} \displaystyle\sum_{\omega\in \operatorname{atom}\mu} \psi_k\big(|\mu(\{\omega\})|\big), & \mu_{\text{cont}} = 0, \\ +\infty, & \mu_{\text{cont}} \neq 0. \end{cases}$$

Lemma 4.7 (Elementary properties of Φ_{ψ_k})**.** *Let $\mu \in \mathcal{M}(\mathbb{S}^d)$. Then*

(1) *the sum in Definition 4.6 has at most countably many nonnegative terms and is therefore independent of the enumeration and well defined in $[0, +\infty]$;*
(2) *$\Phi_{\psi_k}(0) = 0$, and $\Phi_{\psi_k}(\mu) > 0$ for every $\mu \neq 0$ with $\Phi_{\psi_k}(\mu) < \infty$;*
(3) *$\Phi_{\psi_k}(\mu) \geq \|\mu\|^q_{\mathcal{M}(\mathbb{S}^d)}$ whenever $\Phi_{\psi_k}(\mu) < \infty$; and*
(4) *if $\mu = \mu_{\boldsymbol{\omega},c} = \sum_{n=1}^N c_n\delta_{\omega_n}$ with arbitrary, possibly repeated, locations $\omega_n \in \mathbb{S}^d$, then*

$$\Phi_{\psi_k}(\mu) \leq \sum_{n=1}^N \psi_k(|c_n|),$$

with equality whenever the ω_n are distinct.

Proof. (1) The atom set of a finite signed Radon measure is at most countable (Section 2.1), and every summand $\psi_k(|\mu(\{\omega\})|)$ is nonnegative. A series of nonnegative terms has a value in $[0, +\infty]$ that does not depend on the order of summation.

(2) $\operatorname{atom} 0 = \emptyset$, so the empty sum gives $\Phi_{\psi_k}(0) = 0$. If $\Phi_{\psi_k}(\mu) < \infty$ then $\mu_{\text{cont}} = 0$ and hence $\mu = \mu_{\text{atom}}$; if in addition $\mu \neq 0$, then $\operatorname{atom}\mu \neq \emptyset$ and at least one summand $\psi_k(|\mu(\{\omega\})|)$ is strictly positive.

(3) If $\Phi_{\psi_k}(\mu) < \infty$ then $\mu = \mu_{\text{atom}}$ and $\|\mu\|_{\mathcal{M}(\mathbb{S}^d)} = \sum_{\omega\in\operatorname{atom}\mu} |\mu(\{\omega\})|$. Applying (40) to this sum gives $\psi_k(\|\mu\|_{\mathcal{M}(\mathbb{S}^d)}) \leq \Phi_{\psi_k}(\mu)$, which is the assertion.

(4) The measure μ is purely atomic, its atoms are the distinct locations carrying nonzero total mass, and the mass at such a location ω is $\mu(\{\omega\}) = \sum_{n:\,\omega_n=\omega} c_n$. Since ψ_k is nondecreasing and satisfies (39),

$$\psi_k\big(|\mu(\{\omega\})|\big) \leq \psi_k\Big(\sum_{n:\,\omega_n=\omega} |c_n|\Big) \leq \sum_{n:\,\omega_n=\omega} \psi_k(|c_n|).$$

Summing over the distinct locations and adding the nonnegative terms $\psi_k(|c_n|)$ that belong to locations with vanishing total mass gives the inequality. If the ω_n are distinct, every location carries exactly one term and both inequalities above are equalities; the indices with $c_n = 0$ contribute $\psi_k(0) = 0$ to the right-hand side and do not belong to $\operatorname{atom}\mu$, so the two sides still agree. $\square$

Property (4) is the exact counterpart of the merging of repeated nodes: two atoms at one location may always be replaced by their sum without increasing the penalty, and with a strict decrease unless one of them vanishes.

For $\mu \in \mathcal{M}(\mathbb{S}^d)$ we now set

$$J_{\psi_k}(\mu) := L(\mu) + \alpha\Phi_{\psi_k}(\mu), \tag{41}$$

$$J^M_{\psi_k}(\mu) := l^M(\mu) + \alpha\Phi_{\psi_k}(\mu), \tag{42}$$

with the convention that a sum containing $+\infty$ equals $+\infty$. The empirical fidelity (4) was introduced for finite networks; for ReLU^k with $k \geq 2$ it is defined on all of $\{\mu \in \mathcal{M}(\mathbb{S}^d) : \Phi_{\psi_k}(\mu) < \infty\}$, provided the sampled data $V(x^m)$ and $\nabla V(x^m)$ are given. Indeed, such a μ is purely atomic with $\sum_j |c_j| < \infty$, while Lemma 4.5(2)–(3) and compactness of $\mathbb{S}^d$ bound $|[K(\omega)](x^m)|$ and $|\nabla K(\omega)(x^m)|$ uniformly in $\omega \in \mathbb{S}^d$ and m; hence $\sum_j c_j[K(\omega_j)](x^m)$ and $\sum_j c_j \nabla K(\omega_j)(x^m)$ converge absolutely, and $r_\mu(x^m)$, $\nabla r_\mu(x^m)$ and $l^M(\mu)$ are defined. We study the minimization problem

$$(P_{\psi_k}) \qquad \min_{\mu \in \mathcal{M}(\mathbb{S}^d)} J_{\psi_k}(\mu).$$

For a finite network with distinct locations $\omega_n \in \mathbb{S}^d$, Lemma 4.7(4) gives

$$J_{\psi_k}(\mu_{\boldsymbol{\omega},c}) = L(\mu_{\boldsymbol{\omega},c}) + \alpha \sum_{n=1}^{N} |c_n|^q,$$

which is the objective minimized by the finite-dimensional solver in Algorithm 2; the empirical objective (42) restricts in the same way.

Definition 4.8 (Local minimizer in total variation and reduced width)**.** A measure $\bar{\mu} \in \mathcal{M}(\mathbb{S}^d)$ is a *local minimizer of* (P_{ψ_k}) *in total variation* if $J_{\psi_k}(\bar{\mu}) < \infty$ and there is $\varepsilon > 0$ with

$$J_{\psi_k}(\bar{\mu}) \le J_{\psi_k}(\mu) \qquad \text{for all } \mu \in \mathcal{M}(\mathbb{S}^d) \text{ with } \|\mu - \bar{\mu}\|_{\mathcal{M}(\mathbb{S}^d)} < \varepsilon.$$

For μ with $\Phi_{\psi_k}(\mu) < \infty$ we call $N := \#\,\mathrm{atom}\,\mu \in \mathbb{N}_0 \cup \{\infty\}$ the *reduced width* of μ: it counts the distinct locations carrying nonzero mass, that is, the width of the representation obtained after exact repeated locations are merged and zero coefficients are discarded.

Remark 4.9 (A finite penalty does not bound the support)*.* Let $(\omega_j)_{j\ge1}$ be distinct points of $\mathbb{S}^d$ and $\mu := \sum_{j\ge1} 2^{-j}\delta_{\omega_j}$. Then $\|\mu\|_{\mathcal{M}(\mathbb{S}^d)} = 1$ and

$$\Phi_{\psi_k}(\mu) = \sum_{j\ge1} 2^{-jq} = \frac{2^{-q}}{1-2^{-q}} < \infty,$$

while $\mathrm{atom}\,\mu$ is infinite. A finite value of Φ_{ψ_k} therefore does not by itself imply finite support, and no argument below attributes finite support to the penalty alone. Finiteness of the support is obtained in Theorem 4.11 and Corollary 4.14 from a uniform positive lower bound on the atom weights of a stationary measure.

4.3. **Atomwise optimality.**

Proposition 4.10 (Finite truncations and equality of the infima)**.** *Let $V \in H^1(D)$, let $K : \mathbb{S}^d \to H^1(D)$ be continuous with bound C_S, let $\mu \in \mathcal{M}(\mathbb{S}^d)$ satisfy $\Phi_{\psi_k}(\mu) < \infty$, let $(\omega_j)_{j\in\mathcal{I}}$, $\mathcal{I} \subseteq \mathbb{N}$, enumerate* $\mathrm{atom}\,\mu$, *write $c_j := \mu(\{\omega_j\}) \ne 0$, and set*

$$\mu_m := \sum_{j\in\mathcal{I},\ j\le m} c_j \delta_{\omega_j}.$$

Then

1. *$\|\mu_m - \mu\|_{\mathcal{M}(\mathbb{S}^d)} \to 0$;*
2. *$\mathcal{N}\mu \in H^1(D)$ with $\|\mathcal{N}\mu\|_{H^1(D)} \le C_S\|\mu\|_{\mathcal{M}(\mathbb{S}^d)}$, and $\|\mathcal{N}\mu_m - \mathcal{N}\mu\|_{H^1(D)} \le C_S\|\mu_m - \mu\|_{\mathcal{M}(\mathbb{S}^d)} \to 0$;*
3. *$L(\mu_m) \to L(\mu) < \infty$;*
4. *$\Phi_{\psi_k}(\mu_m) \uparrow \Phi_{\psi_k}(\mu)$; and*
5. *$J_{\psi_k}(\mu_m) \to J_{\psi_k}(\mu)$.*

Consequently the all-measures infimum and the finite-network infimum coincide:

$$\inf_{\mu\in\mathcal{M}(\mathbb{S}^d)} J_{\psi_k}(\mu) = \inf_{N\in\mathbb{N}_0} \inf_{\substack{\omega_n\in\mathbb{S}^d,\ c_n\in\mathbb{R}\\ n=1,\dots,N}} \Big[L(\mu_{\boldsymbol{\omega},c}) + \alpha \sum_{n=1}^{N} \psi_k(|c_n|)\Big]. \tag{43}$$

Proof. We prove the five limits and then compare the two infima.

Since $\Phi_{\psi_k}(\mu) < \infty$, Definition 4.6 gives $\mu_{\text{cont}} = 0$, so $\mu = \sum_{j\in\mathcal{I}} c_j\delta_{\omega_j}$ with distinct ω_j and $c_j \neq 0$. Finiteness of the total variation gives

$$\sum_{j\in\mathcal{I}} |c_j| = \|\mu\|_{\mathcal{M}(\mathbb{S}^d)} < \infty. \tag{44}$$

(1) By construction $\mu - \mu_m$ is the purely atomic measure carrying the atoms with index $j > m$, so $\|\mu_m - \mu\|_{\mathcal{M}(\mathbb{S}^d)} = \sum_{j\in\mathcal{I},\, j>m} |c_j|$, which is the tail of the convergent series (44) and therefore tends to 0.

(2) By (44) and $\|K(\omega_j)\|_{H^1(D)} \leq C_S$, the series $\sum_{j\in\mathcal{I}} c_jK(\omega_j)$ converges absolutely in the Banach space $H^1(D)$; its sum is $\mathcal{N}\mu$, and in particular $\mathcal{N}\mu \in H^1(D)$ with $\|\mathcal{N}\mu\|_{H^1(D)} \leq \sum_{j\in\mathcal{I}} |c_j|\, C_S = C_S\|\mu\|_{\mathcal{M}(\mathbb{S}^d)}$. Consequently

$$\begin{aligned}
\|\mathcal{N}\mu_m - \mathcal{N}\mu\|_{H^1(D)} &= \Big\| \sum_{j\in\mathcal{I},\, j>m} c_jK(\omega_j) \Big\|_{H^1(D)} \\
&\leq \sum_{j\in\mathcal{I},\, j>m} |c_j|\, \|K(\omega_j)\|_{H^1(D)} \\
&\leq C_S \sum_{j\in\mathcal{I},\, j>m} |c_j| = C_S\|\mu_m - \mu\|_{\mathcal{M}(\mathbb{S}^d)},
\end{aligned}$$

which tends to 0 by (1).

(3) Each μ_m is a finite atomic measure, so $\mathcal{N}\mu_m \in H^1(D)$, and $\mathcal{N}\mu \in H^1(D)$ by (2). On $H^1(D)$ the map $u \mapsto \frac{1}{2}\|u - V\|^2_{H^1(D)}$ is continuous, and $L(\mu_m) = \frac{1}{2}\|\mathcal{N}\mu_m - V\|^2_{H^1(D)}$, $L(\mu) = \frac{1}{2}\|\mathcal{N}\mu - V\|^2_{H^1(D)}$. Hence (2) gives $L(\mu_m) \to L(\mu)$, and $L(\mu) < \infty$. Because the $H^1(D)$ norm controls both the value and the gradient term, this is convergence of both channels of the gradient-augmented fidelity, not merely of the values.

(4) $\Phi_{\psi_k}(\mu_m) = \sum_{j\in\mathcal{I},\, j\leq m} \psi_k(|c_j|)$ is the m-th partial sum of the series of nonnegative terms whose value is $\Phi_{\psi_k}(\mu)$; partial sums of a nonnegative series increase to its value.

(5) Add α times (4) to (3); both limits are finite.

It remains to compare the infima; call them i_{meas} and i_{disc}, in the order in which they appear in (43). For every finite family $(\boldsymbol{\omega}, c)$, Lemma 4.7(4) gives

$$J_{\psi_k}(\mu_{\boldsymbol{\omega},c}) \leq L(\mu_{\boldsymbol{\omega},c}) + \alpha\sum_{n=1}^{N} \psi_k(|c_n|),$$

and $\mu_{\boldsymbol{\omega},c} \in \mathcal{M}(\mathbb{S}^d)$; taking the infimum over all finite families yields $i_{\text{meas}} \leq i_{\text{disc}}$. Conversely, let $\mu \in \mathcal{M}(\mathbb{S}^d)$. If $J_{\psi_k}(\mu) = \infty$ then trivially $i_{\text{disc}} \leq J_{\psi_k}(\mu)$. If $J_{\psi_k}(\mu) < \infty$, each truncation μ_m is a finite family with distinct locations and nonzero coefficients, so $i_{\text{disc}} \leq J_{\psi_k}(\mu_m)$ for every m, and letting $m \to \infty$ in (5) gives $i_{\text{disc}} \leq J_{\psi_k}(\mu)$. Taking the infimum over μ yields $i_{\text{disc}} \leq i_{\text{meas}}$. □

Parts (2) and (3) show that a measure of finite penalty automatically has finite fidelity. Since conversely $J_{\psi_k}(\mu) < \infty$ contains $\alpha\Phi_{\psi_k}(\mu) < \infty$, the two conditions agree:

$$J_{\psi_k}(\mu) < \infty \quad \Longleftrightarrow \quad \Phi_{\psi_k}(\mu) < \infty \qquad \big(\mu \in \mathcal{M}(\mathbb{S}^d)\big),$$

and either may be used as the standing finiteness hypothesis.

Proposition 4.10 is used only as a bridge between the finite-network problem and (P_{ψ_k}). It equates the two infima; attainment is established in Theorem 4.13 from the finite-dimensional argument and is not claimed here.

Theorem 4.11 (Atomwise optimality and minimum atom weight)**.** *Let $k \geq 2$, $q = 2/(k+1) \in (0,1)$, $\alpha > 0$, $V \in H^1(D)$, and let $K : \mathbb{S}^d \to H^1(D)$ be continuous. Let $\bar{\mu} \in \mathcal{M}(\mathbb{S}^d)$ satisfy*

$J_{\psi_k}(\bar\mu) < \infty$ and be optimal against perturbations of the weights of its own atoms: there is $\varepsilon > 0$ such that

$$J_{\psi_k}(\bar\mu) \le J_{\psi_k}(\bar\mu + t\delta_\omega) \qquad \text{for every } \omega \in \operatorname{atom}\bar\mu \text{ and every } |t| < \varepsilon. \tag{45}$$

Set

$$r_{\bar\mu} := \mathcal{N}\bar\mu - V, \qquad \bar P(\omega) := \langle r_{\bar\mu}, K(\omega)\rangle_{H^1(D)}, \qquad B(\bar\mu) := \|\bar P\|_{C(\mathbb{S}^d)}.$$

Then $\bar\mu$ is purely atomic, $B(\bar\mu) < \infty$, and every atom $\omega_n \in \operatorname{atom}\bar\mu$ with weight $c_n := \bar\mu(\{\omega_n\})$ satisfies

$$\bar P(\omega_n) + \alpha q|c_n|^{q-1}\operatorname{sign}(c_n) = 0. \tag{46}$$

If $B(\bar\mu) = 0$, then $\bar\mu = 0$. If $B(\bar\mu) > 0$, then

$$|c_n| \ge c_{\min}(\bar\mu) := \left(\frac{\alpha q}{B(\bar\mu)}\right)^{1/(1-q)} > 0 \qquad \text{for every } \omega_n \in \operatorname{atom}\bar\mu. \tag{47}$$

Hypothesis (45) holds in particular if $\bar\mu$ is a local minimizer of (P_{ψ_k}) in total variation in the sense of Definition 4.8, or if $\bar\mu$ is a minimizer of the finite-network objective at a fixed width whose locations are distinct.

Proof. We derive (46) from a two-sided perturbation of a single atom weight, and then convert it into the lower bound (47).

Since $J_{\psi_k}(\bar\mu) < \infty$ we have $\Phi_{\psi_k}(\bar\mu) < \infty$, so $\bar\mu_{\text{cont}} = 0$ and $\bar\mu$ is purely atomic. By Proposition 4.10(2), applied to $\bar\mu$, $\mathcal{N}\bar\mu \in H^1(D)$, so $r_{\bar\mu} \in H^1(D)$ is well defined. The map $\bar P$ is continuous on $\mathbb{S}^d$, being the composition of the continuous map K with the continuous linear functional $\langle r_{\bar\mu}, \cdot\rangle_{H^1(D)}$, and $\mathbb{S}^d$ is compact, so $B(\bar\mu) < \infty$.

Fix $\omega_n \in \operatorname{atom}\bar\mu$, write $c_n = \bar\mu(\{\omega_n\}) \neq 0$, let $s \in \{-1, 1\}$, and consider $\mu_t := \bar\mu + ts\delta_{\omega_n}$ for $0 < t < \min\{\varepsilon, |c_n|\}$. The measure μ_t differs from $\bar\mu$ only in the weight of the atom ω_n, which becomes $c_n + ts \neq 0$, so $\operatorname{atom}\mu_t = \operatorname{atom}\bar\mu$ and

$$\Phi_{\psi_k}(\mu_t) - \Phi_{\psi_k}(\bar\mu) = \psi_k(|c_n + ts|) - \psi_k(|c_n|),$$

all other summands being unchanged. For the fidelity, $\mathcal{N}\mu_t = \mathcal{N}\bar\mu + tsK(\omega_n)$, so the residual is $r_{\bar\mu} + tsK(\omega_n)$ and expanding the square gives the exact identity

$$\begin{aligned} L(\mu_t) - L(\bar\mu) &= ts\langle r_{\bar\mu}, K(\omega_n)\rangle_{H^1(D)} + \frac{t^2}{2}\|K(\omega_n)\|^2_{H^1(D)} \\ &= ts\bar P(\omega_n) + \frac{t^2}{2}\|K(\omega_n)\|^2_{H^1(D)}. \end{aligned}$$

Because $c_n \neq 0$, the map $z \mapsto |z|^q$ is differentiable at c_n with derivative $q|c_n|^{q-1}\operatorname{sign}(c_n)$. Hypothesis (45) gives $J_{\psi_k}(\mu_t) - J_{\psi_k}(\bar\mu) \ge 0$; dividing by $t > 0$ and letting $t \downarrow 0$ yields

$$0 \le s\Big[\bar P(\omega_n) + \alpha q|c_n|^{q-1}\operatorname{sign}(c_n)\Big].$$

Choosing $s = 1$ and $s = -1$ gives (46).

Suppose $B(\bar\mu) = 0$, that is, $\bar P \equiv 0$ on $\mathbb{S}^d$. If $\bar\mu$ had an atom ω_n, then (46) would read $\alpha q|c_n|^{q-1}\operatorname{sign}(c_n) = 0$, which is impossible because $\alpha > 0$, $q > 0$, and $|c_n|^{q-1} \in (0, \infty)$ for $c_n \neq 0$. Hence $\operatorname{atom}\bar\mu = \emptyset$, and since $\bar\mu$ is purely atomic, $\bar\mu = 0$.

Suppose $B(\bar\mu) > 0$. Taking absolute values in (46) gives

$$\alpha q|c_n|^{q-1} = |\bar P(\omega_n)| \le B(\bar\mu).$$

Since $q - 1 = -(1-q) < 0$, this reads $|c_n|^{-(1-q)} \le B(\bar\mu)/(\alpha q)$, hence $|c_n|^{1-q} \ge \alpha q/B(\bar\mu)$; raising both sides to the positive power $1/(1-q)$ preserves the inequality and gives (47).

Finally we check the two sufficient conditions for (45). If $\bar\mu$ is a local minimizer in total variation with radius ε, then $\|(\bar\mu + t\delta_\omega) - \bar\mu\|_{\mathcal{M}(\mathbb{S}^d)} = |t| < \varepsilon$ for $|t| < \varepsilon$, so the competitor is admissible. If $\bar\mu = \mu_{\boldsymbol{\omega},c}$ minimizes $L(\mu_{\boldsymbol{\omega},c}) + \alpha\sum_n \psi_k(|c_n|)$ over $(\mathbb{S}^d)^N \times \mathbb{R}^N$ and its locations are distinct, then $\bar{\mu} + t\delta_{\omega_n} = \mu_{\boldsymbol{\omega},c+te_n}$, where e_n is the nth coordinate vector in $\mathbb{R}^N$, is a competitor of the same width for every $t \in \mathbb{R}$, and by Lemma 4.7(4) the value of J_{ψ_k} at each of the two measures equals the corresponding coefficient objective, because both have distinct locations; the hypothesis therefore holds with every $\varepsilon > 0$. $\square$

Remark 4.12 (Consequences of the singular slope). The identity $\psi_k'(0^+) = +\infty$ has three consequences.

(1) Inserting a new atom at $\omega \notin \operatorname{atom}\bar{\mu}$ with weight ts, $s \in \{-1, 1\}$, changes the penalty by αt^q. Since $t^q/t = t^{q-1} \to \infty$ as $t \downarrow 0$, division by t produces no finite limit, and hence no pointwise condition on $\bar{P}(\omega)$ away from $\operatorname{atom}\bar{\mu}$. Theorem 4.11 is restricted to existing atoms for this reason.

(2) At a nonzero atom weight, the same singularity produces the positive lower bound (47). Combined with an upper bound on the objective it yields the width estimates below.

(3) The zero measure is a local minimizer of (P_{ψ_k}) in total variation. Indeed, $J_{\psi_k}(\mu) = \infty$ whenever $\Phi_{\psi_k}(\mu) = \infty$, while for μ with $\Phi_{\psi_k}(\mu) < \infty$ and $\|\mu\|_{\mathcal{M}(\mathbb{S}^d)} = \varepsilon$,
$$\begin{aligned} J_{\psi_k}(\mu) - J_{\psi_k}(0) &= \langle -V, \mathcal{N}\mu\rangle_{H^1(D)} + \tfrac{1}{2}\|\mathcal{N}\mu\|^2_{H^1(D)} + \alpha\Phi_{\psi_k}(\mu) \\ &\geq -\|V\|_{H^1(D)} C_S\, \varepsilon + \alpha\varepsilon^q, \end{aligned}$$
by Lemma 4.7(3), the bound $\|\mathcal{N}\mu\|_{H^1(D)} \leq C_S\varepsilon$ of Proposition 4.10(2), and the Cauchy–Schwarz inequality. Since $q < 1$, the right-hand side is positive for every sufficiently small $\varepsilon > 0$. An insertion criterion based on an infinitesimal weight perturbation is therefore unavailable for this penalty, and Algorithm 2 uses a finite-step decrease test instead.

4.4. **Global existence and a uniform width bound.**

Theorem 4.13 (Existence and width bound)**.** *Let $k \geq 2$, $q = 2/(k+1) \in (0, 1)$, $\alpha > 0$, let $V \in H^1(D)$ with $V \neq 0$, and let $K : \mathbb{S}^d \to H^1(D)$ be continuous with $C_S = \sup_{\omega\in\mathbb{S}^d} \|K(\omega)\|_{H^1(D)} > 0$; the latter holds for* ReLU^k *by Lemma 4.5(4). Then* (P_{ψ_k}) *admits a global minimizer, and its minimum value equals the finite-network infimum on the right-hand side of* (43). *Every global minimizer $\bar{\mu}$ is a finite atomic measure whose reduced width satisfies*
$$N \leq N_{\max} := \frac{\|V\|^2_{H^1(D)}}{2\alpha\, c^q_{\min,\text{glob}}}, \qquad c_{\min,\text{glob}} := \left(\frac{\alpha q}{\|V\|_{H^1(D)}\, C_S}\right)^{1/(1-q)}.$$

If $V = 0$ or $C_S = 0$ the problem degenerates: the fidelity is then either minimized at $\mathcal{N}\mu = 0$ or, since $K \equiv 0$ forces $\mathcal{N}\mu = 0$, constant, so by Lemma 4.7(2) the measure $\bar{\mu} = 0$ is the unique global minimizer, of reduced width 0. Under the hypotheses of the theorem $c_{\min,\text{glob}} \in (0, \infty)$ and $N_{\max} < \infty$ are well defined.

Proof. We first obtain minimizers at each fixed width by compactness, then derive one width estimate for both the existence argument and arbitrary global minimizers.

For $N \in \mathbb{N}_0$, set
$$F_N(\boldsymbol{\omega}, c) := L(\mu_{\boldsymbol{\omega},c}) + \alpha\sum_{n=1}^{N} |c_n|^q, \qquad m_N := \inf F_N,$$
with the empty-family convention $F_0 = m_0 = \frac{1}{2}\|V\|^2_{H^1(D)}$. For $N \geq 1$, continuity of K, scalar multiplication, and finite addition show that $(\boldsymbol{\omega}, c) \mapsto \sum_{n=1}^{N} c_n K(\omega_n)$ is continuous into $H^1(D)$; hence F_N is continuous. Any minimizing sequence may be chosen in the sublevel $F_N \leq m_0 + 1$. Since the fidelity is nonnegative, this sublevel satisfies
$$|c_n| \leq R := \left(\frac{m_0 + 1}{\alpha}\right)^{1/q} \qquad (n = 1, \ldots, N).$$

It therefore lies in the compact set $(\mathbb{S}^d)^N \times [-R, R]^N$. Compactness gives a convergent subsequence, and continuity shows that its limit attains m_N. The case $N = 0$ is immediate, and zero-padding gives $m_{N+1} \le m_N$.

We use the following estimate twice. Suppose that a measure μ satisfies $J_{\psi_k}(\mu) \le m_0$ and the atomwise optimality hypothesis (45). Theorem 4.11 shows that μ is purely atomic. The nonnegativity of the penalty gives $\|r_\mu\|_{H^1(D)} \le \|V\|_{H^1(D)}$, and hence

$$B(\mu) \le \|V\|_{H^1(D)} C_S.$$

If $\mu \ne 0$, Theorem 4.11 gives $B(\mu) > 0$ and

$$|\mu(\{\omega\})| \ge \left(\frac{\alpha q}{B(\mu)}\right)^{1/(1-q)} \ge c_{\min,\text{glob}} \qquad (\omega \in \operatorname{atom}\mu).$$

Consequently, for every finite $A \subseteq \operatorname{atom}\mu$,

$$\#A\, \alpha c_{\min,\text{glob}}^q \le \alpha \sum_{\omega\in A} |\mu(\{\omega\})|^q \le \alpha \Phi_{\psi_k}(\mu) \le J_{\psi_k}(\mu) \le \frac{1}{2}\|V\|^2_{H^1(D)}.$$

An infinite atom set would contain finite subsets of arbitrarily large cardinality. Thus $\operatorname{atom}\mu$ is finite and its cardinality is at most $N_{\max}$; the same conclusion is immediate when $\mu = 0$.

Now fix N and reduce a minimizer of F_N by merging coincident atoms and discarding zero coefficients. If $(\widehat{\boldsymbol{\omega}}, \widehat{c})$ is the resulting family of width $N' \le N$, then Lemma 4.7(4) and zero-padding give

$$m_{N'} \le F_{N'}(\widehat{\boldsymbol{\omega}}, \widehat{c}) \le m_N \le m_{N'}.$$

Hence this family is a reduced minimizer at width N' with value m_N. Its measure $\widehat{\mu}$ satisfies $J_{\psi_k}(\widehat{\mu}) = m_N \le m_0$ and, by the last assertion of Theorem 4.11, the atomwise optimality hypothesis. The estimate above therefore yields $N' \le N_{\max}$.

Set $N_{\text{cut}} := \lfloor N_{\max} \rfloor$. Every m_N thus equals some $m_{N'}$ with $N' \le N_{\text{cut}}$. Since (m_N) is nonincreasing,

$$\inf_{N\in\mathbb{N}_0} m_N = \min_{0\le N\le N_{\text{cut}}} m_N = m_{N_{\text{cut}}}.$$

Reducing a minimizer of $F_{N_{\text{cut}}}$ gives a reduced family of width $N_* \le N_{\text{cut}}$ whose objective value is $m_{N_{\text{cut}}}$. Denote its measure by $\bar{\mu}^* = \mu_{\boldsymbol{\omega}^*, c^*}$. By Lemma 4.7(4), $J_{\psi_k}(\bar{\mu}^*) = m_{N_{\text{cut}}}$; Proposition 4.10 then shows that $\bar{\mu}^*$ is a global minimizer of (P_{ψ_k}) and that its value equals the finite-network infimum.

Finally, any global minimizer $\bar{\mu}$ satisfies $J_{\psi_k}(\bar{\mu}) \le J_{\psi_k}(0) = m_0$ and is a local minimizer in total variation. It therefore satisfies the hypotheses of the common estimate, which proves that $\bar{\mu}$ is finite atomic and has reduced width at most $N_{\max}$. $\square$

Corollary 4.14 (Finite support of total-variation local minimizers)**.** *Let the hypotheses of Theorem 4.11 on k, q, α, V, K hold and let $\bar{\mu} \ne 0$ be a local minimizer of (P_{ψ_k}) in total variation. Then $B(\bar{\mu}) > 0$, the measure $\bar{\mu}$ is finite atomic, and its reduced width satisfies*

$$N \le \frac{J_{\psi_k}(\bar{\mu})}{\alpha\, c_{\min}(\bar{\mu})^q}.$$

Proof. By Theorem 4.11, $B(\bar{\mu}) = 0$ would force $\bar{\mu} = 0$; hence $B(\bar{\mu}) > 0$ and every atom weight satisfies $|c_n| \ge c_{\min}(\bar{\mu}) > 0$. Since ψ_k is nondecreasing and the fidelity is nonnegative,

$$N\alpha\, c_{\min}(\bar{\mu})^q \le \alpha \Phi_{\psi_k}(\bar{\mu}) \le J_{\psi_k}(\bar{\mu}) < \infty$$

with $N = \#\operatorname{atom}\bar{\mu} \in \mathbb{N}_0 \cup \{\infty\}$, which forces $N < \infty$ and gives the stated bound. $\square$

The estimate of Corollary 4.14 depends on the minimizer through both $J_{\psi_k}(\bar{\mu})$ and $B(\bar{\mu})$ and is therefore an a posteriori bound. It is not computable in advance, unlike the a priori bound $N_{\max}$ of Theorem 4.13, which is uniform over all global minimizers and computable from $\|V\|_{H^1(D)}$,

C_S, α and q alone. On a global minimizer the two are comparable and the a posteriori bound is the sharper one, since $J_{\psi_k}(\bar{\mu}) \le \frac{1}{2}\|V\|^2_{H^1(D)}$ and $c_{\min}(\bar{\mu}) \ge c_{\min,\text{glob}}$ there.

Remark 4.15 (The empirical problem). Let ReLU^k with $k \ge 2$ be the activation and let the sampled data $V(x^m)$ and $\nabla V(x^m)$ be given. Then Proposition 4.10, Theorem 4.11, Theorem 4.13 and Corollary 4.14 hold for the empirical objective (42) as well. Their proofs use only two properties: the fidelity is one half of the squared norm of $\mathcal{N}\mu - V$ in a Hilbert space, and the dictionary is continuous and bounded on the compact sphere. The empirical fidelity (4) has the first property with $\mathbb{R}^{M(d+1)}$ carrying its Euclidean inner product, the target being the scaled family $M^{-1/2}\big(V(x^m), \nabla V(x^m)\big)_{m=1}^{M}$ of sampled values and gradients of V, and the dictionary being the value-gradient feature map K^M of Section 5 defined in (61). This map is continuous on $\mathbb{S}^d$ by Lemma 4.5(2)–(3) and hence bounded there, which is the second property. Substituting these data for $H^1(D)$, V and K throughout replaces C_S and $\|V\|_{H^1(D)}$ by the corresponding empirical quantities and leaves every argument unchanged; the population proofs are not repeated. The nondegeneracy hypotheses of Theorem 4.13 become the requirements that the empirical dictionary bound be positive, which holds because $\widehat{\omega} = (0,1) \in \mathbb{S}^d$ has $[K(\widehat{\omega})](x) = 1$ and $\nabla K(\widehat{\omega})(x) = 0$ on D, and that the sampled data not vanish identically. Lemma 4.3 and Lemma 4.5 concern the radial calculus and the dictionary alone and are independent of which fidelity is used.

Because $q < 1$, an infinitesimal insertion at a new location yields no first-order condition (Remark 4.12). The greedy scheme of Section 5 therefore replaces the infinitesimal test used for the finite-slope penalty by a finite-step decrease criterion, evaluated with a conservative surrogate coefficient; this is the content of Algorithm 2.

5. Algorithms

This section describes the computational pipeline: the training data are generated via the open-loop problem, candidate neurons are inserted greedily, and insertion alternates with finite-dimensional optimization of the current network.

Theorem 5.1. *Let $\mathcal{H}$ be a real Hilbert space, let $V \in \mathcal{H}$, and let $K_p \in C_0(\Omega; \mathcal{H})$. For $\mu \in \mathcal{M}_p(\Omega)$, set*

$$J_{\mathcal{H}}(\mu) := \frac{1}{2}\left\|\int_\Omega K_p(\omega)\, d\mu_p(\omega) - V\right\|^2_{\mathcal{H}} + \alpha\Phi_\phi(\mu_p) \tag{48}$$

and define

$$\begin{aligned}
r_\mu &:= \int_\Omega K_p(\omega)\, d\mu_p(\omega) - V,\\
P_{p,\mu}(\omega) &:= \langle r_\mu, K_p(\omega)\rangle_{\mathcal{H}},\\
B_p &:= \sup_{\omega\in\Omega} \|K_p(\omega)\|_{\mathcal{H}},\\
\Delta(\mu,\omega) &:= \max\big\{|P_{p,\mu}(\omega)| - \alpha L_\phi, 0\big\},\\
\Delta(\mu) &:= \sup_{\omega\in\Omega} \Delta(\mu,\omega) = \max\big\{\|P_{p,\mu}\|_\infty - \alpha L_\phi, 0\big\}.
\end{aligned}$$

Suppose that $\alpha > 0$ and that ϕ satisfies Assumptions 3.2 and 3.3. Let μ be finite atomic and $J_{\mathcal{H}}(\mu) < \infty$, and assume $\Delta(\mu) > 0$. Then $B_p > 0$, and for every $\omega \in \Omega$ the insertion

$$\kappa(\omega) := -\frac{\Delta(\mu,\omega)}{B_p^2}\,\mathrm{sign}\big(P_{p,\mu}(\omega)\big), \qquad c(\omega) := \frac{\kappa(\omega)}{w_p(\omega)}, \tag{49}$$

satisfies

$$J_{\mathcal{H}}\big(\mu + c(\omega)\delta_\omega\big) \le J_{\mathcal{H}}(\mu) - \frac{\Delta(\mu,\omega)^2}{2B_p^2}. \tag{50}$$

The condition $\Delta(\mu,\omega) > 0$ is sufficient for a strict decrease but not necessary: at $|P_{p,\mu}(\omega)| = \alpha L_\phi$ the first-order terms cancel and the second-order terms decide. The supremum defining $\Delta(\mu)$ is attained, at any $\omega^* \in \Omega$ with

$$|P_{p,\mu}(\omega^*)| = \|P_{p,\mu}\|_\infty, \qquad \text{so that } \Delta(\mu,\omega^*) = \Delta(\mu). \tag{51}$$

Let $(\mu_k)_{k\geq 0}$ be finite atomic iterates with $J_{\mathcal{H}}(\mu_0) < \infty$, and put $\Delta_k := \Delta(\mu_k)$. Whenever $\Delta_k > 0$, choose an exact global maximizer as in (51), make the insertion (49) at ω^*, and then apply a finite-dimensional correction whose objective is no larger than the post-insertion objective. When $\Delta_k = 0$, require only that the correction not increase the objective. Then

$$\sum_{k=0}^{\infty} \Delta_k^2 \leq 2B_p^2 J_{\mathcal{H}}(\mu_0), \tag{52}$$

$$\Delta_k \longrightarrow 0, \tag{53}$$

$$\min_{0\leq k<n} \Delta_k \leq B_p\sqrt{\frac{2J_{\mathcal{H}}(\mu_0)}{n}} \qquad (n \geq 1). \tag{54}$$

Beyond (52)– (54), these conclusions do not assert convergence of (μ_k) to a global minimizer or an objective-gap rate.

There is also an explicit bounded search region. Suppose, in addition, that there are $C > 0$ and $0 \leq s < p$ such that

$$w_p(\omega)\|K_p(\omega)\|_{\mathcal{H}} \leq C(1+|\omega|)^s \qquad (\omega \in \Omega). \tag{55}$$

Define

$$R(\mu) := \max\left\{1, \left(\frac{2^{s+1}C\|r_\mu\|_{\mathcal{H}}}{\alpha L_\phi}\right)^{1/(p-s)}\right\}. \tag{56}$$

Then

$$|P_{p,\mu}(\omega)| < \frac{\alpha L_\phi}{2} \qquad \text{whenever } |\omega| > R(\mu). \tag{57}$$

Consequently, every point satisfying $|P_{p,\mu}(\omega)| > \alpha L_\phi$, and every maximizer in (51) when $\Delta(\mu) > 0$, belongs to $\overline{B}_{R(\mu)}$.

Proof. We first establish attainment of the certificate maximum. $K_p \in C_0(\Omega;\mathcal{H})$ gives $B_p < \infty$ and $P_{p,\mu} \in C_0(\Omega)$. If $\Delta(\mu) > 0$, then $\|P_{p,\mu}\|_\infty > \alpha L_\phi > 0$. Choose a closed ball outside which $|P_{p,\mu}| < \|P_{p,\mu}\|_\infty/2$; this is possible because $P_{p,\mu}$ vanishes at infinity. The continuous function $|P_{p,\mu}|$ attains its maximum on that compact ball, proving (51). Moreover, $B_p = 0$ would imply $P_{p,\mu} = 0$, so $B_p > 0$.

We now bound the change caused by an arbitrary normalized insertion $\kappa\delta_\omega$ with $\kappa \in \mathbb{R}$. The explicit atomic–continuous representation of Φ_ϕ, the monotonicity and subadditivity of ϕ, and Lemma 3.6 give

$$\begin{aligned}\Phi_\phi(\mu_p + \kappa\delta_\omega) - \Phi_\phi(\mu_p) &\leq \phi(|\kappa|) && \text{by subadditivity at the point } \omega\\ &\leq L_\phi|\kappa| && \text{by the tangent bound at zero.}\end{aligned} \tag{58}$$

Since $(\mu + \kappa w_p(\omega)^{-1}\delta_\omega)_p = \mu_p + \kappa\delta_\omega$, the quadratic fidelity has the exact expansion

$$\begin{aligned}&J_{\mathcal{H}}\left(\mu + \frac{\kappa}{w_p(\omega)}\delta_\omega\right) - J_{\mathcal{H}}(\mu)\\ &\quad\leq \kappa P_{p,\mu}(\omega) + \frac{\kappa^2}{2}\|K_p(\omega)\|_{\mathcal{H}}^2 + \alpha L_\phi|\kappa| && \text{by (58)}\\ &\quad\leq \kappa P_{p,\mu}(\omega) + \frac{B_p^2\kappa^2}{2} + \alpha L_\phi|\kappa| && \text{by the definition of } B_p.\end{aligned} \tag{59}$$

Fix $\omega \in \Omega$. If $\Delta(\mu,\omega) = 0$, then $\kappa(\omega) = 0$ and (50) is trivial. Otherwise $|P_{p,\mu}(\omega)| = \alpha L_\phi + \Delta(\mu,\omega)$, and substitution of $\kappa = \kappa(\omega)$ from (49) into (59) yields

$$\begin{aligned}
&J_{\mathcal{H}}\big(\mu + c(\omega)\delta_\omega\big) - J_{\mathcal{H}}(\mu) \\
&\quad\le -|\kappa(\omega)|\big(\alpha L_\phi + \Delta(\mu,\omega)\big) + \frac{B_p^2|\kappa(\omega)|^2}{2} + \alpha L_\phi|\kappa(\omega)| && \text{since } \operatorname{sign}\kappa(\omega) = -\operatorname{sign} P_{p,\mu}(\omega) \\
&\quad= -|\kappa(\omega)|\Delta(\mu,\omega) + \frac{B_p^2|\kappa(\omega)|^2}{2} && \text{the threshold terms cancel} \\
&\quad= -\frac{\Delta(\mu,\omega)^2}{2B_p^2} && \text{by the choice of } \kappa(\omega).
\end{aligned}$$

This proves (50); at $\omega = \omega^*$ it gives the decrease $\Delta(\mu)^2/(2B_p^2)$. Sufficiency is the strict negativity of that right-hand side. Necessity fails because at a new location ω with $|P_{p,\mu}(\omega)| = \alpha L_\phi$ and $\operatorname{sign}\kappa = -\operatorname{sign} P_{p,\mu}(\omega)$ the increment is exactly

$$\frac{\|K_p(\omega)\|_{\mathcal{H}}^2}{2}\kappa^2 - \alpha\big(L_\phi|\kappa| - \phi(|\kappa|)\big),$$

which is negative whenever the concavity gain $L_\phi z - \phi(z)$ exceeds $\|K_p(\omega)\|_{\mathcal{H}}^2 z^2/(2\alpha)$ at $z = |\kappa|$.

We next telescope the one-step estimate. The correction hypothesis and (50) give, for every $n \ge 1$,

$$\begin{aligned}
\sum_{k=0}^{n-1}\frac{\Delta_k^2}{2B_p^2} &\le J_{\mathcal{H}}(\mu_0) - J_{\mathcal{H}}(\mu_n) && \text{by telescoping the decreases} \\
&\le J_{\mathcal{H}}(\mu_0) && \text{since } J_{\mathcal{H}} \text{ is nonnegative.}
\end{aligned} \tag{60}$$

Letting $n \to \infty$ proves (52), and a convergent series of nonnegative terms gives (53). The smallest of the first n squared violations is at most their average, so

$$\begin{aligned}
\min_{0\le k<n}\Delta_k^2 &\le \frac{1}{n}\sum_{k=0}^{n-1}\Delta_k^2 && \text{by the minimum–average inequality} \\
&\le \frac{2B_p^2 J_{\mathcal{H}}(\mu_0)}{n} && \text{by (60).}
\end{aligned}$$

Taking square roots proves (54). If $B_p = 0$, then $P_{p,\mu_k} = 0$ and $\Delta_k = 0$ for every k, so all three conclusions hold without an insertion.

Finally, we derive the radius directly from the feature growth. If $|\omega| \ge 1$, then (55) and Cauchy–Schwarz give

$$\begin{aligned}
|P_{p,\mu}(\omega)| &\le \|r_\mu\|_{\mathcal{H}}\|K_p(\omega)\|_{\mathcal{H}} && \text{by Cauchy–Schwarz} \\
&\le C\|r_\mu\|_{\mathcal{H}}\frac{(1+|\omega|)^s}{1+|\omega|^p} && \text{by (55)} \\
&\le 2^s C\|r_\mu\|_{\mathcal{H}}|\omega|^{-(p-s)} && \text{since } |\omega| \ge 1.
\end{aligned}$$

If $|\omega| > R(\mu)$, the last expression is strictly smaller than $\alpha L_\phi/2$ by (56). This proves (57) and the claimed localization of every violation and every maximizing insertion. □

Remark 5.2 (Hypotheses used). Of Assumptions 3.2 and 3.3 the proof uses only that ϕ is concave and nondecreasing with $\phi(0) = 0$ and $L_\phi < \infty$; coercivity and $\phi \in C^1((0,\infty))$ are not needed. Finite atomicity and the finiteness of $J_{\mathcal{H}}$ are assumed because the algorithms below produce finite atomic iterates, not because the estimate requires them: for a finite atomic $\mu \in \mathcal{M}_p(\Omega)$ the normalized measure μ_p is again finite atomic, so $\Phi_\phi(\mu_p)$ is a finite sum of finite values and $\|\int_\Omega K_p\,d\mu_p\|_{\mathcal{H}} \le B_p\|\mu_p\|_{\mathcal{M}(\Omega)} < \infty$, whence $J_{\mathcal{H}}(\mu) < \infty$ is automatic.

Remark 5.3 (Population specialization). For the population problem, take $\mathcal{H} = H^1(D)$ and the population K_p. Then $J_{\mathcal{H}} = J$, and at the candidate $\bar{\mu}$ the abstract function $P_{p,\bar{\mu}}$ is $\bar{P}_p$. Lemma 3.17 gives (55) with $C = C_K$ and $s = s_1$ whenever $p > s_1$.

Remark 5.4 (Empirical specialization). For the empirical problem, take $\mathcal{H} = \mathbb{R}^{M(d+1)}$ with its Euclidean inner product, assume that the following feature is continuous, and define

$$K^M(\omega) := \frac{1}{\sqrt{M}}\big(K(\omega)(x^m), \nabla K(\omega)(x^m)\big)_{m=1}^M, \qquad K_p^M(\omega) := \frac{K^M(\omega)}{w_p(\omega)}. \tag{61}$$

If $K_p^M \in C_0(\Omega; \mathcal{H})$, use $K_p = K_p^M$ in (48) and take the abstract target V there to be

$$\frac{1}{\sqrt{M}}\big(V(x^m), \nabla V(x^m)\big)_{m=1}^M.$$

Then the fidelity in (48) is l^M. Denote the corresponding function in this specialization by

$$P_{p,\mu}^M := P_{p,\mu}.$$

For a finite measure $\mu_{\boldsymbol{\omega},c} = \sum_{n=1}^N c_n \delta_{\omega_n}$ with distinct nodes, Theorem 3.13 gives the exact identity

$$J_{\mathcal{H}}(\mu_{\boldsymbol{\omega},c}) = l^M(\mu_{\boldsymbol{\omega},c}) + \alpha \sum_{n=1}^N \phi\big(w_p(\omega_n)|c_n|\big).$$

Repeated nodes are first merged; the inequality in the proof of Corollary 3.22 shows that merging cannot increase the penalty. Any bound

$$\|K^M(\omega)\|_{\mathcal{H}} \le C(1 + |\omega|)^s, \qquad s < p,$$

implies $K_p^M \in C_0(\Omega; \mathcal{H})$ and gives the empirical search radius (56).

For value-only fitting the same specialization applies with $\mathcal{H} = \mathbb{R}^M$, dropping the gradient entries from K^M and from the target.

5.1. **Data generation via the open-loop problem.** Set $U := L^2(t_0, T; \mathbb{R}^{m_u})$ and assume that for every $u \in U$ the state equation in (P) has a unique solution $y(u)$ on $[t_0, T]$ and that the control-to-state map $u \mapsto y(u)$ is continuously Fréchet differentiable from U into $C([t_0, T]; \mathbb{R}^d)$. The objective of ($P$) is then the reduced functional $u \mapsto \mathcal{C}(u; t_0, x)$ on U, and the standard adjoint (Lagrangian) calculus — see, e.g., [13] — gives its gradient

$$\nabla_u \mathcal{C}(u; t_0, x) = 2\eta u + g(y(u))^\top p \quad \text{in } U,$$

where p solves the costate equation of (TPBVP) backward in time along $(y(u), u)$ in place of (y^*, u^*). Each gradient evaluation therefore costs one forward solve of the state equation and one backward solve of the costate equation, and the implicit derivative of $u \mapsto y(u)$ is never formed. For the data-driven approach it is essential that the initial dataset $\{x^j, V(0, x^j), \nabla V(0, x^j)\}_{j=1}^M$ is *causality-free*: the samples are generated without approximate values at neighboring points. [14] provides a comprehensive survey of generation methods. As in [1], we minimize $\mathcal{C}(\cdot; t_0, x)$ with Barzilai–Borwein step sizes. Since (P) is not convex, the iteration returns a stationary control rather than a certified global minimizer; the converged costate supplies the gradient sample ∇V subject to that reservation and to the differentiability qualification of Section 1.

5.2. **Greedy insertion.** The Gâteaux derivative of (4) in an unnormalized Dirac direction is

$$P_{\mu_t}^M(\omega) := Dl^M(\mu_t)[\delta_\omega] = \frac{1}{M}\sum_{m=1}^M \left([K(\omega)](x^m) r_{\mu_t}(x^m) + \nabla K(\omega)(x^m) \cdot \nabla r_{\mu_t}(x^m)\right).$$

Again, the second summand is omitted for value-only fitting, in which case the value-only substitution of Remark 5.4 is in force. For $\boldsymbol{\omega} = (\omega_n)_{n=1}^N$, $c = (c_n)_{n=1}^N$, and $\mu_{\boldsymbol{\omega},c} = \sum_{n=1}^N c_n \delta_{\omega_n}$, define

$$J^M(\boldsymbol{\omega}, c) := l^M(\mu_{\boldsymbol{\omega},c}) + \alpha \sum_{n=1}^N \phi_\gamma\big(w_p(\omega_n)|c_n|\big). \tag{62}$$

Throughout this subsection we impose the growth bound of Remark 5.4 and apply Theorem 5.1 with $\mathcal{H} = \mathbb{R}^{M(d+1)}$, the empirical target, $K_p = K_p^M$ from (61), and $\phi = \phi_\gamma$, for which $L_\phi = 1$ by (31). For distinct nodes J^M is then the objective (48) of that theorem, with repeated nodes merged first; its derivative function $P^M_{p,\mu}$ of Remark 5.4 is P^M_μ / w_p, and its uniform bound $B_p^M := \sup_\omega \|K_p^M(\omega)\|$ is finite. Every estimate below remains valid when B_p^M is replaced by a computable upper bound $\widehat{B} \geq B_p^M$, since the increment bound (59) only becomes weaker.

At the current iterate $\mu_t = \mu_{\boldsymbol{\omega}^{(t)},c^{(t)}}$ a candidate ω is therefore accepted when

$$\Delta(\mu_t, \omega) > 0, \qquad \text{that is,} \qquad |P^M_{p,\mu_t}(\omega)| > \alpha L_\phi, \tag{63}$$

and inserted with the coefficient $c(\omega)$ of (49), which by (50) decreases J^M by at least $\frac{\Delta(\mu_t,\omega)^2}{(2(B_p^M)^2)}$. This holds at every accepted candidate, not only at a global maximizer of $|P^M_{p,\mu_t}|$, so a local search suffices for the one-step decrease. The increment bound (59) also holds at an existing atom by subadditivity, so an exact repetition may be merged with the existing coefficient instead of being added as a second atom.

For a nonhomogeneous activation, evaluate $\omega = 0$ explicitly. Every nonzero parameter can be written as $\omega = \lambda\widehat{\omega}$ with $\lambda > 0$ and $|\widehat{\omega}| = 1$. The search first samples and refines directions on the unit sphere and then maximizes

$$\lambda \longmapsto |P^M_{p,\mu_t}(\lambda\widehat{\omega})|$$

over $0 \leq \lambda \leq R(\mu_t)$ along each retained direction. Under the standing growth bound, Theorem 5.1 gives the radius $R(\mu_t)$ and attainment of the global maximum of $|P^M_{p,\mu_t}|$.

Finding a global maximizer is generally intractable; for ReLU dictionaries it is NP-hard [2, Section 3.3]. The practical search therefore uses multiple initial directions and local L-BFGS refinement, followed by the removal of near-duplicates: writing $\widehat{\omega}$ for the unit representative of a refined candidate, two candidates are *near-duplicates* when $\widehat{\omega}_i \cdot \widehat{\omega}_j > 1 - \theta_{\text{merge}}$ for a fixed tolerance $\theta_{\text{merge}} > 0$, and of each such group only the first in the enumeration is retained. For these candidates, the preceding local decrease estimate applies. The rate in Theorem 5.1 additionally requires that the selected candidate attain the global maximum and that the subsequent correction not increase the post-insertion objective. It does not cover the multistart heuristic without a quantitative search-accuracy condition.

With M value samples and Md gradient components, the convex total-variation problem underlying a conditional-gradient step has $M(d+1)$ scalar measurements and admits a minimizer with at most $M(d+1)$ atoms [2, Section 2.2]. This observation does not assert convexity of the nonconvex coefficient problem used here.

5.3. **The nonconvex penalized algorithms.** We use two greedy insertion algorithms of primal–dual active point (PDAP) type, in the sense of the accelerated conditional-gradient method for measure spaces of [25]. Algorithm 1 follows the nonconvex variant of [24], whose scalar penalty satisfies $\phi'(0) < \infty$ as in Assumption 3.2; Algorithm 2 is an adaptation of it to the penalty ψ_k, whose infinite slope at the origin is explicitly excluded by that reference. Algorithm 1 uses the scalar penalty ϕ_γ from (31) and the objective (62), with the insertion condition (63). At a nonzero coefficient, the penalty contribution to the coefficient derivative is

$$\alpha w_p(\omega_n) \phi'_\gamma\big(w_p(\omega_n)|c_n|\big)\, \mathrm{sign}(c_n).$$

Algorithm 2 uses $\psi_k(z) = z^q$, $q = 2/(k+1) < 1$, on the unit sphere. Because $\psi_k'(0^+) = +\infty$, it uses a finite-step decrease test instead of (63).

In both algorithms the correction step freezes the locations $\boldsymbol{\omega}$ and solves the resulting finite-dimensional coefficient problem by the semismooth Newton method applied to the proximal residual

$$c = \operatorname{prox}_{\sigma\alpha\phi}\bigl(c - \sigma\nabla_c l^M(\mu_{\boldsymbol{\omega},c})\bigr), \qquad \operatorname{prox}_{\sigma\alpha\phi}(v) := \operatorname*{arg\,min}_{z\in\mathbb{R}}\Bigl\{\tfrac{1}{2}|z-v|^2 + \sigma\alpha\phi(|z|)\Bigr\}, \tag{64}$$

with a fixed $\sigma > 0$, the scalar proximal acting coordinatewise, and ϕ standing for the penalty in force, ϕ_γ or ψ_k. For ϕ_γ this is the construction of [24], whose penalty has finite slope at the origin. For ψ_k with $q < 1$ the map $c \mapsto |c|^q$ is not locally Lipschitz at the origin and that construction does not apply: the scalar proximal is then a one-dimensional problem of the form treated in (65) below, vanishing beneath a threshold and equal to the larger stationary point above it, and the correction is used as a local method for which no convergence guarantee is claimed.

Both algorithms evaluate every candidate of one iteration against the same residual r_{μ_t} and insert up to a prescribed number $N_{\text{ins}} \geq 1$ of them together. The decrease estimates above are one-candidate statements: for a joint insertion $\sum_{i=1}^r c_i\delta_{\omega_i}$ at distinct new locations the penalty is additive, but the fidelity change carries the additional cross terms

$$\sum_{i<j} c_i c_j \bigl\langle K^M(\omega_i), K^M(\omega_j)\bigr\rangle_{\mathbb{R}^{M(d+1)}},$$

which no individual test controls. For $N_{\text{ins}} = 1$ the guarantees of Theorem 5.1 and of the finite-step criterion apply as stated; for $N_{\text{ins}} > 1$ they do not, and neither a decrease per outer iteration nor monotonicity of J^M along the iterates is claimed. Restoring the guarantee requires recomputing the residual after each insertion, or testing the joint increment of the whole batch.

Algorithm 1 Iterative node insertion for J^M

Choose T_{out}, N_{trial}, N_{ins}, $\varepsilon_{\text{prune}}$, and an initial network $\boldsymbol{\omega}^{(0)}, c^{(0)}$ of width $N(0)$; set $t = 0$.
while $t < T_{\text{out}}$ **do**
    Evaluate $\omega = 0$ and sample N_{trial} random directions $\widehat{\omega} \in \mathbb{S}^d$.
    Refine the directions on $\mathbb{S}^d$ by constrained L-BFGS applied to $|P^M_{p,\mu_t}(\widehat{\omega})|$, and remove near-duplicate candidates.
    For each retained direction, optimize $0 \leq \lambda \leq R(\mu_t)$ in $|P^M_{p,\mu_t}(\lambda\widehat{\omega})|$ over the search region from Theorem 5.1.
    Keep the candidates satisfying (63); terminate if there is none.
    Retain the at most N_{ins} candidates with the largest $\Delta(\mu_t, \omega)$, merge any exact repetition, and insert each with coefficient $c(\omega)$.
    Apply the semismooth Newton correction (64) to (62); accept it only if its objective is no larger than the post-insertion objective, and otherwise retain the post-insertion coefficients.
    Remove nodes with $|c_n| \leq \varepsilon_{\text{prune}}$ (resulting in width $N(t+1)$).
    $t = t + 1$.
end while

For the q-penalty with $q < 1$, the zero network is always a local minimizer, and an infinitesimal insertion test gives no condition away from the existing support (Remark 4.12). We replace it with a *finite-step decrease criterion*: given a candidate ω, we evaluate the objective change

$$\Delta J^M_{\psi_k}(c;\omega) = cP^M_{\mu_t}(\omega) + \tfrac{1}{2}c^2\|K(\omega)\|_M^2 + \frac{\alpha}{q}|c|^q,$$

where

$$\|K(\omega)\|_M^2 := \frac{1}{M}\sum_{m=1}^M \left(|[K(\omega)](x^m)|^2 + |\nabla K(\omega)(x^m)|^2\right),$$

and we accept ω if $\min_c \Delta J^M_{\psi_k}(c;\omega) < 0$.

The minimizing coefficient has sign $-\operatorname{sign}(P^M_{\mu_t}(\omega))$, and only that sign can decrease the objective; writing $A := \|K(\omega)\|^2_M > 0$, its magnitude solves

$$Az + \alpha z^{q-1} = |P^M_{\mu_t}(\omega)|. \tag{65}$$

The left-hand side is strictly convex with minimum at

$$z_{\text{turn}} := \left(\frac{\alpha(1-q)}{A}\right)^{1/(2-q)}.$$

Below that minimum value (65) has no positive root, $\Delta J^M_{\psi_k}(\cdot;\omega)$ is nondecreasing and ω is rejected; otherwise there are two roots, and the larger one z^* is the only positive local minimizer of the increment. We compute z^* by Newton's method to the right of z_{turn}, where the left-hand side is increasing and convex, so the iterates decrease monotonically to it, and we accept ω with initial outer weight $c^* := -\operatorname{sign}(P^M_{\mu_t}(\omega))z^*$ when $\Delta J^M_{\psi_k}(c^*;\omega) < 0$. The subsequent coefficient solve minimizes (42), whose penalty coefficient is α rather than α/q. Since $\alpha/q > \alpha$ for $q < 1$, a candidate accepted by the displayed finite-step test and inserted alone strictly decreases (42). The value c^* is an initialization for that solve, not a stationary point of its objective.

Algorithm 2 Iterative node insertion for $J^M_{\psi_k}$

1: Choose T_{out}, N_{trial}, N_{ins}, $\varepsilon_{\text{prune}}$, and an initial network $\boldsymbol{\omega}^{(0)}, c^{(0)}$ of width $N(0)$; set $t = 0$.
2: **while** $t < T_{\text{out}}$ **do**
3: Sample N_{trial} random nodes $\omega \in \mathbb{S}^d$.
4: Refine them on $\mathbb{S}^d$ by constrained L-BFGS applied to $|P^M_{\mu_t}(\omega)|$, and remove near-duplicate candidates.
5: For each candidate compute c^* from (65) when it exists, and keep the candidate when $\Delta J^M_{\psi_k}(c^*;\omega) < 0$; terminate if there is none.
6: Retain the at most N_{ins} kept candidates with the smallest $\Delta J^M_{\psi_k}(c^*;\omega)$ and insert each with outer weight c^*.
7: Minimize (42) in the coefficients by the semismooth Newton method (64); accept the result only if its objective is no larger than the post-insertion objective, and otherwise retain the post-insertion coefficients.
8: Remove nodes with $|c_n| \le \varepsilon_{\text{prune}}$ (resulting in width $N(t+1)$).
9: $t = t + 1$.
10: **end while**

6. Numerical examples

The examples compare the activation functions and regularizers on a smooth value function and on a value function with a discontinuous gradient. All runs use seed 42. Each reported parameter choice is run once, and the validation sets are random subsets of the PMP samples. Input normalization is computed from the full sample before the split.

For a fitted value function $\widehat{V}$, the relative errors are

$$\mathrm{Err}_{L^2} := \frac{\|\widehat{V} - V\|_{L^2}}{\|V\|_{L^2}},$$

$$\mathrm{Err}_{H^1} := \frac{\left(\|\widehat{V} - V\|^2_{L^2} + \|\nabla\widehat{V} - \nabla V\|^2_{L^2}\right)^{1/2}}{\left(\|V\|^2_{L^2} + \|\nabla V\|^2_{L^2}\right)^{1/2}}.$$

Both norms are computed empirically on the validation or evaluation set specified for each experiment.

6.1. **Van der Pol Oscillator.** We consider the optimal control of the Van der Pol oscillator:

$$\min_{u\in L^2(0,T;\mathbb{R})} \mathcal{C}(u;0,x) = \int_0^T \big(y_1^2(t) + y_2^2(t) + \eta u^2(t)\big)\,dt$$

subject to

$$\begin{cases} \frac{dy_1}{dt} = y_2, \\ \frac{dy_2}{dt} = -y_1 + y_2(1-y_1^2) + u, \\ (y_1(0), y_2(0)) = (x_1, x_2), \end{cases}$$

In this setting, $d = 2$, $m_u = 1$, $\eta = 0.1$, $T = 3$, and

$$f(y) = \begin{pmatrix} y_2 \\ -y_1 + y_2(1-y_1^2) \end{pmatrix}, \quad g(y) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad h(y) = y_1^2 + y_2^2.$$

By the optimality condition, $u^* = -\frac{1}{2\eta} g(y^*)^\top p^* = -\frac{1}{2\eta} p_2^*$. The two-point boundary value problem (TPBVP) becomes

$$\begin{cases} \dot{y}_1^* = y_2^*, \\ \dot{y}_2^* = -y_1^* + y_2^*(1 - y_1^{*2}) + u^*, \\ -\dot{p}_1^* = 2y_1^* - p_2^*(1 + 2y_1^* y_2^*), \\ -\dot{p}_2^* = 2y_2^* + p_1^* + p_2^*(1 - y_1^{*2}), \\ y^*(0) = x, \quad p^*(T) = 0. \end{cases} \tag{66}$$

For each initial condition $x = (x_1, x_2)$ sampled on the grid $[-3,3]^2$, we solve (66) using a Barzilai–Borwein gradient method with backtracking line search. The control is represented in a Legendre polynomial basis $u(t) = \sum_{i=0}^{N_b-1} c_i P_i(t)$ with $N_b = 30$ coefficients. The optimization yields the dataset $\{x^m, V(x^m), \nabla V(x^m)\}_{m=1}^M$, where $V(x^m) = \mathcal{C}(u^*;0,x^m)$ and $\nabla V(x^m) = p^*(0)$.

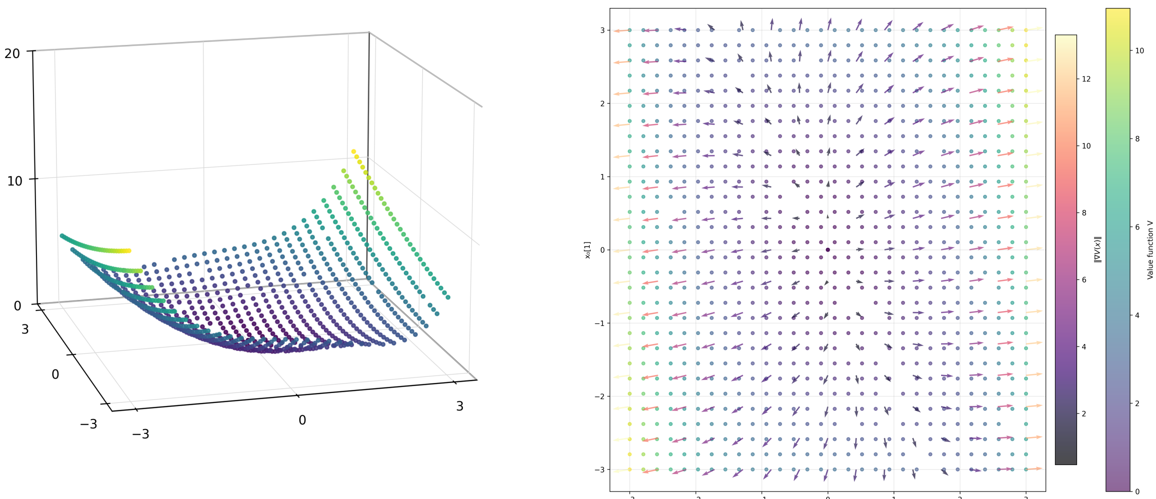


FIGURE 2. Left: value function $V(x)$ over $[-3,3]^2$. Right: $V(x)$ (dot color) and $\nabla V(x)$ (arrows); arrow color indicates $\|\nabla V(x)\|$.

We solve the open-loop problem on a 30×30 grid over $[-3,3]^2$ ($M = 900$ data points). All experiments use $T_{\text{out}} = 10$ outer iterations, $N_{\text{trial}} = 50$ sampled candidate nodes per iteration, $N_{\text{ins}} = 15$, and $\varepsilon_{\text{prune}} = 10^{-8}$. The insertion batches are therefore of the heuristic kind discussed in Section 5.

6.1.1. *Effect of β and p.*

Dependence on α and β. We first investigate the dependence of the validation error on the two regularization coefficients. We use Algorithm 1 with $\gamma = 1$, equal value and gradient weights, and seed 42. The tested values are

$$\alpha \in \{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}, \quad \beta \in \{0, 10^{-10}, 10^{-5}, 10^{-2}, 10^{-1}\}, \quad p \in \{2.01, 2.5, 3, 4\},$$

for tanh, softplus, and the Gaussian. Since p has no effect when $\beta = 0$, the same fit is used for every displayed value of p.

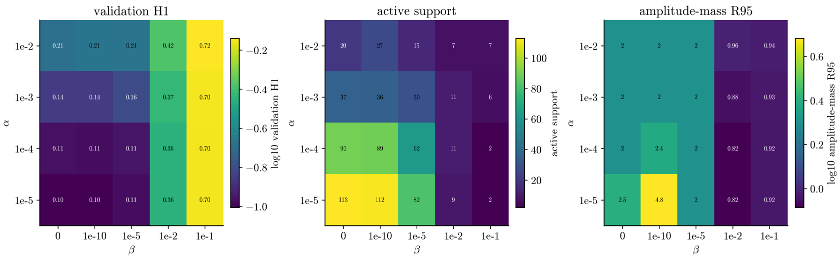


FIGURE 3. Relative H^1 validation error over the $\alpha \times \beta$ grid for the Gaussian at $p = 3$. At $\alpha = 10^{-5}$ the error changes little for $\beta \le 10^{-5}$ and then increases; at $\alpha = 10^{-2}$ it is high for all tested values of β.

Effect of β on validation error, support size, and $R_{0.95}$. For a discrete measure $\mu = \sum_j c_j \delta_{\omega_j}$, define

$$R_{0.95}(\mu) := \inf \{R \ge 0 : |\mu|(B_R) \ge 0.95\, |\mu|(\Omega)\}\,.$$

For the Gaussian with $p = 3$ and $\alpha = 10^{-5}$, moving from $\beta = 0$ to $\beta = 10^{-5}$ changes the validation H^1 error from 0.0994 to 0.1053 while the support falls from 113 to 82 neurons and $R_{0.95}$ from 2.49 to 2.01. At $\beta = 10^{-2}$, the error is 0.3556, the support contains 9 neurons, and $R_{0.95} = 0.82$. The smallest error among the runs with $\beta > 0$ is 0.0989, obtained for the Gaussian at $\alpha = 10^{-5}$, $\beta = 10^{-10}$, and $p = 3$.

For tanh and $\beta = 10^{-1}$, the algorithm returns the zero measure for every tested value of α and p. At $\mu = 0$, no candidate satisfies the insertion condition.

Effect of p on $R_{0.95}$. Lemma 3.20 gives

$$R_* = \max\left\{1, \left(\frac{2^{s_1} C_P}{\beta}\right)^{1/(p-s_1)}\right\}.$$

For softplus ($s_1 = 1$), the β-dependent factor $\beta^{-1/(p-s_1)}$ is of order 10^{10} at $p = 2.01$, $\beta = 10^{-10}$, but only of order 10^3 at $p = 4$ and the same β. The unknown residual-dependent factor $2^{s_1} C_P$ prevents an absolute value of R_* from being inferred from these experiments. For softplus with $\alpha = 10^{-4}$ and $\beta = 10^{-2}$, the observed $R_{0.95}$ values are 1.74, 1.43, and 1.09 for $p = 2.5$, 3, and 4, respectively. This observed decrease is consistent with the dependence of R_* on p, although C_P differs between the fitted networks.

Upper bound on the radial parameter. The algorithm restricts λ to $\lambda \le e^5$. For the Gaussian runs reported above, $R_{0.95} \le 2.49 \ll e^5$; hence at least 95% of the total variation lies well inside this upper bound.

6.1.2. *Algorithm 1: log penalty and moment norm.* We investigate the effect of including gradient data for softplus, tanh, and the Gaussian. Table 1 compares fitting the values alone with fitting both values and gradients.

TABLE 1. Effect of gradient-augmented fitting ($\gamma = 1$, $\alpha = 10^{-5}$, $\beta = 10^{-10}$; $p = 2.01$ for softplus and tanh, and $p = 3$ for the Gaussian).

Activation	L^2 trained	H^1 trained
softplus	8.70×10^{-2}	4.31×10^{-1}
tanh	4.46×10^{-2}	4.43×10^{-1}
Gaussian	3.28×10^{-2}	4.15×10^{-1}

(A) Effect of gradient training on Err_{L^2}

Activation	L^2 trained	H^1 trained
softplus	5.63×10^{-1}	1.25×10^{-1}
tanh	4.90×10^{-1}	1.67×10^{-1}
Gaussian	4.71×10^{-1}	9.89×10^{-2}

(B) Effect of gradient training on Err_{H^1}

Choice of activation function in gradient training. We compare softplus, tanh, and the Gaussian. Because the gradient-augmented loss contains both the value and gradient errors, $K(\omega)$ contributes to the approximation of V and $\nabla K(\omega)$ contributes to the approximation of ∇V. Figure 4 contrasts the three activations through their value, first derivative, and second derivative. The softplus derivative increases from 0 to 1, the derivative of tanh tends to zero at both ends of the real line, and the Gaussian derivative is localized and changes sign. These derivatives generate different columns $a\,\rho'(a \cdot x + b)$ in the gradient fitting problem.

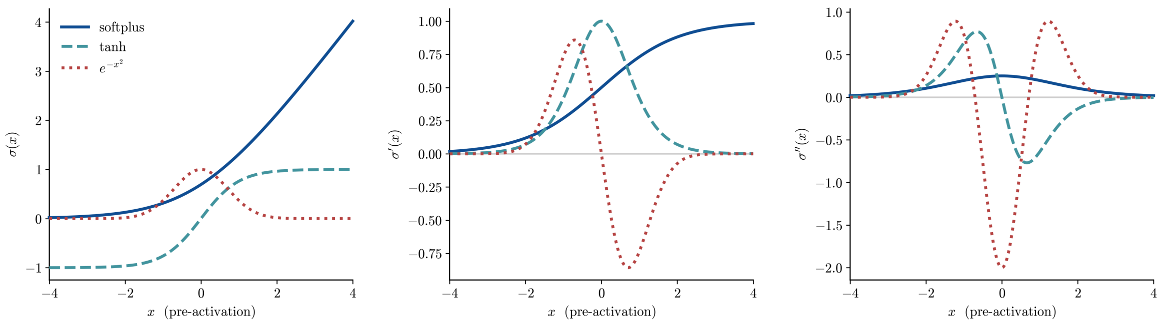


FIGURE 4. The three smooth profiles used by Algorithm 1 — softplus, tanh, and the Gaussian e^{-z^2} — shown as ρ, ρ', and ρ''. Under the gradient-augmented loss, the first derivative is the basis that reconstructs ∇V.

For the tested parameters, the Gaussian has the smallest relative H^1 error. All three approximate the smooth value function, with different support sizes.

TABLE 2. Support size and validation error for the parameter values in Table 1.

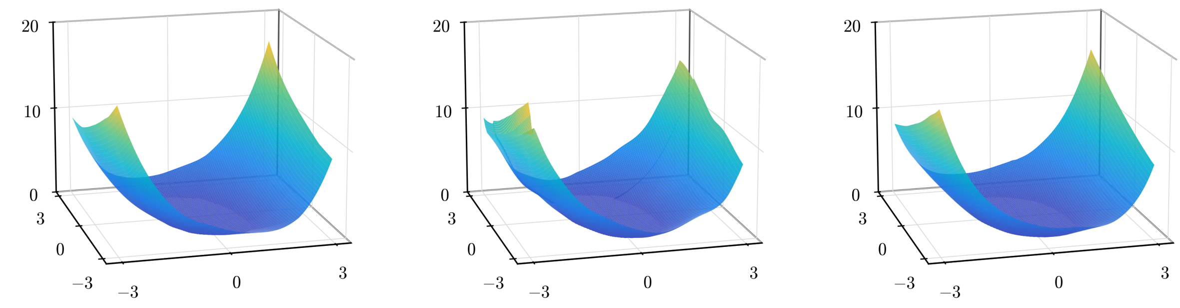
Activation	N	Err_{L^2}
softplus	62	8.70×10^{-2}
tanh	109	4.46×10^{-2}
Gaussian	97	3.28×10^{-2}

(A) L^2 trained

Activation	N	Err_{H^1}
softplus	35	1.25×10^{-1}
tanh	76	1.67×10^{-1}
Gaussian	112	9.89×10^{-2}

(B) H^1 trained

With the H^1 loss, all three gradient errors fall substantially. The Gaussian is most accurate (9.89×10^{-2}) and uses 112 atoms. Softplus reaches 1.25×10^{-1} with only 35 atoms, while tanh reaches 1.67×10^{-1} with 76.



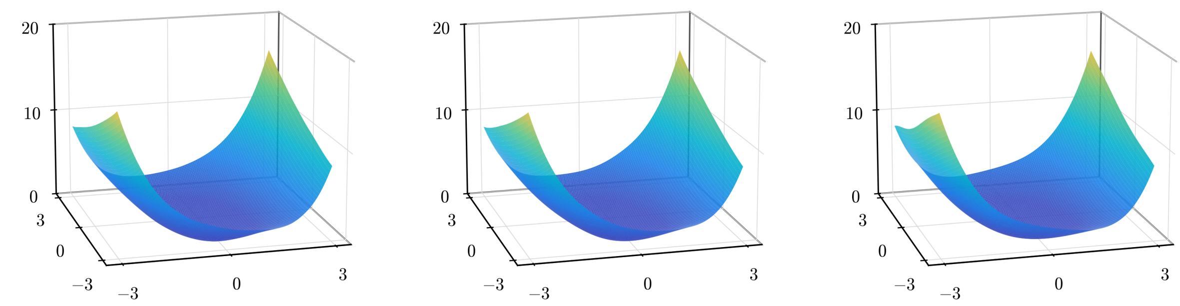
(A) softplus
($N = 35$, $\mathrm{Err}_{H^1} = 0.125$).

(B) tanh
($N = 76$, $\mathrm{Err}_{H^1} = 0.167$).

(C) Gaussian
($N = 112$, $\mathrm{Err}_{H^1} = 0.099$).

FIGURE 5. Learned value functions under the H^1 loss using the parameter values in Table 1. All three approximate the smooth value surface; they differ in support size and accuracy.

6.1.3. *Algorithm 2: $|c|^q$-penalty with* ReLU^k *activation.* Algorithm 2 uses the profile $\rho(z) = \mathrm{ReLU}(z)^k$ with the penalty ψ_k, namely $\alpha \sum_i |c_i|^q$ with $q = 2/(k+1)$. The swept parameter is the power k; all fits below use $\alpha = 10^{-5}$.



(A) $k = 2$
($N = 41$, $\mathrm{Err}_{H^1} = 0.10$).

(B) $k = 3$
($N = 20$, $\mathrm{Err}_{H^1} = 0.10$).

(C) $k = 5$
($N = 21$, $\mathrm{Err}_{H^1} = 0.10$).

FIGURE 6. H^1-trained networks with $\alpha = 10^{-5}$ for $k \in \{2, 3, 5\}$.

Under the L^2 loss the gradient error stays large for every power ($\mathrm{Err}_{H^1} \approx 4$–$5 \times 10^{-1}$): the value-only fit ignores ∇V, so the atoms are placed without regard to the gradient field. The H^1 loss drives the gradient error down to $\approx 1.0 \times 10^{-1}$ for *all* powers (Table 3).

Increasing k changes both the activation and the induced exponent $q = 2/(k+1)$. The support drops from 41–42 atoms for $k = 2$ and 2.01 to 20–22 atoms for $k = 3, 4, 5$, while the error remains between 0.098 and 0.104. Under the L^2 loss the fits use 10–14 atoms and have $\mathrm{Err}_{H^1} \approx 4$–$5 \times 10^{-1}$ (Table 3 (a)).

TABLE 3. Algorithm 2: network size and validation error vs. the atom power k ($\alpha = 10^{-5}$).

Power	N	Err_{L^2}	Err_{H^1}
$k = 2$	14	5.05×10^{-2}	4.83×10^{-1}
$k = 2.01$	14	4.83×10^{-2}	4.81×10^{-1}
$k = 3$	12	5.98×10^{-2}	4.94×10^{-1}
$k = 4$	12	5.48×10^{-2}	4.61×10^{-1}
$k = 5$	10	4.77×10^{-2}	4.04×10^{-1}

(A) L^2 trained

Power	N	Err_{L^2}	Err_{H^1}
$k = 2$	41	4.16×10^{-1}	9.84×10^{-2}
$k = 2.01$	42	4.17×10^{-1}	9.93×10^{-2}
$k = 3$	20	4.17×10^{-1}	1.00×10^{-1}
$k = 4$	22	4.16×10^{-1}	9.97×10^{-2}
$k = 5$	21	4.14×10^{-1}	1.04×10^{-1}

(B) H^1 trained

6.1.4. *Summary.* Figure 7 plots the running best relative H^1 error. The Gaussian reaches 9.89×10^{-2} with 112 atoms, while ReLU^2 and ReLU^5 reach comparable errors with roughly 40 and 20 atoms. Softplus stops at 1.25×10^{-1} with 35 atoms, and therefore uses fewer atoms than the Gaussian at a larger error.

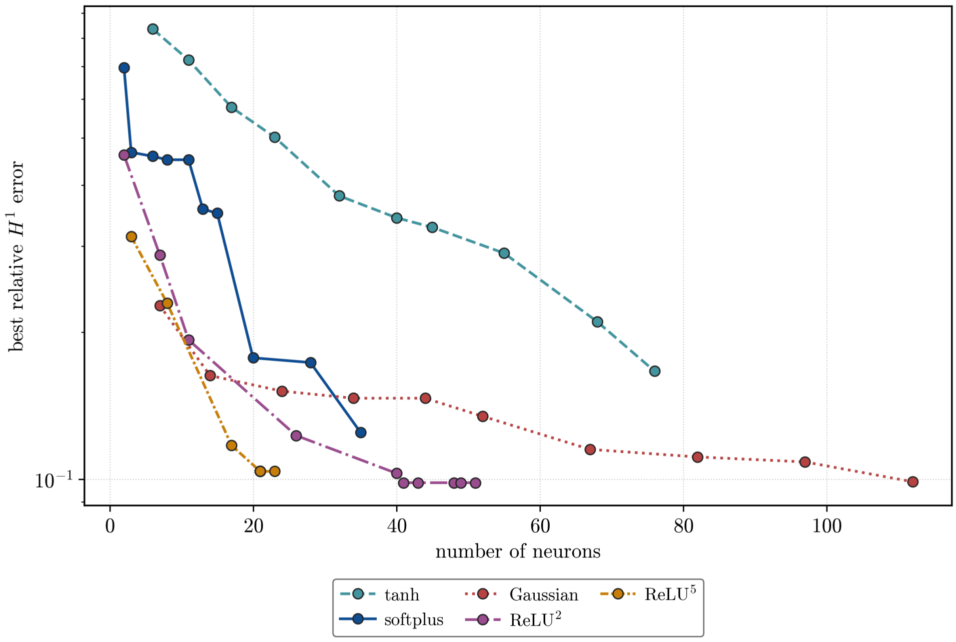
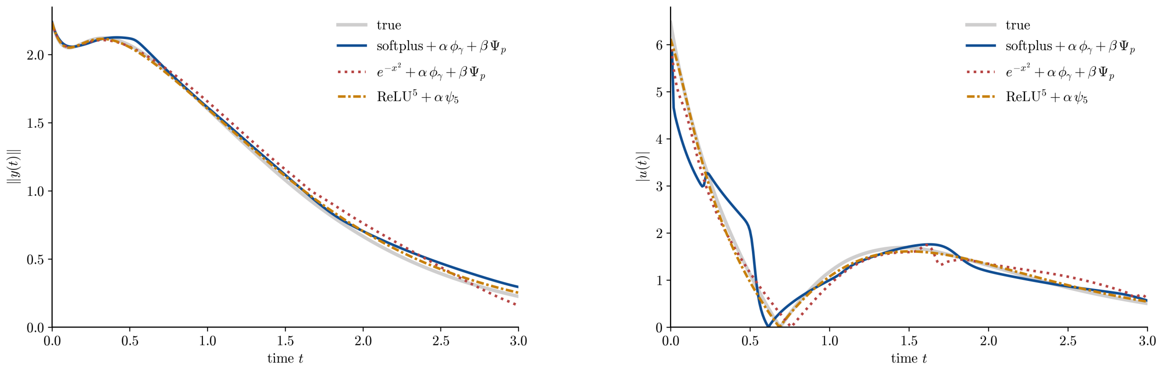


FIGURE 7. Running best relative H^1 validation error against support size. Non-homogeneous profiles use the parameter values stated in Table 1; homogeneous profiles use $\alpha\psi_k$ at $\alpha = 10^{-5}$.

We synthesize feedback laws from $y_0 = (2, 1)$ and roll them out over the horizon $T = 12$ with fixed-step RK4, zero-order-hold controls clipped to $|u| \leq 50$ as a numerical guard. The reference law interpolates the time-zero costate component of the dataset with a C^1 Clough–Tocher interpolant and applies it as a stationary feedback. The finite-horizon optimal feedback is time-dependent and the rollout horizon exceeds the data horizon $T = 3$. Therefore its rollout cost 6.48 is not $V(0, y_0)$, and the comparison is not a test of finite-horizon optimality. Figure 8 shows $\|y(t)\|$ and $|u(t)|$ for softplus, the Gaussian, and ReLU^k ($k = 5$), beside this reference. All three drive the state to the origin and track the reference control, at essentially its closed-loop cost (6.63, 6.58, 6.49 vs. 6.48).

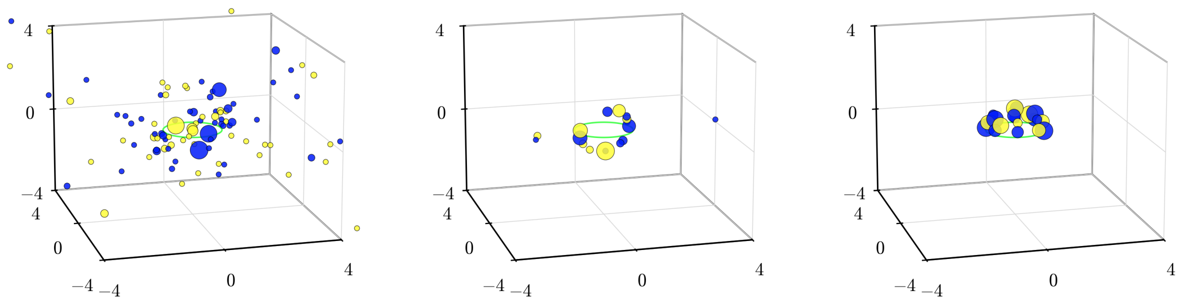


(A) State norm $\|y(t)\|$.

(B) Control $|u(t)|$.

FIGURE 8. Evolution of the state and control from $y_0 = (2, 1)$ under the feedback synthesized from each fitted value function and under the stationary reference law obtained from the time-zero costates.

Figure 9 shows the inner parameters for the Gaussian, softplus, and ReLU^k $(k = 5)$. The two algorithms produce structurally different representations: Algorithm 2 constrains its atoms to the unit sphere $\mathbb{S}^2$, while Algorithm 1 leaves them free — the Gaussian atoms in particular span a wide range of norms.



(A) Gaussian, $p = 3$.

(B) softplus, $p = 2.01$.

(C) ReLU$^5 + \alpha\,\psi_5$.

FIGURE 9. Inner parameters (a_1, a_2, b), with the unit circle in the plane $b = 0$ shown in light green. The two nonhomogeneous fits use $\alpha = 10^{-5}$, $\beta = 10^{-10}$, $\gamma = 1$; dot color gives the sign and dot size the magnitude of the outer weight.

6.2. **Pendulum Swing-Up.** The question in this example is how a sparse gradient-trained network approximates a continuous value function whose gradient jumps across a switching set. A finite network with a continuously differentiable activation has a continuous gradient and therefore cannot reproduce the jump exactly. Rectified homogeneous activations can place changes of slope along hyperplanes. We compare these two classes at the support sizes returned by Algorithms 1 and 2.

Figure 10 contrasts three activation functions. The rectified linear unit ReLU has a step derivative, so a single neuron can represent a gradient jump on its hyperplane. The derivative of ReLU^2 is continuous and piecewise linear, whereas the softplus derivative is smooth. The positive homogeneity of the rectified functions allows the fractional-exponent formulation of Section 4.

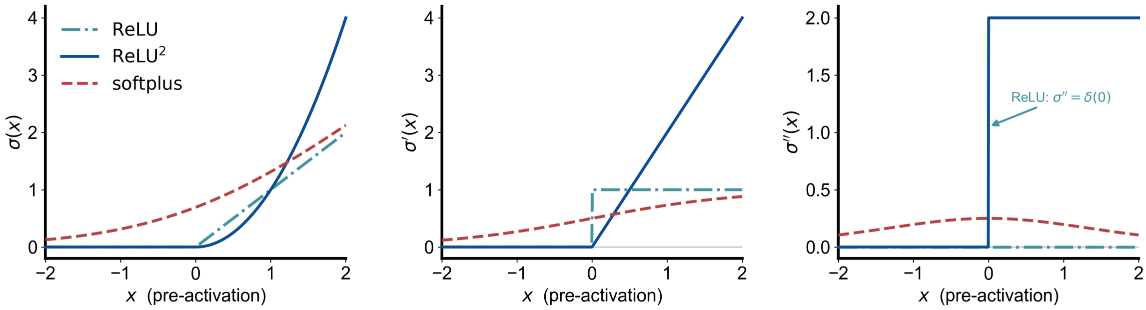


FIGURE 10. The profiles ρ, ρ', and ρ'' for ReLU, ReLU^2, and softplus. The derivative of ReLU jumps, the derivative of ReLU^2 has a kink, and the softplus derivative is smooth. The second distributional derivative of ReLU is a Dirac measure at the origin.

The example we consider follows from [11]. We write the state as

$$x = (x_1, x_2)^\top = (\theta, \dot\theta)^\top,$$

the pendulum angle and angular velocity. (The letter ω remains reserved for the network parameters.) We consider the optimal control problem

$$\min_{u \in L^2(0,\infty;\mathbb{R})} \mathcal{C}(u; 0, x) = \int_0^\infty \left(\sin^2\theta(t) + (\cos\theta(t) - 1)^2 + \dot\theta^2(t) + \eta u^2(t)\right) dt$$

subject to

$$\begin{cases} m\ell^2\,\ddot\theta = -b\dot\theta + mg\ell\sin\theta + u, \\ (\theta(0), \dot\theta(0)) = (x_1, x_2). \end{cases}$$

The target equilibria are $x_{U,j} := (2j\pi, 0)$, $j \in \mathbb{Z}$; for a selected target we write $x_U = x_{U,j}$.

In this setting the state dimension is $d = 2$ and the control is scalar. The physical and cost parameters are $m = \ell = 1$, $b = 0.1$, $g = 9.8$, and $\eta = 1$. To initialize the backward integration, we use the local linear-quadratic regulator (LQR) approximation near x_U. Let $\widetilde{x} = x - x_U$ be the local coordinate around the upright equilibrium. Linearizing the dynamics at $(x_U, 0)$ gives

$$\dot{\widetilde{x}} = A\widetilde{x} + Bu, \qquad A = \begin{pmatrix} 0 & 1 \\ \frac{g}{\ell} & -\frac{b}{m\ell^2} \end{pmatrix}, \qquad B = \begin{pmatrix} 0 \\ \frac{1}{m\ell^2} \end{pmatrix}.$$

The running cost is locally approximated by

$$|\widetilde{x}|^2 + \eta u^2.$$

The local infinite-horizon LQR value function is

$$V^\infty(\widetilde{x}) = \widetilde{x}^\top P \widetilde{x},$$

where P is the positive definite solution of the algebraic Riccati equation

$$A^\top P + PA - \frac{1}{\eta} PBB^\top P + I_2 = 0.$$

The superscript ∞ refers to the infinite-horizon LQR problem. For $\varepsilon > 0$, we define the local LQR sublevel set and boundary by

$$\mathcal{E}_\varepsilon = \{\widetilde{x} : V^\infty(\widetilde{x}) \le \varepsilon\}, \qquad \partial\mathcal{E}_\varepsilon = \{\widetilde{x} : V^\infty(\widetilde{x}) = \varepsilon\}.$$

The corresponding state-costate system becomes

$$
\begin{cases}
m\ell^2\,\ddot{\theta}^* = -b\dot{\theta}^* + mg\ell\sin\theta^* + u^*, \\
-\dot{p}_1^* = 2\sin\theta^* + \dfrac{g}{\ell}\cos\theta^*\,p_2^*, \\
-\dot{p}_2^* = 2\dot{\theta}^* + p_1^* - \dfrac{b}{m\ell^2}p_2^*, \\
x^*(0) = x, \quad x^*(T_x) - x_U \in \partial\mathcal{E}_\varepsilon, \quad p^*(T_x) = \nabla V^\infty(x^*(T_x) - x_U).
\end{cases}
\tag{67}
$$

In the experiment we use $\varepsilon = 2\times 10^{-4}$, and on $\partial\mathcal{E}_\varepsilon$ the terminal costate is

$$p^*(T_x) = \nabla V^\infty(\widetilde{x}^*(T_x)) = 2P\widetilde{x}^*(T_x).$$

For the data generation, we integrate 2000 PMP paths backward from the local LQR boundary $\partial\mathcal{E}_\varepsilon$. Along each path, the accumulated running cost plus the boundary value ε gives $V(x)$, and the costate gives $\nabla V(x) = p$. The backward integration stops when the accumulated value reaches 100; the training samples are restricted to values below 50. This yields the dataset

$$\{x^m, V(x^m), \nabla V(x^m)\}_{m=1}^M.$$

The construction provides samples on both sides of the gradient discontinuity. For $j\in\mathbb{Z}$, let $V_{2j\pi}(x)$ be the fixed-target value obtained by minimizing over paths that reach the local LQR neighbourhood of $(2j\pi, 0)$. The stationary value is

$$V(x) = \min_{j\in\mathbb{Z}} V_{2j\pi}(x).$$

In the region where the targets $(0,0)$ and $(2\pi,0)$ are minimizing, this reduces to $V(x) = \min\{V_0(x), V_{2\pi}(x)\}$. Define the representative switching set between these adjacent targets by

$$S_{0,2\pi} := \{x\in\mathbb{R}^2 : V_0(x) = V_{2\pi}(x) = V(x)\}.$$

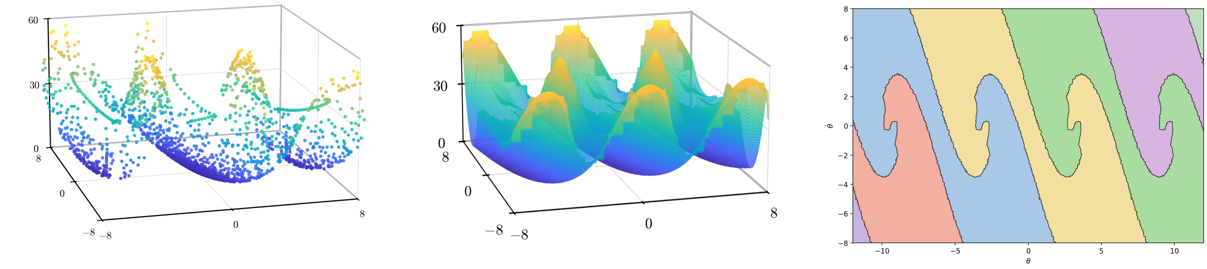
On the numerical domain $D = [-8,8]^2$, the periodic switching set and its closed δ-neighbourhood are

$$S := D\cap\bigcup_{j\in\mathbb{Z}}\big(S_{0,2\pi} + (2j\pi,0)\big), \qquad S_\delta := \{x\in D : \operatorname{dist}(x,S)\le\delta\}, \qquad \delta>0.$$

The regional errors below use $\delta = 0.3$. At points of S, two adjacent fixed-target values agree while their gradients may differ; the sampled PMP costates provide the corresponding one-sided gradients.

The generator locates $S_{0,2\pi}$ by intersecting equal-value contours with their 2π-shifted copies and uses periodic translates in D. To obtain samples on both sides of S, the raw paths are shifted by $\pm 2\pi$ in θ. A shifted point within distance 0.5 of S is retained when its value agrees with the corresponding fixed-target value V_0 or $V_{2\pi}$ and is at least 0.3 below the first-order extrapolation of the other value. The latter is computed from the 32 nearest raw samples within radius 0.1. This adds 900 samples, 300 on one side of S and 600 on the other. The final dataset contains 3,900 samples: 3,000 from the backward PMP paths and 900 near S. It therefore contains gradient samples from both sides of the discontinuity. Figure 11 shows the dataset generated.

For 23 tested parameter choices, at least 95% of the total variation is carried by atoms at the upper radial bound $\lambda = e^5$. These runs are retained in the numerical record but excluded when selecting the nonhomogeneous networks reported below.



(A) value samples $\{x^m, V(x^m)\}$. (B) value surface $V(x)$. (C) regions associated with the periodic targets.

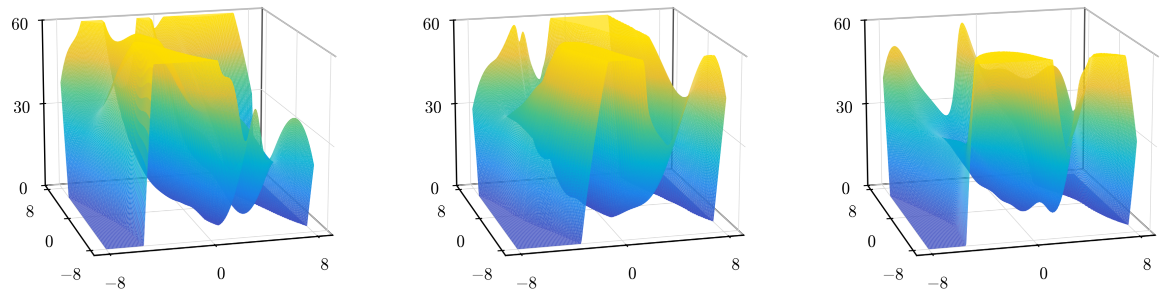
FIGURE 11. Optimal value function $V(x)$ over the $(\theta, \dot{\theta})$ plane, 2π-periodic in θ. (a) The raw (x, V) samples produced by the PMP generator. (b) The reconstructed value surface. (c) Regions where V_0 or another periodic fixed-target value is minimal; their boundaries form S, across which ∇V may be discontinuous.

6.2.1. *Approximation near the switching set.* Figure 12 shows the fitted surfaces on the same physical window and value range as the value function in Figure 11b. The cross-section below compares the approximations near S.

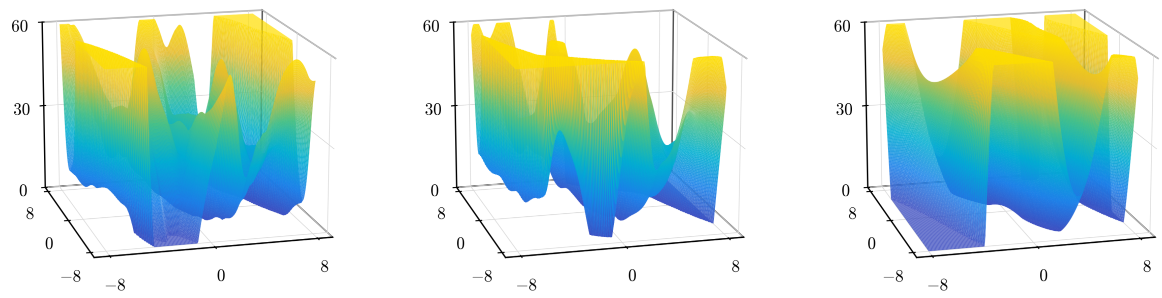
Let x_S be the selected regular point of S in the region with the highest sample density, and let n be a unit normal to S at x_S. Figure 13 shows the cross-section $x(s) = x_S + sn$. Along this section, $V = \min\{V_0, V_{2\pi}\}$. It has a kink at $s = 0$, and the normal gradient $n \cdot \nabla V$ jumps there. The training set contains samples on both sides of this point. No fitted model reproduces the magnitude of the jump: all undershoot the steep gradient on the side $s < 0$. ReLU^2 has the smallest error on $S_{0.3}$ and develops the sharpest change of slope near $s = 0$; the smooth Gaussian, softplus, and tanh fits interpolate through the discontinuity. The higher rectified powers remain smooth enough in their first derivative to miss the jump at the fitted budgets.

We next ask whether adding samples near S reduces the error on $S_{0.3}$. We compare 6,000-sample sets containing 23%, 40%, or 60% of their samples near S with a set obtained by adding 2,000 such samples to the first set. The Gaussian and ReLU^2 networks are evaluated on the same disjoint set of approximately 9.4×10^5 points.

For the tested sets, increasing the proportion of samples near S does not reduce the error on $S_{0.3}$ (Table 4). Thus the observed difficulty is not resolved by additional local samples.

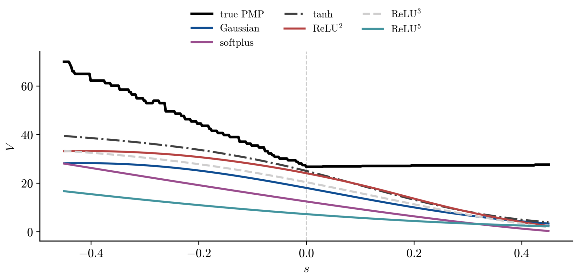


(A) Gaussian (B) softplus (C) tanh

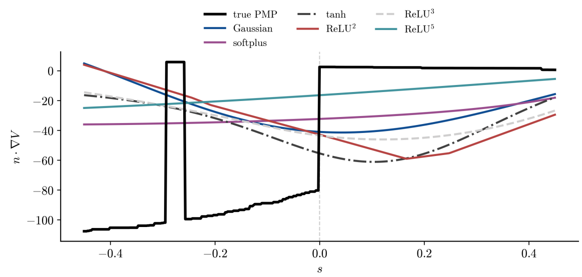


(D) ReLU^2 (E) ReLU^3 (F) ReLU^5

FIGURE 12. Learned value surfaces $\widehat{V}(x)$ over the $(\theta, \dot{\theta})$ window $[-8, 8]^2$, displayed on the common range $\widehat{V} \in [0, 60]$. The ReLU^k models are trained with the fractional-exponent penalty ψ_k, and the nonhomogeneous models with (62) using the parameter values reported in Table 5.



(A) value V along the normal.



(B) normal gradient $n \cdot \nabla V$ along the normal.

FIGURE 13. Normal cross-section of the switching curve, parametrized by the normal distance s. Left: the true value (dashed) is $\min\{V_0, V_{2\pi}\}$ on this cross-section and has a kink at $s = 0$. Right: the true normal gradient jumps at $s = 0$; ReLU^2 develops the sharpest fitted change of slope, while the smooth models interpolate through the discontinuity. The rectangular excursion of the reference curve near $s \approx -0.3$ is a nearest-neighbor artifact in the evaluation of $\min\{V_0, V_{2\pi}\}$, not a feature of V.

Model	Training set	$S_{0.3}$	$D \setminus S_{0.3}$	N
Gaussian	6k, 23% near S	6.25×10^{-1}	6.12×10^{-1}	69
	6k, 40% near S	6.05×10^{-1}	6.02×10^{-1}	106
	6k, 60% near S	6.17×10^{-1}	6.99×10^{-1}	91
	6k + 2k near S	6.17×10^{-1}	6.53×10^{-1}	88
ReLU^2	6k, 23% near S	2.46×10^{-1}	1.56×10^{-1}	108
	6k, 40% near S	2.89×10^{-1}	1.72×10^{-1}	131
	6k, 60% near S	3.46×10^{-1}	2.35×10^{-1}	109
	6k + 2k near S	2.88×10^{-1}	1.84×10^{-1}	131

TABLE 4. Relative H^1 error on $S_{0.3}$ and $D \setminus S_{0.3}$. For each model and training set, the table reports the result with the smallest error on $S_{0.3}$ among the three tested values of α, with every entry in that row taken from the same run. The Gaussian uses $(\beta, p, \gamma) = (10^{-4}, 2.01, 0)$ and $\alpha \in \{10^{-3}, 10^{-4}, 10^{-5}\}$; ReLU^2 uses ψ_2, $\beta = 0$, and $\alpha \in \{10^{-4}, 10^{-5}, 10^{-6}\}$. The symbol N denotes the support size.

6.2.2. ReLU^2 *fits the discontinuous gradient best.* Among the tested activation functions, ReLU^2 has the smallest error in the value and gradient near S.

Figure 14 tracks the best relative H^1 validation error reached as neurons are inserted, for each activation paired with its penalty. The three nonhomogeneous activations use the parameter values in Table 5; the rectified powers use ψ_k, with $\alpha = 10^{-6}$ for $k \in \{2, 3\}$ and $\alpha = 10^{-4}$ for $k = 5$. The error for ReLU^2 becomes smaller than the other errors within the first insertions and reaches $\approx 3.1 \times 10^{-1}$. ReLU^3 is second, while softplus, tanh, and the Gaussian remain above 5×10^{-1} and ReLU^5 has the largest final error. Among the rectified powers, the error increases with k; ReLU^3 remains second overall, while the advantage is lost by $k = 5$.

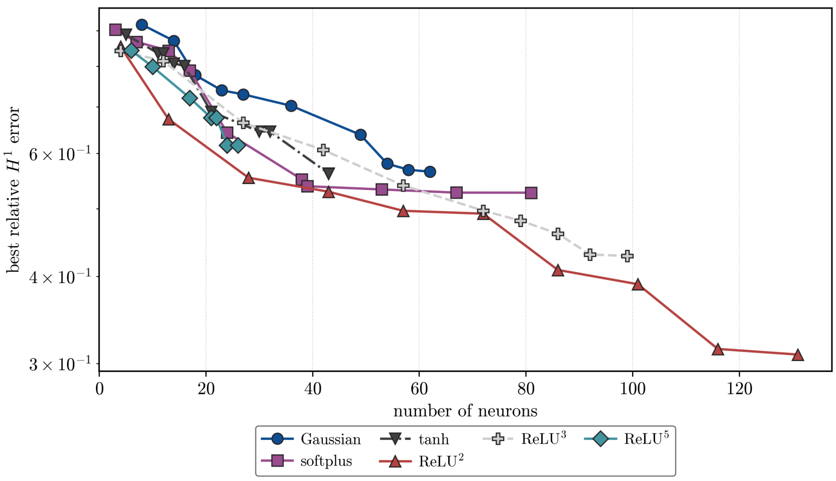


FIGURE 14. Best relative H^1 validation error against neuron count. The nonhomogeneous activations use (62); the homogeneous activations use $J^M_{\psi_k}$ with $\alpha = 10^{-6}$ for $k \in \{2, 3\}$ and $\alpha = 10^{-4}$ for $k = 5$.

The same ordering holds region by region. Figure 15 reports the relative H^1 error on a separate evaluation set of approximately 9.6×10^5 points. Training rows are excluded, and for each run we use the recorded iterate with the smallest training objective. Errors are reported on

$S_{0.3}$ and $D \setminus S_{0.3}$. ReLU^2 has the smallest error in both sets, followed by ReLU^3. Among the moment-regularized profiles, softplus has the smallest errors, followed by tanh and the Gaussian. ReLU^5 is the one model whose two regional errors are nearly identical and largest.

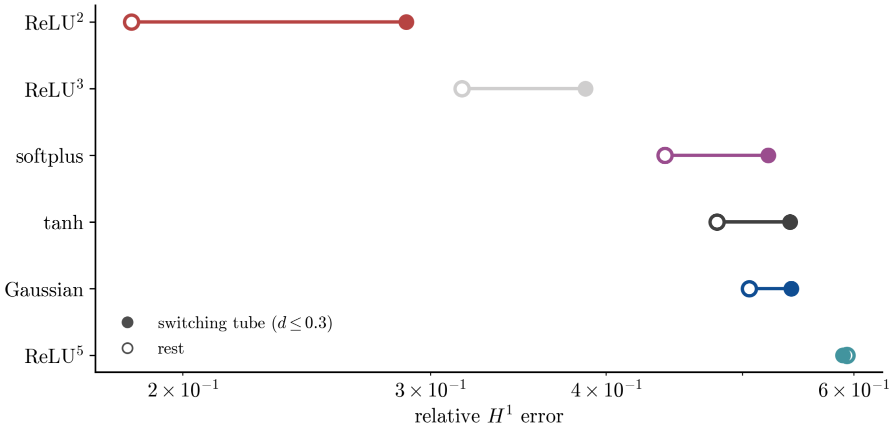


FIGURE 15. Relative H^1 error on $S_{0.3}$ (filled markers) and $D\setminus S_{0.3}$ (open markers), with rows ordered by the latter error.

Figure 16 draws the line $\{a \cdot x + b = 0\}$ associated with each atom in the physical $(\theta, \dot\theta)$ plane, with opacity proportional to the outer weight $|c_n|$, for ReLU^2 (left) and Gaussian (right); the switching curve is shown in black. The lines with the largest coefficients for ReLU^2 are approximately parallel to the diagonal parts of S. The Gaussian lines have a wider range of orientations. Each Gaussian atom $e^{-(a\cdot x+b)^2}$ is constant along its line.

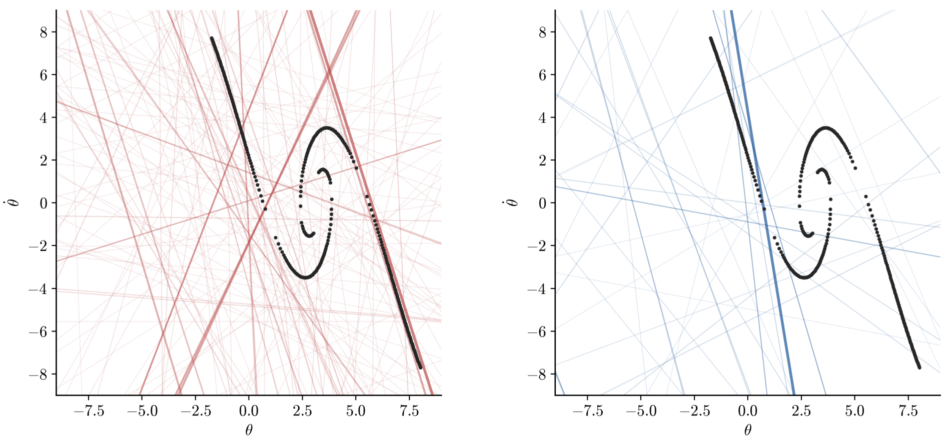
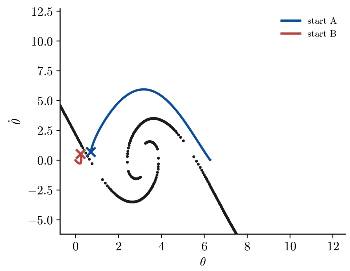


FIGURE 16. Lines $\{a \cdot x + b = 0\}$ in the $(\theta, \dot\theta)$ plane (opacity $\propto |c_n|$) for ReLU^2 (left) and Gaussian (right); the switching curve is black. ReLU^2's strongest ridges run approximately parallel to the switching set; the Gaussian atoms have a wider range of orientations.

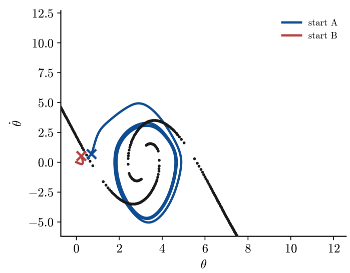
6.2.3. *Feedback synthesis across the switching set.* Minimizing the Hamiltonian in u gives the optimal control in feedback form,

$$u(x) = -\frac{1}{2\,\eta\, m\ell^2}\,\partial_{\dot\theta} V(x);$$

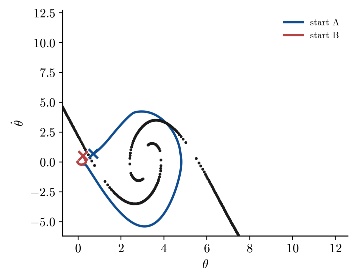
replacing V by a fitted $\widehat{V}$ yields a candidate feedback law. We test each law in closed loop from two initial states placed on either side of the switching curve, $A = (0.71, 0.68)$ and $B = (0.23, 0.53)$, integrating the dynamics over the horizon $T = 10$ with fixed-step RK4 ($\Delta t = 5 \times 10^{-3}$), zero-order-hold controls, and the numerical bound $|u| \leq 30$ (the OCP itself is unconstrained). The optimal paths from the two starts are different. From A, the optimal law swings the pendulum over the top to the upright at $\theta = 2\pi$, while from B it brakes directly to the upright at $\theta = 0$. Thus it must select the path associated with $V_{2\pi}$ at A and the path associated with V_0 at B. The reference law uses the 40 nearest raw PMP samples to compare V_0 and $V_{2\pi}$ and applies the feedback formula using the costate associated with the smaller value.



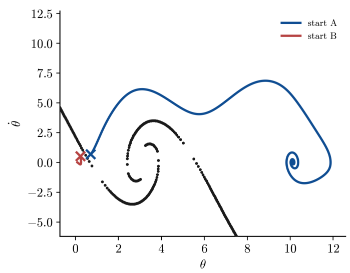
(A) reference (PMP)



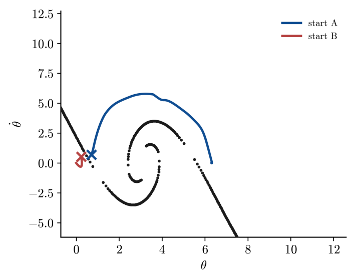
(B) Gaussian, $\gamma = 0$ and moment norm



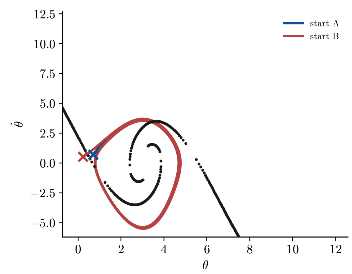
(C) softplus, $\gamma = 1$ and moment norm



(D) tanh, $\gamma = 1$ and moment norm



(E) ReLU^2 + ψ_2



(F) ReLU^5 + ψ_5

FIGURE 17. Closed-loop paths in the $(\theta, \dot{\theta})$ plane from the two starts A (blue) and B (red) on either side of the switching curve (black), one panel per feedback law; all panels share the same axes. The reference law swings over the top from A and brakes to $\theta = 0$ from B; ReLU^2 reproduces both behaviors.

Two laws reach the upright neighbourhood from both starts, ReLU^2 and softplus, and they do so at very different costs (Table 5). From B, ReLU^2 brakes to the upright at essentially the reference cost (10.3 vs 10.2); from A it follows the path associated with $V_{2\pi}$, with cost 57.9 rather than the reference cost 26.2. Softplus reaches the upright from both starts as well, but at 89.2 and 28.2, three times the reference from A and nearly three times from B. The Gaussian, tanh, and ReLU^3 satisfy the terminal criterion only from B, while ReLU^5 satisfies it from neither start. Figure 18 shows the control signal from B: the ReLU^2 law tracks the reference almost exactly, while the others oscillate or saturate.

Effect of β with α, γ, and p fixed. The parameter values in Table 5 differ in α and γ as well as in β. To vary only β, we fix softplus at $\alpha = 10^{-5}$, $\gamma = 10$, $p = 2.01$ with the H^1 loss and vary only β (Table 6). The $\beta = 0$ fit has 77 neurons and relative H^1 error 0.515.

	start $A = (0.71, 0.68)$		start $B = (0.23, 0.53)$	
Feedback law	cost	reaches upright	cost	reaches upright
reference (PMP)	26.2	yes	10.2	yes
Gaussian	186.0	no	15.2	yes
softplus	89.2	yes	28.2	yes
tanh	434.5	no	18.9	yes
$\mathrm{ReLU}^2 + \psi_2$	57.9	yes	10.3	yes
$\mathrm{ReLU}^3 + \psi_3$	29261.8	no	10.3	yes
$\mathrm{ReLU}^5 + \psi_5$	235.9	no	232.4	no

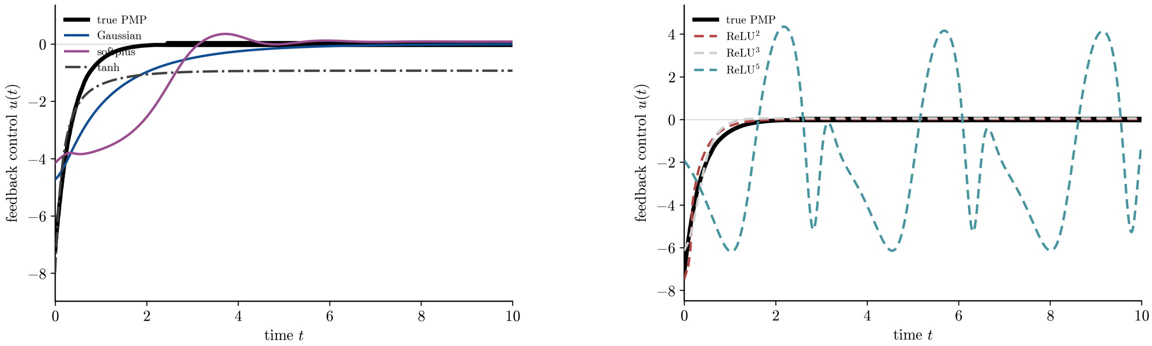
TABLE 5. Closed-loop rollout cost over the horizon $T = 10$ and terminal outcome for each feedback law from the two starts on either side of the switching curve; "reaches upright" means the final state satisfies $\mathrm{dist}(\theta, 2\pi\mathbb{Z}) < 0.4$ and $|\dot{\theta}| < 0.4$ at $T = 10$. This condition concerns the terminal state of one finite rollout and does not assert asymptotic stability. The reference law compares V_0 and $V_{2\pi}$ using the nearest PMP samples. The middle block uses Algorithm 1 (Gaussian $\alpha = 10^{-4}$, $\beta = 10^{-4}$, $p = 2.01$, $\gamma = 0$; softplus $\alpha = 10^{-4}$, $\beta = 10^{-10}$, $p = 2.01$, $\gamma = 1$; tanh $\alpha = 10^{-4}$, $\beta = 10^{-5}$, $p = 3$, $\gamma = 1$), the lower block uses Algorithm 2 at $\beta = 0$, with $\alpha = 10^{-6}$ for $k \in \{2, 3\}$ and $\alpha = 10^{-4}$ for $k = 5$.

β	N	Err_{H^1}	$R_{0.95}$	cost A	cost B
0	77	0.515	1.00	73013.3	66996.7
10^{-10}	69	0.530	1.00	168.3	170.2
10^{-5}	34	0.711	1.00	87.5	92.2
10^{-2}	7	0.857	2.20	82.6	86.0

TABLE 6. Softplus with fixed parameters ($\alpha = 10^{-5}$, $\gamma = 10$, $p = 2.01$, seed 42), varying only β. No row reaches the upright from either start.

Every tested value $\beta > 0$ reduces the rollout costs relative to $\beta = 0$; at $\beta = 10^{-2}$ they are 82.6 and 86.0. None of these four networks reaches the upright from either start. The softplus network in Table 5 also uses different values of α and γ, so its result cannot be attributed to β alone.

In summary, ReLU^2 with the fractional-exponent penalty gives the lowest error for this nonsmooth target and the rollout closest to the reference: it reaches the upright from both starts, at cost 10.3 from B and 57.9 from A. The softplus network in Table 5 also reaches the upright from both starts, at the larger costs 28.2 and 89.2. The Gaussian, tanh, and ReLU^3 succeed only from B, while ReLU^5 succeeds from neither. Adding samples near S does not reverse this ordering of the validation errors.



(A) Algorithm 1. (B) ReLU^k models.

FIGURE 18. Feedback control $u(t)$ from start B, per law, beside the reference (black). The vertical axis is truncated; the softplus control reaches the numerical bound ± 30. The reference brakes to $\theta = 0$ with u rising from ≈ -7 to 0; ReLU^2 tracks it almost exactly.

7. Conclusion and outlook

We developed sparse gradient-augmented neural-network approximations of value functions from PMP data on an unbounded parameter space. For nonhomogeneous profiles, the adopted objective is

$$J(\mu) = L(\mu) + \alpha \Phi_\phi(\mu_p), \qquad \mu_p = w_p \mu, \qquad K_p = \frac{K}{w_p}.$$

The finite right slope $L_\phi = \phi'(0+)$ determines the cost of the continuous part in the measure penalty. Theorem 3.9 gives its disjoint-test representation, and Corollary 3.10 gives weak-* lower semicontinuity and bounded total-variation sublevels. The normalization isometry in Theorem 3.13 places the parameter weight inside the scalar penalty and makes the normalized dictionary vanish at infinity under a polynomial growth gap.

For value growth $s_0 < p$, normalized weak-* compactness gives a global minimizer (Theorem 3.16). Under full H^1 growth $s_1 < p$, Theorem 3.18 gives the global inequality

$$|\bar{P}_p(\omega)| \leq \alpha L_\phi.$$

With strict bounded-interval curvature, every finite-objective local minimizer has no continuous part, distinct normalized atoms satisfy the quantitative separation in Theorem 3.21, and its number of neurons is bounded by a covering number. Corollary 3.22 shows that every global minimizer is finite atomic and that the measure-theoretic minimum equals an attained finite-network minimum. For exact global maximization of $|P_{p,\mu}|$ followed by nonincreasing corrections, Theorem 5.1 gives squared one-step descent, convergence of the excess above the insertion threshold to zero, and a best-iterate rate of order $n^{-1/2}$. The sufficient local condition is given in Theorem 3.24.

For positively k-homogeneous profiles, radial scaling is absorbed into the outer coefficient. The equivalent problem on the unit sphere has the fractional-power penalty $|c|^{2/(k+1)}$, together with the existing first-order conditions, global existence result, and explicit width bound. This infinite-slope case is structurally linked to, but is not obtained by substitution into, the finite-slope theory of Section 3.

For the Van der Pol oscillator, the displayed Gaussian comparison shows the trade-off between validation error, support size, and $R_{0.95}$ as β changes. The Gaussian gives the smallest H^1 error among the tested nonhomogeneous profiles, while softplus uses fewer atoms. The homogeneous ReLU^k networks obtain comparable H^1 errors with smaller supports for larger k.

For the pendulum swing-up, the ReLU^2 errors in Table 4 are larger on $S_{0.3}$ than on its complement, whereas the Gaussian errors are nearly equal for the first two training sets and larger on the complement for the other two. Adding training samples near S does not reduce the error on $S_{0.3}$ at the tested network widths. ReLU^2 gives the smallest error and the closed-loop path closest to the reference from both initial states. The softplus network in Table 5 also reaches the upright from both states, at a larger cost. In the softplus comparison with α, γ, and p fixed, every tested positive value of β reduces the rollout cost relative to $\beta = 0$, but none of those networks reaches the upright.

The experiments are two-dimensional, and the behavior in higher dimensions remains to be studied. For targets with a gradient discontinuity, different weights on samples near and away from S may also be considered.

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